A Linear Exponent Algorithm for Three Tensor Isomorphism

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Abstract

This note develops a candidate deterministic algorithm for isomorphism of arbitrary three-tensors over a finite field of odd characteristic. If the three dimensions have sum N, the proposed bound is $q^{O(N)}$ times a polynomial in the explicit input length. The algorithm returns a complete transporter coset and generators of the full stabilizer. Its recursive objects are literal physical flag layers with their exact correlated affine fibers, rather than arbitrary cubic quotient modules.

The principal changes from a near-linear exponent analysis are a fixed-source-prefix multiset algorithm, geometric normalization of new multiplicities, a row-relative low-rank exit, and a weighted count of the entire recursion tree. Technical appendices give the finite-group transfer, affine module, and group-operation arguments on which the construction depends. This manuscript presents a proposed new research result for mathematical review. It is unpublished and has not been externally peer reviewed. Established external inputs are cited separately from the new arguments.

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1 Problem statement and the recursive algorithm

Let $\mathbb{F}_q = \mathbb{F}_q$, where $q = p^e$ and p is odd. Let V_1, V_2, V_3 be finite-dimensional $\mathbb{F}_q$-spaces, and put

$$N = \dim_{\mathbb{F}_q} V_1 + \dim_{\mathbb{F}_q} V_2 + \dim_{\mathbb{F}_q} V_3.$$

A tensor is a trilinear form $T : V_1 \times V_2 \times V_3 \to \mathbb{F}_q$. The product $G = \mathrm{GL}(V_1) \times \mathrm{GL}(V_2) \times \mathrm{GL}(V_3)$ acts by

$$(gT)(v_1, v_2, v_3) = T(g_1^{-1} v_1, g_2^{-1} v_2, g_3^{-1} v_3).$$

The isomorphism problem asks whether $gT = T'$ for some $g \in G$. For any left action, write $\mathrm{Trans}_H(x, y) = \{h \in H : hx = y\}$. A nonempty transporter is a left coset $h_0 \,\mathrm{Stab}_H(x)$.

We work over $k = \mathbb{F}_p$ whenever defining fields need not contain the original field $\mathbb{F}_q$. The original $\mathbb{F}_q$-linearity equations are retained. Fix once and for all an original faithful physical representation of degree

$$D_0 = O(eN),$$

including the primal and dual rows needed to realize adjoints. All exponential estimates use D_0, even when an auxiliary tensor or field space is much larger. Thus $p^{O(D_0)} = q^{O(N)}$. The parameter L is always the original explicit input length, including original coefficients, generators, supported label actions, and their given maps. It is never redefined to be a current auxiliary description length. An arbitrary original label length enters only a uniform polynomial factor.

Theorem 1.1 (All-input algorithm)**.** *There is a deterministic algorithm which, for arbitrary T, T' over $\mathbb{F}_q$ of odd characteristic, returns either the empty set or a representative and generators of the complete transporter, in time*

$$q^{O(N)} \,\mathrm{poly}(\mathsf{L}, \log q).$$

The same recursive construction returns exact transporters for every carried physical state it generates, in the class specified by Definition 1.2. The construction and complete-return induction are proved in Theorem 6.3; Appendix N specifies the established external inputs and the scope of the new arguments.

No regularity, nonsingularity, slice independence, or balance among the three slot dimensions is assumed. Even characteristic, unbounded tensor arity, arbitrary supplied cubic subquotients, and an unsupported abstract label action are outside the statement.

1.1 The carried input and complete output

A *physical module* means an original row, a retained invariant flag grade of that row, or an explicitly represented invariant subquotient of such a grade. An auxiliary encoding is not a new physical row. A *supported label module* is a module supplied together with an exact numerator–denominator realization on one current constant-degree cover of the original physical arguments. Being trivial on the current large classical factors alone is not a support certificate.

Definition 1.2 (Carried physical state)**.** A recursive state consists of the following explicit data.

1. An actual subgroup $H \le \mathrm{GL}_{D_0}(p)$ or a coset of it, with generators, original matrix lifts, and compressed straight-line words. All original field, form, scalar, and previously imposed conditions are part of this group.

2. Three active physical argument modules Q_1, Q_2, Q_3, with dimensions d_1, d_2, d_3, and one whole ordered tensor family, possibly with a supported moving label space L.

3. A current exact affine fiber $x + U/P$, where $P \leq U$ are invariant prime-field spaces inside a literal tensor layer. There is one correlated denominator P: on an outer cubic grade it is exactly the image $\eta(R)$ of the actual current physical-prefix translation map, with its full kernel and lifting words. Cubic coefficient pullbacks are injective; unmatched-type and field-balancing relations are numerator equations. Representative denominators from retired deleted-module covers occur only inside completed at-most-two-critical calls, never as arbitrary outer cubic quotients.

4. An invariant physical refinement forest, its already completed whole observations, and a cursor specifying the current tensor grade. A refinement is made inside old flag intervals. Completed unordered observations retain their whole orbit family and full triangular action.

5. A structural certificate: the actual physical normal p-kernel, standard natural derived classical factors, their fields, duals, twists and diagonal ties, supported retired covers, and exact outside actions. Criticality uses one threshold $p^{C_* D_0}$ throughout the computation. The hereditary outside certificate makes every later huge critical core act trivially on the saved supported label module. Active cores supported only on unprocessed siblings remain invisible ancestor kernels, not part of a purported bounded active outside image.

6. A bit recording whether the present insertion denominator has been consumed. During an in-node batch, retain the entry dimensions and all ordinary natural-type links, including paired duals.

The output of a complete call is empty or one full coset in the original physical group, its full source stabilizer, every invisible physical or additive kernel, all lift words, and a reconstructed structural certificate.

The initial state has $H = G$, $P = 0$, the original tensor, empty completed flags, and no active batch. A source/target dimension mismatch in a corresponding slot rejects. Rectangular formats within a tensor are allowed.

1.2 Priority order

The following order is essential. Each local procedure must finish with a complete coset before a caller proceeds, unless it is an explicitly charged outer branch.

1. **Prepare the actual group.** Construct the physical p-kernel, primitive blocks, standard large cores, and supported retired covers by Appendix I. Retain earlier flags and observations. When an actual standard section is required, use the whole-Hom gauge of Appendix H.

2. **Process physical layers.** Refine by actual physical augmentation flags. Compute the exact translation image on the current literal grade. A value with at most two critical natural positions is completed by Appendix K.

3. **Normalize a component axis.** If an active argument has a genuine group-permuted direct sum, compute each component-pair transporter once in the same parent group and use Section 3. Keep the unique possible heavy component pair in the current batch.

4. **Normalize new multiplicity.** An isotypic row $A \otimes_K M$ with $\dim_K M \geq 2$ is reduced by its compatible-frame orbit. This is a charged geometric exit. Reapply this test immediately after every restriction and preparation.

5. **Force a fixed-tag packet.** Choose a smallest coefficient slot. Apply the complete labelled off-diagonal or same-row test of Section 4, including its insertion variants. Proper physical conductors are unbranched. Field-phase lists close inside complete terminals. A literal low-rank cover invokes Section 5.

6. **Retain the batch until real slot loss.** Point or tuple marks stay imposed until the whole cubic observation has been solved. No same-size unrestricted restart is allowed. Export only after a corresponding slot has lost a fixed fraction, or complete the state in-node.

7. **Return exactly.** Lift every complete child, correct its actual translation error, restore the full kernel, and reunite complete branches. Reconstruct the structural certificate from the actual returned matrices.

Every local search is bounded by $p^{O(D_0)}\,\mathrm{poly}(\mathsf{L})$. That fact alone does not establish the global bound: Sections 6–6.3 count the entire outer recursion.

2 Exact affine fibers and physical translations

The use of an exact affine fiber is necessary even when the original action is linear. Choosing lifts of already matched quotient coordinates produces an affine action on the next grade. Its translations must remain correlated across all labels and components.

Lemma 2.1 (Sequential fiber)**.** *Let $P \le U$ be vector spaces and let $\epsilon : U \to E$ be linear. Put $U_0 = U \cap \ker \epsilon$ and $P_0 = P \cap \ker \epsilon$. There is an exact sequence*

$$0 \longrightarrow U_0/P_0 \longrightarrow U/P \longrightarrow \epsilon(U)/\epsilon(P) \longrightarrow 0.$$

After a first observation has been matched and its target normalized by an actual witness, the remaining direction is U_0/P_0. Any discrepancy in $\epsilon(P)$ is corrected by an actual lift in P.

Proof. The displayed last map sends $u + P$ to $\epsilon(u) + \epsilon(P)$. It is well defined and onto. If its value is zero, choose $p \in P$ with $\epsilon(p) = \epsilon(u)$. Then $u - p \in U_0$, and two such representatives differ by P_0. This identifies the kernel exactly. The witness correction is the same subtraction performed with a recorded physical translation lift. □

Updating only the denominator is incorrect. For example, for $U = \{(a, a, t)\}$ and $P = \{(a, a, 0)\}$, fixing the first coordinate leaves one direction, not the larger space obtained by keeping old U and replacing P by zero.

2.1 A fixed origin from physical splittings

Lemma 2.2 (Literal layer origin)**.** *Let H preserve flags in its original physical rows. Let $R \lhd H$ act trivially on all their grades, and let an actual subgroup $D \le H$ have semisimple natural/dual/twisted-natural physical representations. Assume the image of D is normal in H/R. At a truncated tensor-filtration stage whose earlier coefficients have been matched exactly, the current fiber has an origin b fixed by D. Its action is*

$$h \cdot (b + m) = b + \rho(h)m + c(h), \qquad c(h) = hb - b,$$

and $c(d) = 0$ for $d \in D$.

Proof. Every D-invariant physical subspace has a D-invariant complement. Compute a projection by requiring it to commute with the supplied generators, have image in that subspace, and restrict to its identity. The three physical splittings induce a D-equivariant splitting of the tensor coefficient filtration. Lift the already matched coefficient through this splitting. It is fixed by D because D lies in its actual stabilizer. This supplies b. The formula and cocycle identity follow from the original linear action in the current truncated quotient. Later coefficients are retained; they are not declared zero. No semisimplicity of a tensor cube was used. □

Lemma 2.3 (Actual translation support)**.** *Under Lemma 2.2, the map*

$$\eta : R \to M_+, \qquad \eta(r) = rb - b$$

is a homomorphism. Its image P is H-invariant. Every D-composition factor of P occurs in a physical degree-two Hom module.

Proof. Since R acts trivially on the current associated-graded coefficient space M, increments add. Conjugation in the affine action gives $\eta(hrh^{-1}) = \rho(h)\eta(r)$. Let I be the nilpotent algebra of strictly flag-lowering physical endomorphisms. The filtration $R_i = R \cap (1 + I^i)$ has additive grades embedded by $r \mapsto r - 1$ into I^i/I^{i+1}. These are invariant sections of physical Hom spaces. The filtration $\eta(R_i)$ proves the composition-factor assertion. The entire image and its kernel are computed from actual generators; only $\mathbb{F}_p$-linearity is automatic. □

2.2 The three-critical part

The finite image-algebra transfer in Appendix C applies jointly to all physical Hom modules, the literal cubic layer, their invariant sections, and the natural comparison modules. In particular, comparisons made for D are comparisons for its canonical full classical cover L.

Proposition 2.4 (Insertion form of the translation image)**.** *In a three-critical literal packet, P is contained in the sum of the permitted natural pairing-insertion images. For three distinct masters it is zero. For two paired occurrences and a free occurrence, it is contained in the invariant pairing of the former times the latter. For three occurrences of one master, it is the sum of the three permitted positionwise insertions, with exactly the natural/dual and Frobenius matching conditions of the actual representation. All additive and cross-label correlations remain in the actual subspace P.*

Proof. Central parity in the full cover excludes a physical degree-two constituent involving three independent odd natural positions. In a two-plus-one packet only the free natural type can occur, and the repeated factor is trivial on it. In a one-master odd packet the possible factors are natural/dual/twisted-natural types. The whole-Hom H^1 calculation in Appendices G and H splits their successive extensions. Thus the relevant actual image is a sum of natural types. The complete natural-to-cube Hom computation in Appendix M identifies each such inclusion with a permitted pairing insertion. Its proof uses currying to whole degree-two modules, including their nonsplit chains, and is valid in characteristic three. The actual image is never enlarged to the full insertion space. □

Lemma 2.5 (No residual cubic cocycle)**.** *On the three-critical projection M_3/P_3, the current H-action is linear.*

Proof. Write c for the projected cocycle modulo P_3. It is zero on R and on D. For $h \in H$ and $d \in D$, write $hdh^{-1} = rd'$ with $r \in R$ and $d' \in D$. The cocycle identity yields

$$c(h) = c(hd) = c(rd'h) = \rho(d')c(h).$$

Thus $c(h)$ is D-fixed. Joint transfer, with a trivial summand, identifies the D-fixed vectors with the full-cover fixed vectors. Each genuine cubic packet has an odd natural master; its central $-I$ acts by -1, so it has no fixed vector. Hence $c(h) = 0$. □

Lower-critical packets need not have zero cocycle. They remain in the degree-two problem with their correlations.

2.3 Canonical literalization and the joint residual

In the scalar-value branch, after all relevant quotient-value orientations have vanished as proved in Appendix M, let $J : S \to M_3$ be the actual permitted insertion map. This extraction is not asserted for arbitrary tensors outside that branch. Use its faithful parameter module

$$S_0 = S/\ker J, \qquad J_0 : S_0 \hookrightarrow M_3, \qquad R_0 = J^{-1}(P_3)/\ker J.$$

The extraction constructed in Appendix M is an equivariant map $\bar{s}$ into S_0 satisfying

$$\bar{s}(t + Jr) = \bar{s}(t) + (r + \ker J).$$

Here $r \in J^{-1}(P_3)$ and the translated tensor is in the current scalar-value class. Equivalently, any chosen lift to S obeys this identity modulo $\ker J$. The exact pair is

$$\big(t - J_0\bar{s}(t),\ \bar{s}(t) + R_0\big). \tag{1}$$

It is independent of a representative modulo P_3 and equivariant under the actual residual action. If h matches both entries, then

$$ht - t' = J_0\big(h\bar{s}(t) - \bar{s}(t')\big) \in J_0R_0 = P_3.$$

Conversely, if $ht - t' \in P_3$, extraction equivariance and its shift law make the parameter classes agree and the literal entries equal. This proves exact equivalence to the original quotient transporter. The canonical literal tensor need not itself lie in the original affine class; the second entry retains the difference. The full parameter kernel is still retained in original lifts.

If the current grade also has a lower-critical part M_2, keep the single map

$$F : M_2 \oplus S \to M_2 \oplus M_3, \qquad F(x, s) = (x, J(s)),$$

and the single denominator $F^{-1}(P)$. Solving M_2 and S_0/R_0 separately could lose correlations. After solving this joint degree-two residual, lift to the actual parent, correct its error through $\eta : R \to P$, and restore $\ker\eta$. Projection plus kernel proves a complete original return.

A completed physical observation remains its exact predicate after a refinement or gauge. In finer coordinates its full value tuple has the induced triangular action, including shears. There is generally no equivariant refinement map to a purely diagonal associated-graded action. This is why completed unordered families are retained as whole families, rather than marking their new tags pointwise fixed.

3 One component axis and new multiplicities

A tensor with a decomposed argument has one component object for each block of that argument. It does not have one object for each triple of tensor indices. The other arguments and the entire label family stay together.

Theorem 3.1 (Fixed-source-prefix multiset transport)**.** *Let Γ act on a set Ω, and let $x_1, \dots, x_b$ and $y_1, \dots, y_b$ be lists, allowing repetitions. Suppose every pair transporter $C_{ij} = \mathrm{Trans}_\Gamma(x_i, y_j)$ is supplied as empty or a complete coset, and exact coset intersection is available. The full multiset transporter can be computed using 2^b states, at most $b2^{b-1} - b$ intersections, and $O(b2^b)$ additional group operations.*

Proof. For $J \subseteq [b]$ define

$$D(J) = \mathrm{Trans}_\Gamma(\{x_1, \dots, x_{|J|}\}, \{y_j : j \in J\}),$$

where braces denote multisets. Set $D(\varnothing) = \Gamma$ and $D(\{j\}) = C_{1j}$. For $k = |J| \geq 2$,

$$D(J) = \bigcup_{j \in J} (C_{kj} \cap D(J \setminus \{j\})). \tag{2}$$

A solution maps the last source occurrence to some target occurrence. Removing that occurrence proves one inclusion; adding it proves the other, including repeated values.

Let H_k stabilize the source prefix multiset and let G_k stabilize x_k. Every nonempty branch in (2) is $b_j K_k$, where

$$K_k = G_k \cap H_{k-1}$$

is independent of j and J. Choose a nonempty branch representative b and let

$$L = \langle K_k,\ b^{-1}b_j : \text{nonempty branches} \rangle.$$

Every generator fixes the source prefix, so $L \leq H_k$. Conversely $bh \in D(J)$ for every $h \in H_k$, hence $bh = b_j u$ for some $u \in K_k$, and $h \in L$. Thus $L = H_k$ and the answer is the full coset bL. This generation is justified for the entire successful union, not a partial union of unrelated probes.

The number of intersections is at most $\sum_{k=2}^{b} k\binom{b}{k} = b2^{b-1} - b$. Only one branch subgroup list is needed at a state. All nonempty level-k states have the same H_k, so its generators may be reused. □

The genuine common-action premise cannot be dropped: arbitrary pairwise cosets need not have a coset as their assembled union.

3.1 Pair calls in the original group

Suppose one physical argument is $Q_1 = \bigoplus_{i \in I} A_i$ and the current linear action is block-monomial. If $b(g)$ is the actual affine cocycle, the action on tagged fibers is

$$g(i, t_i) = (gi, \rho_i(g)t_i + b_{gi}(g)).$$

Normalize an incoming coset once, so both source and target use this same action of Γ. For a pair (i, j), first compute a tag transporter r_{ij} and the full tag stabilizer Γ_i. Solve the whole fiber equation

$$h(i, t_i) = r_{ij}^{-1}(j, t'_j), \qquad h \in \Gamma_i.$$

The child retains the full ancestor word carrier, the two unchanged arguments, all labels, and every invisible evaluation kernel. If its complete result is $h_{ij}S_i$, return $r_{ij}h_{ij}S_i$ in the original Γ.

For example, a global translation space $P = \langle (1, 1) \rangle$ has a full one-dimensional image on either coordinate. The two pair calls must still return cosets in that same diagonal group. Replacing them by independent marginal shift groups would permit incompatible translations.

The exact group-coset intersection used by (2) is a consequence of the degree-two backend, not a permutation-group primitive. For $G_1, G_2 \leq \mathrm{GL}_d(k)$, the product action on $\mathrm{Hom}(k^d, k^d)$ is

$$(g_1, g_2)X = g_1 X g_2^{-1}.$$

The complete transporter of I to t projects onto $G_1 \cap tG_2$. Its source stabilizer is the diagonal copy of $G_1 \cap G_2$. Preparation of the actual product group and the degree-two solver therefore give exact intersection in $p^{O(d)}$; Appendix K establishes the required noncircular dependency order.

There are at most b^2 pair calls, each computed once. Their costs add. The 2^b dynamic-program states use these completed results and add local work; they do not multiply every child cost. There is at most one source block larger than half its argument, and at most one dimension-compatible target block of that size. Hence there is at most one heavy pair. Keep it in-node with the inherited marks and ties. All other pair calls shrink that argument by at least half.

3.2 Compatible multiplicity frames

Lemma 3.2 (Paid multiplicity normalization)**.** *Suppose a critical isotypic physical row is $A \otimes_K M$, where $K = \mathbb{F}_{p^f}$, $\dim_K A = a$, $\dim_K M = d \geq 2$. Supply the actual natural–multiplicity factorization cover, its scalar kernel, field and duality records, and original projection words. A compatible-frame orbit has at most p^{fd^2} elements. Within each frame branch the active row becomes A, with all multiplicity coordinates retained in one label family. If the permanent criticality threshold implies $a \geq cd$ for an absolute $c > 0$, then*

$$fd^2 = O\big(fa(d-1)\big),$$

so the branch entropy is paid by the actual row loss. The row at least halves.

Proof. Choose a reference basis e of M and its evaluation-dual basis e^*. Compatible frames are

$$\mathcal{F} = \{(eB, e^*B^{-T}) : B \in \mathrm{GL}_d(K)\}.$$

The formal contragredient is used even if an original physical dual row carries an additional character; that character stays separate. A factor-cover element acts on frame coordinates by

$$B \longmapsto m\theta(B),$$

where θ is a field map or a field map followed by inverse transpose. The semidirect multiplication makes this a genuine action and $\theta(I) = I$. It is not a homomorphism obtained by forgetting the outer record.

Breadth-first orbit search and Schreier generators on the orbit of I use at most $|\mathrm{GL}_d(K)| \leq p^{fd^2}$ points. The stabilizer has multiplicity matrix part I, while compatible field and exchange records remain. Keep the full central factorization kernel until complete original returns are projected.

Expand M in the fixed frame. Its d coordinates are one whole tensor family on A, with common natural maps, field-balancing equations, old frames, and supported label action. Conversely these retained cover equations are exactly the lifting conditions to the original row. The new natural row is realized physically as $A \otimes e_1$ inside the current grade, with its paired dual tag when an exchange occurs. Its ancestry is therefore still a physical invariant subquotient under the frame stabilizer. The new degree is fa, the old degree fad, and their difference is $fa(d-1)$. Since $d/(d-1) \leq 2$ and $a \geq cd$, the displayed bound follows. □

Criticality is measured against original degree: $fa^2 > C_* D_0$ and original mass $fad_{\mathrm{total}} = O(D_0)$ imply $a > C'd$, for $d \leq d_{\mathrm{total}}$. Only the new local multiplicity is enumerated. Prior frame conditions remain imposed. When $d = 1$, no matrix-frame list is opened; scalar, field, duality and similitude records remain in the actual prefix.

4 Exhaustive forcing on a fixed physical packet

After component and multiplicity normalization, each active slot has one critical natural occurrence, with its ordinary type, field, duality and common gauge fixed as records. Keep the actual inherited group. The words "fixed tag" do not mean that its field automorphisms, similitudes, value characters or supported outside label action have been independently enumerated.

Choose a smallest physical coefficient slot C. The other two slots give a whole labelled operator pencil $M_x : B \to V^* \otimes L$. Distinct and equal ordinary operator-row types require different tests. This distinction includes natural/dual occurrences of a general-linear exchange factor, even if they belong to one abstract master.

Lemma 4.1 (Off-diagonal rank pays a mark)**.** *Let the two operator rows be distinct ordinary simple components of the reduced active star-algebra E. Put $c = \dim_k C$. If the complete stack at a coefficient point x has rank $\rho \geq c/64$, marking x costs at most p^c targets and reduces the ordinary core degree by at least $\rho/2$, after its complete degree-two comparison and structural preparation.*

Proof. In the reduced physical realization, generate the algebra B_x by $\mathrm{End}_E(W)$, all labelled off-diagonal $M_j(x)$, and their adjoints. The original row projections lie in B_x, so its commutator ideal $I = B_x[B_x, B_x]B_x$ contains every $M_j(x)$ and its adjoint. The sum of adjoint images has dimension equal to the rank of the whole stack, hence $\dim IW \geq \rho$.

Write the Krull–Schmidt decomposition as $W = \bigoplus_i Q_i^{m_i}$, with residue division fields K_i of degrees f_i, and $\dim_k Q_i = f_i j_i$. The semisimple quotient of $\mathrm{End}_{B_x}(W)$ has simple factors $\mathrm{M}(m_i, K_i)$, whose ordinary natural degree sum is $\sum_i f_i m_i$. If $j_i = 1$, the B_x-action factors through a field and I kills that summand. Therefore

$$\dim IW \leq \sum_{j_i \geq 2} f_i j_i m_i \leq 2 \sum_i f_i (j_i - 1) m_i.$$

This proves the degree drop at least $\rho/2$ in the actual centralizer algebra. The full labelled contraction is one degree-two observation, so the exact backend retains its radicals, actual parent and kernels. New critical factors act trivially on the old bounded outside image and embed in this old diagonal centralizer; subsequent conductors cannot increase its ordinary natural degrees. There are at most p^c images of the one marked point. □

It is generally false that the two adjoint image spaces have total dimension 2ρ: labels may share a small common output. The factor $1/2$ avoids that error.

If every stack has rank $r < c/64$, the canonical cover has boundary at most $4r$, and Theorem 5.1 applies. All labels enter one stack; no labelwise rank sum is substituted.

4.1 The same-row alternative

Suppose the operator rows are the same ordinary natural type $A = K^a$, with $[K : k] = f$. Since C was chosen smallest, $\dim_k C \leq fa$. For each tuple S of at most three coefficient points, form the raw algebra

$$B_S = \mathrm{alg}_k(K, I, \{M_j(x) : x \in S,\ j \in L\}).$$

When the operators commute with K, this is equivalently the K-algebra they generate. Otherwise K remains a distinguished constant subalgebra. New word-vector contractions are not inserted into this raw algebra.

For $a < 64$ use the bounded-rank complete terminal. Otherwise put $r_K = \lfloor a/32 \rfloor$. In the Krull–Schmidt decomposition of the raw module, a centralizer natural degree greater than $f(a-r_K) > fa/2$ forces its indecomposable dimension over the residue field to be one: a larger dimension would exceed the whole row. Such a type is unique, and the raw algebra has a field quotient whose annihilator core has dimension at least $a - r_K$ over K.

Thus either one raw triple has every centralizer natural degree at most $f(a - r_K)$, or every raw triple has a unique dominant field quotient. In the first case mark that tuple and compare its entire labelled degree-two data. Its at most three coefficient marks cost $p^{3 \dim C} = p^{O(fa)}$, and both operator rows fission into natural constituents of degree at most $f(a - r_K)$, with all radicals and multiplicities retained. The geometric row loss pays the list. In an insertion quotient the analogous test uses the projected raw algebras and at most six marks; Appendix M proves the same conclusion with the actual inherited normalizer and scalar-shift action.

4.2 Common field values and their complete successors

Appendix L glues the dominant fields into one field F/K of degree at most $\operatorname{lcm}(1, \dots, a) = \exp(O(a))$. It produces a genuine prime-field linear value

$$\lambda : C \longrightarrow \operatorname{Hom}_k(L, F),$$

unique up to one common Frobenius, and equivariant with all original field and label actions. The field is auxiliary. Its elements are not enumerated, and its degree is not a new faithful group degree.

Use one uniform rule for a nonzero value, or for the nonzero quotient value in an insertion packet.

1. If $0 < \ker \lambda < C$, match that actual physical prime-field kernel completely. Its pure stabilizer is the unitary group of the explicit star-subalgebra

   $$E_W = \{a \in E : aW \subseteq W,\ a^*W \subseteq W\}.$$

   The current full natural coefficient core is irreducible and therefore cannot preserve a nonzero proper W. This is a strict unbranched conductor. The whole returned diagonal prefix normalizes E_W, and its outside quotient embeds the old one. Pull back the entire original p-extension. No claim that W is linear over an old coefficient field is needed.

2. If λ is injective, enumerate each admissible global Frobenius phase. For a fixed phase, match the complete value by one supported degree-one/whole-Hom call, with F as passive multiplicity. Injection kills the residual old pure coefficient action. The permanent hereditary outside bound excludes a new critical projection hidden in that old outside action. The whole remaining observation now has at most two critical positions and is completed by Theorem K.2. Only after that complete terminal is finished is the next phase considered. Reunite all complete terminals; no phase list escapes to outer recursion.

If the literal value is zero, every operator kills the same dominant annihilator core at that coefficient point. The whole stack rank is at most fr_K, so the row-only bound in Theorem 5.1 applies. A projected scalar-zero value does not imply this literal rank bound. Instead extract all insertion parameters, form the exact pair (1), match its natural parameter class completely, and run ordinary literal forcing on its first entry. The current denominator is consumed once.

Proposition 4.2 (Exhaustive packet transitions)**.** *Every fixed-tag three-critical packet admits one of the following exact successors: a paid off-diagonal high-rank mark; a paid raw-triple geometric fission; a strict unbranched physical conductor; a completed field-phase or pure-insertion terminal;*

canonical paired literalization; or a literal row-relative low-rank exit. A wholly zero observation is a terminal.

Proof. Distinct versus equal ordinary operator-row types is exhaustive. In the distinct case Lemma 4.1 or the canonical low cover applies. In the equal case the raw-triple Krull–Schmidt alternative and field gluing give the stated cases. Proposition 2.4 classifies every actual cubic translation image. Appendix M handles all its pairing/variance/twist patterns, including the unbalanced one-pair augmented pencil, without a large coefficient mark of the wrong scale. Proper kernels, injective values and zero values exhaust each field-code case. Every subspace/value call is complete, with the old group and all kernels; every open point list remains imposed until the whole cubic observation has been solved. □

5 The unequal-slot low-rank exit

Let the entire labelled observation be $T : C \times B \times V \to L$ over $k = \mathbb{F}_p$, and put $c = \dim C$, $b = \dim B$, $v = \dim V$. For $x \in C$ define the whole contraction

$$M_x : B \to V^* \otimes L.$$

Suppose a canonical cover $M_x(X) \subseteq E$ for all x has been matched completely, with the actual parent and every label action retained.

Theorem 5.1 (Row-relative geometric exit)**.** *Assume*

$$h = \operatorname{codim}_B X + \dim E \leq \frac{1}{8} \min(b, v). \tag{3}$$

Assume each new critical natural factor acts trivially on the supported label space L, and its physical row representation is semisimple of the recognized natural types. Then the full tensor transporter reduces, with exact shared gauges and affine lifts, to completed degree-two calls and three-critical children for which at least one corresponding natural slot has degree at most h. The other slots do not increase. No condition involving c is needed.

Proof. Set

$$V_0 = \operatorname{im}(E \otimes L^* \longrightarrow V^*), \qquad U = \operatorname{ann}_V(V_0).$$

Evaluation is equivariant for the whole cover-stabilizing parent. Hence X and U are actual invariant physical subspaces, and

$$T(C, X, U) = 0. \tag{4}$$

One must not infer $\dim V_0 \leq \dim E$. Instead fix a new critical factor D'. Since D' acts trivially on L, evaluation makes V_0 a quotient of copies of E as a D'-module. Every natural constituent occurring in V_0, hence in V/U, has degree at most $\dim E \leq h$. Also $\dim(B/X) \leq h$.

The two lower arms are intrinsic tensors

$$T_1 : C \times (B/X) \times U \to L, \qquad T_2 : C \times X \times (V/U) \to L.$$

Changing either quotient lift changes its value only by (4). Match T_1 completely and then T_2 inside its full returned parent coset. Afterward the remaining top on $C \times (B/X) \times (V/U)$ is an exact affine fiber. If physical shears are $s : B/X \to X$ and $t : V/U \to U$, its change is

$$T_2(-, s(-), -) + T_1(-, -, t(-));$$

the prospective quadratic term $T(-, s(-), t(-))$ is zero. Use the actual prefix image and kernel of these shears, rather than all formal Hom parameters. Lemma 2.1 supplies the exact current numerator and denominator at each step.

The first and top arms already have a physical argument B/X of degree at most h. For the second arm, prepare the actual V/U row. Its noncritical pieces have at most two remaining critical positions and are complete local calls. A critical isotypic piece is $A_\tau \otimes_{K_\tau} M_\tau$ with $\dim_k A_\tau \le h$. Match its genuine one-axis component family by Theorem 3.1, with all C and X maps shared. Apply Lemma 3.2 immediately if needed. The resulting active natural row is A_τ, not its full isotypic physical mass. The pullback along

$$A_\tau \otimes_k M_\tau \twoheadrightarrow A_\tau \otimes_{K_\tau} M_\tau$$

is injective on tensor coefficients, with unmatched-type and field-balancing relations imposed exactly. Thus each unresolved child has the required small row. Its degree is at most one eighth of the corresponding old operator row by (3).

Finally complete the affine top, correct by actual physical translation lifts, and restore all invisible kernels. The two arms and top determine the entire original tensor, so the resulting coset is exact. □

5.1 Why the coefficient dimension is absent

For an off-diagonal low branch, the smallest coefficient dimension gives maximum stack rank $r < c/64$. The canonical cover of Appendix B has $h \le 4r < c/16$, which implies (3). For a same-row field branch, let both operator rows be K^a, where $[K : k] = f$, and put $r_K = \lfloor a/32 \rfloor$. Zero field value means that every original labelled operator annihilates the same core of codimension at most fr_K over k. Consequently the whole stack has rank at most fr_K, and

$$h \le 4fr_K \le fa/8 = \tfrac{1}{8}\min(b, v),$$

even when $c \ll fa$. Requiring $h \le c/16$ here would not be supplied by the field argument and is unnecessary for either arm.

The zero identity (4) is retained while the smaller physical packets are formed. A correction $T - J(s)$ with s outside the actual allowed translation parameters can refill this zero block. Therefore canonical literalization is performed inside a child only after that child's smaller physical format has been certified.

5.2 The augmented one-pair case is different

A low-rank cover for an insertion-covariant augmented pencil does not imply (4). Its output contains an anchor and may contain a nonzero graph representing a pure insertion. Appendix M proves separate alternatives: a proper physical domain, a proper shear-fixed physical output subspace, or a pure-insertion degree-one terminal. Theorem 5.1 is applied only to a literal tensor or a tensor made literal by an exact prior correction.

6 Retained batches and the global running time

The algebraic degree of a classical core may fall while the physical slot size stays unchanged. It therefore cannot justify an unrestricted same-size recursive restart. We retain the actual restricted group until physical dimensions decrease geometrically.

6.1 Six ordinary natural types

Freeze entry dimensions (d_1, d_2, d_3) and let $c_0 = \min_i d_i$. Fix $\epsilon = 1/128$. This is no larger than the direct raw-triple loss $1/64$ after bounded-rank exceptions, nor than the component, multiplicity or low-rank contraction factors. A continuation is *all-large* if every corresponding active slot exceeds $(1-\epsilon)d_i$. Components and new multiplicities are normalized before this test.

Count ordinary natural types among the three rows and their paired duals. There are at most six. Abstract classical factor names are insufficient: a general-linear natural and dual pair may merge into one orthogonal or symplectic type.

Lemma 6.1 (Batch charge)**.** *Suppose a high point or bounded-tuple mark of entropy β has a certified loss of distinct ordinary core degree paying β up to a constant; its marking cost is at most $p^{C_0 c_0}$. Suppose bad raw triples and new multiplicity frames have their direct physical-loss bounds. Along a retained batch ending in a geometric exit,*

$$\sum \beta \le C \left(\sum_i d_i - \sum_i d_i' \right)$$

for an absolute C. A terminal batch has total entropy $O(\sum_i d_i)$.

Proof. An old ordinary type can supply at most one all-large constituent: two would have combined physical mass greater than the old row. Hence ordinary ties only merge on the internal all-large spine, at most five times. Without a merger, the loss of paired core degree is at most twice actual slot-sum loss, since each physical occurrence has at most its paired dual copy. These charges telescope. Each merger costs at most $C_0 c_0$, so all merger charges total $O(c_0)$.

The final exiting restriction may split an old type. Do not apply internal monotonicity to it. Its last high mark is paid directly by $O(c_0)$. At exit some slot loses at least ϵ of its entry degree, so total loss is at least ϵc_0 and pays these boundedly many charges. Bad triples already lose a fixed fraction of an operator row; Lemma 3.2 pays every new frame by its actual slot loss. Proper-kernel conductors are unbranched, and all injective-code phases are complete terminals. The same argument bounds a final terminal batch by its whole entry size. □

Immediate multiplicity normalization is indispensable. Restricting SL_a to a diagonal copy acting on two copies of $k^{a/2}$ leaves isotypic mass a unchanged. Its successor must have active natural degree $a/2$ and two labels; it is a geometric exit, not a zero-loss continuation.

6.2 Finite same-size work

All flags refine inside old physical intervals. A new normal p-subgroup refines an old interval by its intersections with the new augmentation flag; old cross-block shears remain in the exact translation kernels. Along a live path there are $O(D_0)$ physical intervals and polynomially many induced tensor grades. A completed whole observation stays completed under the full triangular change of coordinates.

An unbranched same-size transition must strictly restrict the actual physical prefix group, properly refine a retained physical flag, advance a previously incomplete physical observation, or consume the current literalization phase. The last phase cannot be regenerated without one of the preceding changes. A strict subgroup chain in $\mathrm{GL}_{D_0}(p)$ has length at most $\log_2 |\mathrm{GL}_{D_0}(p)| = O(D_0^2 \log p)$. Thus such runs have polynomial length. Auxiliary additive chains close inside the degree-two solver; they do not create an outer call for every auxiliary coordinate. A unique heavy component continuation has strict slot decrease and contributes a polynomial-length spine with additive side calls.

6.3 A weighted count of the whole tree

Theorem 6.2 (Weighted outer recursion)**.** *Suppose the outer recursive construction satisfies the following conditions.*

1. *Every open target family has at most p^{β} members, and the total β on every root-to-terminal path is $O(D_0)$.*
2. *Every geometric export reduces one corresponding slot by a fixed factor and increases none. Between geometric exports there is a polynomial-length run with at most one heavy continuation and polynomially many additive side calls.*
3. *Complete component-pair costs add; each pair is computed once. Every other local list closes inside a complete $p^{O(D_0)}\,\mathrm{poly}(\mathsf{L})$ operation.*
4. *Label length creates no outer fanout, and all per-node polynomial exponents are uniform.*

Then the total running time is $p^{O(D_0)}\,\mathrm{poly}(\mathsf{L}, \log p)$.

Proof. Give every edge in a target-family node weight $p^{-\beta}$. The sum of outgoing weights is at most one. Induction down a finite prefix tree shows that the sum of product weights of any prefix-free family is at most one. For a fixed additive-choice address, every terminal target prefix has weight at least p^{-CD_0}, hence there are at most p^{CD_0} such terminals. This is the needed whole-tree argument; a path estimate alone would be insufficient without the outgoing-weight inequality.

Each of three integer slot dimensions can shrink by a fixed factor only $O(\log(D_0 + 2))$ times. Contract all polynomial-length unary/heavy runs. If $P(D_0, \log p)$ bounds their side-call count, the number of additive-choice addresses is at most

$$P(D_0, \log p)^{O(\log(D_0+2))} = p^{O(D_0)}.$$

The equality uses $(\log(D_0 + 2))^2 = O(D_0)$ and $\log(D_0 + 2) \log\log p = O(D_0 \log p)$. This argument can be formalized by fixing the finite sequence of side-choice indices before exposing target choices; absent indices simply give no leaf. Therefore adaptive target lists cause no additional factor.

The polynomial run lengths also bound internal nodes. Multiplying the total node count by the complete local $p^{O(D_0)}$ work changes only the constant in the exponent. Uniform polynomial label processing contributes one polynomial input factor, not a new power at each depth. □

Lemma 6.1 telescopes across geometric exits and supplies condition 1, including the final terminal batch. Conditions 2 and 3 are why a heavy child is retained in-node and why the subset dynamic program is additive after pair precomputation.

6.4 Uniform auxiliary size and complete returns

For each original argument, rebuild its current factored cover from one combined natural row and one combined multiplicity/retired row. A direct sum of pieces $A_\tau \otimes_{K_\tau} M_\tau$ is a quotient of $(\bigoplus A_\tau) \otimes_k (\bigoplus M_\tau)$ by unmatched-type and field-balancing relations. Pullback is injective on coefficients. Thus at most six physical factor positions are required for three original arguments; one supported character/anchor and one temporary frame give arity at most eight. Old auxiliary repairs are never repeatedly tensor-multiplied. Their total faithful physical cover degree is $O(D_0)$. Each exported child coefficient/label description has size $\mathsf{L}\,\mathrm{poly}(D_0)$ by this reconstruction from original rows; no current auxiliary description becomes a new input-size parameter.

The degree-two backend allows any fixed arity, with cost $p^{O_r(D_0)}$; only two positions are critical. An exponentially long field-coordinate array is passive inside a completed call and is never a faithful physical degree. Its size and polynomial arithmetic are absorbed in that call's $p^{O(D_0)}$ work; it is not exported as a recursively growing label family.

For a complete successful union, choose one representative b_0 and generate the source stabilizer by the complete branch stabilizers and $b_0^{-1}b_j$. Each generator stream is pruned by membership in the original vector-permutation representation: retain an element only when it enlarges the generated subgroup. Each retained element at least doubles its order, so $O(D_0^2 \log p)$ suffice. Words remain shared straight-line programs. Apply structural preparation to this actual returned group before any subsequent structured operation.

Theorem 6.3 (Assembly)**.** *The priority-order construction of Section 1 is exhaustive, returns the complete original transporter and stabilizer, and satisfies the bound in Theorem 1.1.*

Proof. Induct on geometric slot shrink, with the finite physical group/flag/cursor ordering for in-node work. We first verify applicability for every state generated by the algorithm.

Initially the physical rows are literal, $P = 0$, and the product group has its explicit standard certificate. Every complete return is an actual original matrix subgroup, so Appendix I applies anew within the retained parents. Its actual-section gauge preserves all flags and transports the entire representation. Physical flag splittings then give the literal grades and their actual $\eta(R)$ images of Lemmas 2.2 and 2.3. Hence Proposition 2.4 applies to precisely these cubic denominators. Retired deleted-cover denominators remain inside completed degree-two calls. The permanent outside certificate makes every newly critical factor trivial on the saved label realization, supplying the label hypothesis in the geometric and field-value steps.

A component pair is solved under the actual index stabilizer with its full physical preimage; the child therefore has the same correlated action and all invisible sibling kernels. Its smaller argument is a genuine physical block. Multiplicity normalization uses a genuine factorization cover, the physical $A \otimes e_1$ row, and an injective field-balanced coefficient pullback, so its natural child has the asserted physical ancestry and does not re-expand. Both operations keep the whole ordered label family.

Forcing begins only after these priorities. Its kernel conductor is an actual physical subspace stabilizer. Each marked contraction is a completed supported degree-two observation under the actual prefix; marks remain constraints on every unresolved cubic child. Projected scalar-zero literalization is used only after the insertion extraction has been established, and retains its one joint low/high residual and original lifts. The augmented one-pair pencil is used only for its proved conductor or pure-insertion alternatives. A literal low-stack call retains the zero numerator while forming the two intrinsic arms and affine top, so their small-row certificates and actual shear fibers satisfy Theorem 5.1. Thus every generated successor remains in Definition 1.2; no separately supplied cubic quotient or unsupported action has entered.

The forcing alternatives are exhaustive. Every at-most-two-critical observation closes by Theorem K.2, every exported nonheavy cubic child shrinks a corresponding slot, and the unique heavy continuation retains its batch. Complete pair returns, exact fibers, physical translation correction, and whole-branch reunion prove that each call returns one full original transporter with its entire kernel. Preparation of that actual group closes the return induction.

The numerical hypotheses of Theorem 6.2 now follow from Lemma 6.1, the physical event count, one-axis pair precomputation, and the rebuilt fixed-arity representation. Therefore total cost is $p^{O(D_0)} \operatorname{poly}(\mathsf{L}, \log p)$. Finally $D_0 = O(eN)$ proves the claimed $q^{O(N)}$ bound.

□

The scope of this induction is exactly Definition 1.2. In particular it does not admit arbitrary cubic quotient modules or arbitrary unrepresented label actions.

A Finite algebra and original-degree operations

All externally invoked algorithms act either on an algebra represented on the original physical rows, or on an explicit permutation domain of size $p^{O(D_0)}$. An auxiliary coefficient dimension is never substituted for the original faithful degree.

Proposition A.1 (Represented algebra operations)**.** *For a finite-dimensional star-algebra represented faithfully in degree d over an odd finite field k, its radical and semisimple structure, a star-stable Levi complement, module isomorphism, star-norm correction, and generators of its unitary group are computable in $|k|^{O(d)}$ times polynomial explicit-input work.*

Proof. Use the finite-algebra algorithms of Rónyai, the module algorithms of Chistov–Ivanyos–Karpinski, and the star-algebra algorithms of Brooksbank–Wilson and Ivanyos–Qiao [5, 2, 1, 3]. Their permitted polynomial dependence on the characteristic is absorbed by the stated bound. A direct original-degree implementation finds a composition flag by enumerating vectors and cyclic spans. The radical is the kernel on all simple factors; faithfulness ensures every simple algebra type occurs. Commutants recover their actual fields. Separability and matrix-unit lifting give a Levi complement [4], with the odd-characteristic star correction.

For $J = \operatorname{rad} E$ and a star-Levi F, one has $\mathrm{U}(E) = \mathrm{U}(1+J) \rtimes \mathrm{U}(F)$. The radical unitary group has explicit layer lifts from the Cayley parametrization $a \mapsto (1-a)(1+a)^{-1}$ for $a^* = -a$ in J. Norm correction reduces on the semisimple quotient to finite classical forms and lifts by nilpotent odd-characteristic corrections. None of these steps enumerates $|k|^{\dim E}$. □

Lemma A.2 (Exact conductor equations)**.** *Transport of a vector in a genuine algebra bimodule, or of a finite family of subspaces in one-sided modules, reduces to module isomorphism and star-norm correction. A nonempty answer is $u\,\mathrm{U}(F)$ for an explicitly computed star-subalgebra F.*

Proof. For subspaces L_j, L'_j of equal dimensions put

$$F = \{a : aL_j \subseteq L_j,\ a^*L_j \subseteq L_j\}, \quad X = \{c : cL_j \subseteq L'_j,\ c^*L'_j \subseteq L_j\}.$$

The conditions run over all j. The space X is a right F-module. If c_0 is a unit in X, equal dimensions make the inclusions equalities and $X = c_0F$. Test $F_F \cong X_F$. For a module generator c, testing invertibility in the original representation is conclusive: any unit generator differs from it by an F-unit. Then $c^*c \in F$, and norm correction decides whether cF contains a unitary element. For bimodule vectors use

$$F = \{a : av = va,\ a^*v = va^*\}, \qquad X = \{c : cv = wc,\ c^*w = vc^*\}.$$

The identical argument proves the claim, without an invertible-element oracle for an arbitrary matrix space. □

A.1 Allowed permutation operations

On an explicitly represented domain of size m, deterministic stabilizer-chain algorithms provide group order, constructive membership, generated subgroups, point orbits and transporters, point stabilizers, and pointwise kernels [6]. Normal-subgroup and quotient computations used for the

physical structural decomposition are those of Kantor–Luks [9]. Normal closure is obtained by adjoining generator conjugates until membership says the subgroup is normal. Each strict addition doubles the order.

A stabilizer-chain presentation can be built with compressed Schreier relations. Reducing a transversal–generator product at each level gives a normal form; the presented group has at most the product of orbit lengths, namely the actual group order, and surjects onto the image. It is therefore an exact presentation. For a supplied homomorphism with an explicit original-degree image action, lift these relations and the image generator expressions; their normal closure is the full kernel.

No arbitrary subgroup intersection, arbitrary set stabilizer, or unspecified huge-image kernel is included here. The later group-intersection theorem is derived from the complete degree-two transporter and may not be used to establish its own dependencies.

Lemma A.3 (Additive image kernel)**.** *For an actual homomorphism $\eta : R \to T_+$ into an explicit prime-field vector space, its entire kernel can be computed with original lifts without enumerating T.*

Proof. Compute a basis of the span of generator images and choose actual lifts $b_1, \dots, b_t$. Correct every original generator by products of these lifts to have zero image. Take the normal closure of the corrected generators, all $[b_i, b_j]$, and all b_i^p. The quotient is elementary abelian of dimension at most t and maps onto the t-dimensional image, so it is exactly that image. The constructed normal subgroup is the full kernel. □

For a symbolic subgroup of $T_+ \rtimes G$, the translation kernel is similarly the span of lifted image-presentation relations, closed under conjugation. Membership first matches the physical image word and then tests the remaining translation in that span. These are polynomial in auxiliary dimension and $p^{O(D_0)}$ in physical degree.

B Canonical covers for bounded-rank matrix spaces

In this appendix, k is the stated base field, $q_k = |k|$, and N denotes the faithful degree of the represented algebra or physical rows over k. In the main algorithm $k = \mathbb{F}_p$, so $q_k = p$ and this degree is $O(D_0)$. Auxiliary module dimensions are not included in that faithful degree.

Lemma B.1. *Let X, Y be finite-dimensional k-spaces and $R \geq 0$ an integer. Let $\mathcal{S} \leq \mathrm{Hom}_k(X, Y)$ be a matrix space whose base-field elements have rank at most R. There are subspaces $U \subseteq X$ and $B \subseteq Y$ such that*

$$\mathcal{S}U \subseteq B, \qquad \operatorname{codim}_X U + \dim B \leq 4R.$$

No lower bound on q_k is required.

Proof. Choose a base-field $A \in \mathcal{S}$ of maximum rank a, and use coordinates in which A has an a-dimensional identity block and zeros elsewhere. For $B \in \mathcal{S}$, let b be the rank of its complementary lower-right block. Reduce that block to an identity of size b. Its row and column pivots can clear the adjoining blocks of B without changing A, since the corresponding rows and columns of A vanish. Thus the two matrices B and $A + B$ have a common separate identity block of size b, and their remaining parts differ by a rank-a matrix. The rank triangle inequality gives

$$\operatorname{rank} B + \operatorname{rank}(A + B) \geq 2b + a.$$

Both matrices are base-field elements of $\mathcal{S}$, so $b \le a/2$. In particular the complementary projected matrix space has maximum base-field rank at most $a/2$.

Delete the a pivot-domain directions and allow the a pivot-image directions in the cover, then repeat on that projected space. The successive maximum ranks are at most $R, R/2, R/4, \ldots$. At termination the remaining projected space is zero. The sum of deleted domain and allowed image dimensions is at most $2\sum_{j\ge 0} R/2^j = 4R$. This argument only uses A, B, and $A+B$ over the original field. □

The preceding proof establishes existence; the algorithm does not need to find its maximum-rank matrix. The next proposition constructs the canonical pair by noncommutative rank and a Wong sequence, using only explicit linear-algebra operations and the cited deterministic algorithm.

For a positive integer e, the blow-up of $\mathcal{S} \le \mathrm{Hom}_k(X, Y)$ is

$$\mathcal{S}^{[e]} = \mathcal{S} \otimes_k \mathrm{M}(e, k) \le \mathrm{Hom}_k(X \otimes_k k^e, Y \otimes_k k^e).$$

Its noncommutative rank is $\mathrm{ncrk}(\mathcal{S}) = \sup_{e\ge 1} \mathrm{mrk}(\mathcal{S}^{[e]})/e$. For a subspace $U \le X$, its deficiency is $\dim U - \dim \mathcal{S}U$. After padding to a common square dimension d, a subspace of maximum deficiency has minimum cover cost $d - \max_{U\le k^d}(\dim U - \dim \mathcal{S}U)$. The noncommutative-rank theorem identifies this minimum with $\mathrm{ncrk}(\mathcal{S})$ and computes the necessary blow-up.

Proposition B.2. *There is a deterministic canonical choice of a pair $(U, \mathcal{S}U)$ with*

$$\mathrm{codim}_X U + \dim \mathcal{S}U \le 4R.$$

For the physical pencils here its time bound is $q_k^{O(N)} \mathrm{poly}(\log q_k)$. More generally the construction has only polynomial dependence on matrix dimensions, apart from a conservative $q_k^{O(1)}$ finite-field factor.

Proof. Write ρ for noncommutative rank. Every cover in Lemma B.1 bounds the rank of every blow-up $\mathcal{S} \otimes \mathrm{M}(e, k)$ by $4Re$: on $U \otimes k^e$ its image lies in $B \otimes k^e$, and the complementary domain has dimension $e \,\mathrm{codim}\, U$. Hence $\rho \le 4R$.

The deterministic noncommutative-rank algorithm [7, Theorem 1.5 and Corollary 1.7(2)] supplies an optimal-rank base-field blow-up

$$B \in \mathcal{S} \otimes \mathrm{M}(e, k), \qquad \mathrm{rank}\, B = e\rho.$$

Its small-field construction passes through an extension and then uses the regular representation to return a base-field blow-up; an arbitrary extension-field subspace is not used as the output here. The standard shrunk-space characterization gives minimum cover cost ρ.

For completeness, canonicity follows by the following Wong sequence in the blow-up domain. After zero-padding rectangular spaces to a common size d, put $\mathcal{S}_e = \mathcal{S} \otimes \mathrm{M}(e, k)$ and

$$U_0 = \ker B, \qquad U_{j+1} = B^{-1}(\mathcal{S}_e U_j).$$

If U has maximum deficiency $e(d-\rho)$, then

$$e(d-\rho) \le \dim(U \cap \ker B) \le \dim \ker B = e(d-\rho).$$

Thus $\ker B \subseteq U$ and $BU = \mathcal{S}_e U$, whence $B^{-1}(\mathcal{S}_e U) = U$. Every sequence term is consequently contained in every maximum-deficiency subspace. The increasing sequence stabilizes in polynomially many steps. At its limit its image lies in $\mathrm{im}\, B$, because it is contained in a maximum-deficiency

subspace, and $BU_\infty = \mathcal{S}_e U_\infty$. Its deficiency is therefore $\dim\ker B$, so it is itself maximum and is the unique smallest such space.

Right changes of basis on the blow-up index preserve $\mathcal{S}_e$. Uniqueness makes U_∞ invariant under all $I_X \otimes \mathrm{GL}(e, q_k)$. These matrices span $I_X \otimes \mathrm{M}(e, k)$ in odd characteristic, giving $U_\infty = U_* \otimes k^e$. Recover U_* by the linear equations $x \otimes e_1 \in U_\infty$. It is the unique smallest original maximum-deficiency subspace. If padding added a zero domain summand, that summand belongs to every maximum-deficiency space and is discarded after descent. The image $\mathcal{S}U_*$ is computed by linear algebra. The resulting original pair is therefore covariant under every left-right equivalence, independently of the blow-up witness chosen.

The finite-field algorithms in the cited theorem are used with their deterministic finite-field implementation. It suffices here to retain a possible polynomial factor in p or q_k, rather than assert an unjustified polynomial dependence on $\log q_k$ alone. A small field is extended only to size polynomial in the matrix dimension, and direct deterministic field construction has that same bound. These costs are absorbed in $q_k^{O(N)}$ on the physical modules. The rank R in the statement is the maximum rank over the original field; no large-field commutative-rank identity is assumed. □

C Finite image-algebra transfer and actual unipotent subgroups

We prove transfer for finite groups themselves, including small odd fields. No assertion of semisimplicity for low-degree tensor modules is needed.

Theorem C.1 (Joint bounded-degree transfer)**.** *Let L be a product of full natural classical groups over odd finite fields and D the product of their natural derived groups. The pairs are $(\mathrm{GL}_a(K), \mathrm{SL}_a(K))$, $(\mathrm{U}_a(K/K_0), \mathrm{SU}_a(K/K_0))$, $(\mathrm{O}_a(K), \Omega_a(K))$, and $(\mathrm{Sp}_a(K), \mathrm{Sp}_a(K))$. On a joint sum M of natural, dual, and specified Frobenius tensor words of degree at most r, allow arbitrary multiplicities, trivial summands, and prime-field descent. If $a > 2r$ for general-linear and unitary factors and $a \geq 4r + 3$ for orthogonal factors, then*

$$\mathrm{span}_{\mathbb{F}_p} \rho(D) = \mathrm{span}_{\mathbb{F}_p} \rho(L).$$

The equality passes to all invariant subquotients and is simultaneous on repeated, dual and twisted occurrences. Natural cutoff 36 suffices for $r = 8$.

Proof. We first use a finite abelian interpolation fact. If T is a prime-to-p abelian group, $S \leq T$, and restriction is injective on the occurring T-characters of M, then each full weight projector is

$$e_\chi = |S|^{-1} \sum_{s \in S} \chi(s)^{-1} \rho(s).$$

Consequently every $\rho(t) = \sum_\chi \chi(t) e_\chi$ is in the S-image algebra. This is character orthogonality on S, permitting arbitrary weight multiplicities. An equality of spans after scalar extension descends by solving a linear system over the ground field.

For $\mathrm{GL}_a(K)$ take the diagonal torus $T = (K^*)^a$. Its intersection with $\mathrm{SL}_a(K)$ is the product-one kernel. Characters trivial on that kernel have constant exponent vectors modulo $|K| - 1$. A degree-r natural/dual/twisted weight has support at most r; a difference of two has support at most $2r$. Since $a > 2r$, such a difference can be a constant vector only when it is zero. Also $\mathrm{GL} = \mathrm{SL}\, T$, so interpolation proves the claim. For unitary groups choose an orthogonal basis and the diagonal norm-one torus. The same proof works modulo $|K_0| + 1$, including the smallest odd fixed field.

For $\mathrm{O}_a(K)$ diagonalize the form with norms 1 and a fixed nonsquare ξ. Arrange that both norm classes occur more than $2r$ times. To justify this, every nonzero ξ is a sum x^2+y^2, since the sets of squares and ξ minus the squares both have size $(|K|+1)/2$. The matrix with rows (x,y) and $(-y,x)$ takes I_2 to ξI_2. Thus the number of nonsquare diagonal norms can be changed by two. The interval $[2r+1, a-2r-1]$ has two consecutive integers when $a \geq 4r+3$, providing the required parity.

Let J_0, J_1 be the two norm classes and let T be the diagonal sign group. A sign change in coordinate i is reflection in that basis vector. Its determinant is -1 and its spinor norm is the norm class of that vector. Hence T maps onto the full determinant/spinor quotient and $\mathrm{O} = \Omega T$. The annihilator of $S = T \cap \Omega$ is the span over $\mathbb{F}_2$ of $\mathbf{1}$ and $\mathbf{1}_{J_1}$. Its nonzero supports are all coordinates, J_0, and J_1, each larger than $2r$. Differences of occurring weights have support at most $2r$, since duals and odd Frobenius twists have the same sign weights. Interpolation again applies. The standard determinant/spinor facts are recalled in [16, Section 2.2] and [17, Section 6].

Symplectic transfer is immediate because $D = L$. Products use the product of these tori; character separation holds on every factor. Prime-field scalar extension decomposes a defining-field natural module into its Frobenius conjugates without changing the tensor-degree bound, so descent applies. Finally every D-invariant subspace is preserved by the same linear combinations representing L. Equality therefore passes to any invariant section and to represented central products. □

Include a trivial summand in a joint transfer. The coefficient sum in an expression $\rho(l) = \sum c_j \rho(d_j)$ is then one. It follows that L acts trivially on every D-fixed vector in any included section, and on every D-trivial composition factor. Intertwiners also transfer by using the joint sum of their domain and codomain, including an allowed outer twist.

Theorem C.2 (Normalization of actual p-subgroups)**.** *Suppose $D \lhd L$ has prime-to-p index and $V = V_1 \geq \cdots \geq V_h = 1$ is an L-invariant central elementary-abelian series of a finite p-group. Write $B_i = V_i/V_{i+1}$. Assume every D-invariant subspace of B_i is L-invariant, and every D-trivial L-composition factor of $B_j^* \otimes B_i$ is L-trivial. Then every actual D-normalized subgroup $R \leq V$ is L-normalized.*

Proof. Induct on the image of R in V/V_i. Suppose its image $\overline{R}$ is L-invariant. In V/V_{i+1} put $P = (R \cap V_i)V_{i+1}/V_{i+1} \leq B_i$. It is L-invariant. The inverse image E of $\overline{R}$ gives the central extension

$$1 \to B_i/P \to E/P \to \overline{R} \to 1.$$

The actual image of R is a complement. Thus group sections form a nonempty affine torsor for $W = \mathrm{Hom}_{\mathbb{F}_p}(\overline{R}/\Phi(\overline{R}), B_i/P)$. The actual section is D-fixed. Since $D \lhd L$, the D-fixed section set is L-stable, with difference space W^D.

Filter $\overline{R}/\Phi(\overline{R})$ by the images of $\overline{R} \cap (V_j/V_i)$. Its factors are sections of B_j. Thus the composition factors of W occur among those of $B_j^* \otimes B_i$. Every factor of W^D is L-trivial by hypothesis. As an L/D-module it is semisimple, so its entire L-action is trivial. Taking the actual section as origin, the remaining affine action is a homomorphism $L/D \to (W^D, +)$, necessarily zero. Hence the actual section is L-fixed and the induction advances. □

For a flag on a module of tensor degree at most s, its Hom grades have degree at most $2s$ and their pairwise Hom spaces at most $4s$. Theorem C.1 through degree $4s$, with a trivial summand, supplies both hypotheses. In the graph application $s = 2$, so degree eight suffices.

Lemma C.3 (Fixed cosets)**.** *Under these hypotheses, if L acts trivially on $(B_i/P)^D$ for every invariant $P \leq B_i$, then $(V/R)^D = (V/R)^L$ as right-coset sets. Normality of R in V is unnecessary.*

Proof. Induct on $V/(RV_i)$. Over an L-fixed point, the next fiber is an affine torsor for B_i/P_i, with $P_i = (R \cap V_i)V_{i+1}/V_{i+1}$. If it has a D-fixed point, the D-fixed locus is a torsor for $(B_i/P_i)^D$. Its difference action is trivial and its residual translations factor through the prime-to-p group L/D, so vanish. Starting with the singleton at $i = 1$ proves the result. This starts with a given fixed point in a fiber and does not assume that all fixed base points lift. □

All transfer expressions can be computed by matrix-span closure, or by enumerating the selected tori of size $p^{O(D_0)}$. No auxiliary vector set is enumerated.

D Unitary transport modulo fixed additive shifts

In this appendix, k is the stated base field, $q_k = |k|$, and N denotes the faithful degree of the represented algebra or physical rows over k. In the main algorithm $k = \mathbb{F}_p$, so $q_k = p$ and this degree is $O(D_0)$. Auxiliary module dimensions are not included in that faithful degree.

The affine transporter algorithm will reduce to conjugation modulo a subspace fixed pointwise by the algebra. Such a subspace may have very large dimension, so enumerating all possible shifts is not feasible. The useful observation is that a successful shift is a tuple of common scalar eigenvalues of operators on the original simple matrix factors. Only a small number of such tuples can survive. We first enumerate them, then solve each fixed-shift equation by the conductor method of Section A.

For an A-bimodule B, write $B^A = \{b \in B : ab = ba \text{ for all } a \in A\}$.

Proposition D.1. *Let A be a semisimple $*$-algebra of faithful degree N over a finite field k of odd order. On an A-bimodule B, conjugacy of vectors $v, w \in B$ modulo $P \leq B^A$ can be solved deterministically in $|k|^{O(N)}$ times polynomial auxiliary work. The algorithm lists every nonempty fixed-shift coset $u_p\,\mathrm{U}(A_v)$, with one representative per actual shift, where*

$$A_v = \{a \in A : av = va,\ a^*v = va^*\}.$$

It also returns generators of $\mathrm{U}(A_v)$ and the full modulo-P stabilizer.

D.1 Representing the bimodule by operators

The enveloping algebra $A^e = A \otimes_k A^{\mathrm{op}}$ is semisimple. A basis of B gives an explicit surjection from a finite direct sum of copies of A^e onto B. Solve linear equations for an A^e-linear section, giving a polynomial-size embedding into $(A^e)^d$. Choose the nondegenerate symmetric Frobenius functional

$$\tau(a) = \sum_i \mathrm{Tr}_{K_i/k}(\mathrm{tr}(a_i)), \qquad A = \prod_i \mathrm{M}(a_i, K_i).$$

The map

$$x \otimes y \longmapsto [z \longmapsto x\tau(yz)]$$

identifies A^e, as an A-bimodule, with $\mathrm{End}_k(A)$, with left and right actions given by composition with left multiplication. Place the d resulting operators in distinct $(j, 0)$ blocks on A^{d+1}. This embeds B into an endomorphism space, equivariantly for conjugation by $\mathrm{U}(A)$, and sends B^A into the commutant of A. All dimensions are polynomial. The controlling exponential degree remains the original N.

It suffices, therefore, to solve the problem for $M, M' \in \mathrm{End}_k(W)$ and $P \leq \mathrm{Cent}(\rho(A))$, with an explicit A-module W.

D.2 Enumerating all shifts

In an isotypic component, an algebra-central shift is the identity on the natural simple module and an arbitrary matrix on its multiplicity space. We enumerate its coefficients jointly as common eigenvalues; we do not enumerate the multiplicity matrices independently.

Write $A = \prod_i \mathrm{M}(a_i, K_i)$, with $K_i = \mathbb{F}_{q_k^{f_i}}$. On an occurring isotypic component use coordinates

$$W_i = K_i^{a_i} \otimes_{K_i} Z_i,$$

where Z_i is the K_i-multiplicity space and A_i acts on the first tensor factor. A commutant shift is zero between distinct isotypic components and is K_i-linear on W_i. Decompose an arbitrary k-linear diagonal block into Frobenius components

$$M_{ii} = \sum_{\sigma \in \mathrm{Gal}(K_i/k)} M_{ii}^{(\sigma)} \sigma.$$

This is a direct-sum decomposition computable by linear algebra. Conjugation preserves its summands. Expand the identity components as

$$M_{ii}^{(1)} = \sum_t A_{it} \otimes Q_{it}, \qquad M'^{(1)}_{ii} = \sum_t B_{it} \otimes Q_{it},$$

where the Q_{it} form a K_i-basis of $\mathrm{End}_{K_i}(Z_i)$. A shift has the form $I \otimes \sum_t \nu_{it} Q_{it}$. The conjugacy condition forces

$$X A_{it} - B_{it} X = \nu_{it} X \quad \text{for all } t, \qquad X = u_i. \tag{5}$$

For operators $L_1, \ldots, L_s \in \mathrm{End}_K(X)$ on a d-dimensional K-space X, start with $W = X$ and list all joint scalar weights by successive intersections

$$W_\lambda = W \cap \ker(L_j - \lambda), \qquad \lambda \in K.$$

No invariance of W under L_j is needed. At each stage the sum of the surviving dimensions is at most d, since distinct eigenvalues give independent spaces. Thus at most d joint weights survive. Applied to (5), there are at most a_i^2 weights on factor i. Enumerating field elements costs $q_k^{f_i}$, and

$$\sum_i f_i a_i \le N, \qquad \prod_i a_i^2 \le 4^N.$$

Hence every complete candidate shift can be enumerated within $q_k^{O(N)}$ and filtered by membership in P. Factors absent from the module impose no shift coordinates and remain in the subsequent algebra computation. All off-diagonal blocks and nonidentity Frobenius components are imposed together in the next step; their witnesses are not chosen independently.

D.3 Completing one fixed shift

Once a shift is fixed, all remaining equations are simultaneous intertwining equations. The two equations below also impose their involution partners, so their solution space is a principal module whenever it contains an invertible solution.

For a candidate shift $p \in P$ put $y = w + p \in B$ and compute

$$A_v = \{a : av = va,\ a^* v = v a^*\}, \qquad X_y = \{c : cv = yc,\ c^* y = v c^*\}.$$

These equations may be imposed in the original bimodule or its faithful embedding. The space X_y is a right A_v-module. If it contains an A-unit c_0, then $X_y = c_0 A_v$: writing $c = c_0 a$ turns the two intertwining equations into the two centralizer equations. Deterministic module isomorphism tests $(A_v)_{A_v} \cong (X_y)_{A_v}$. If an isomorphism ϕ exists, take $c = \phi(1)$ and test whether c is a unit of A. This is conclusive: if any unit c_0 existed, every module generator would be $c_0 a$ for $a \in A_v^\times$ and would be a unit.

For an invertible c in X_y, the element $h = c^* c$ is selfadjoint and lies in A_v. The standard $*$-norm algorithm solves

$$d^* h d = 1, \qquad d \in A_v^\times,$$

or rejects. A successful $u_p = cd$ is unitary, and the complete fixed-shift coset is $u_p \,\mathrm{U}(A_v)$. The algebra A_v may have radical; the deterministic norm and unitary-generator routines apply to it. The module-isomorphism spaces have dimension at most $\dim A \le N^2$; large multiplicity spaces enter only the linear equations.

For self transport the shift map

$$u \longmapsto uvu^{-1} - v$$

is additive, because every unitary fixes P pointwise. Its kernel is $\mathrm{U}(A_v)$. Representatives for the distinct successful shifts together with this kernel generate the full stabilizer. The same representatives, or their ratios to one successful transporter, give the corresponding full transporter and source stabilizer for a distinct target. This proves Proposition D.1.

E Bounding repeated semisimple shift choices

Let $A = \prod_i A_i$ be semisimple, with

$$A_i = \mathrm{M}(a_i, \mathbb{F}_{q_k^{f_i}}), \qquad D_i = f_i a_i^2, \quad \delta_i = f_i a_i.$$

Let H be the modulo-central-space stabilizer in the preceding problem, let $\eta : H \to P$ be its additive shift map, and let $B = A_v$ be its exact source $*$-centralizer. For a factor i occurring in the module, project the shift to its complete diagonal coefficient tuple, denoting the image by $\eta_i(H)$. Each weight space contains $u_i \pi_i(B)$ for a suitable unitary u, where $\pi_i : A \to A_i$ is projection. Distinct weight spaces are independent, so

$$|\eta_i(H)| \dim_k \pi_i(B) \le D_i. \tag{6}$$

For a factor absent from the module take a singleton shift image. The full shift is determined by its occurring-factor tuples, and hence $|\eta(H)| \le \prod_i |\eta_i(H)|$.

Write

$$B/\operatorname{rad}(B) = \prod_j B_j, \qquad B_j = \mathrm{M}(b_j, \mathbb{F}_{q_k^{g_j}}), \qquad D'_j = g_j b_j^2, \quad \delta'_j = g_j b_j.$$

Choose a Levi complement of B and join i to j when B_j has nonzero image in A_i. Every vertex has a neighbor. For the neighbor set J_i, mutually annihilating simple images and restriction of the natural A_i-module give

$$\dim_k \pi_i(B) \ge \sum_{j \in J_i} D'_j, \qquad \sum_{j \in J_i} \delta'_j \le \delta_i, \qquad \sum_{j \in J_i} b_j \le a_i. \tag{7}$$

For the last inequality, the projected central idempotents have K_i-ranks adding to a_i. Every nonzero constituent of $\mathrm{M}(b_j, \mathbb{F}_{q_k^{g_j}})$ after scalar extension to K_i has K_i-dimension at least b_j. The radical of B can only increase $\dim \pi_i(B)$, so these inequalities do not discard it.

Define, for a simple algebra,

$$\mu\big(\mathrm{M}(a,\mathbb{F}_{q_k^f})\big)=\frac{8^{fa+a-2}}{fa^2},\tag{8}$$

and define μ multiplicatively on semisimple products. A faithful degree-N representation gives $1\le\mu(A)\le 8^{2N}$.

Lemma E.1. *Suppose positive integer parameters satisfy $\delta=fa$, $D=\delta a$, $\delta_j=g_jb_j$, $D_j=\delta_jb_j$, $\sum_j\delta_j\le\delta$, and $\sum_j b_j\le a$. Then*

$$\mu(\mathit{old})\ge\frac{D}{\sum_j D_j}\prod_j\mu(\mathit{new}_j).$$

Proof. Put $d=\sum_j\delta_j$, $b=\sum_j b_j$, $t=\#\{j\}$, and $h=(\delta-d)+(a-b)$. The inequalities $\delta\le 2^{\delta-d}d$ and $a\le 2^{a-b}b$ give $D^2\le 4^h(db)^2$. Weighted Cauchy–Schwarz gives

$$d^2=\Big(\sum_j g_jb_j\Big)^2\le\Big(\sum_j D_j\Big)\Big(\sum_j g_j\Big).$$

For positive integers $x_1,\dots,x_t$, $\sum_j x_j\le 2^{t-1}\prod_j x_j$. Apply this to the g_j and twice to the b_j to obtain

$$(db)^2\le 8^{t-1}\Big(\sum_j D_j\Big)\prod_j D_j.$$

Therefore

$$D^2\le 8^{h+2t-2}\Big(\sum_j D_j\Big)\prod_j D_j.$$

On the other hand, $\mu(\mathrm{old})/\prod_j\mu(\mathrm{new}_j)=8^{h+2t-2}\prod_j D_j/D$, proving the claim. □

Apply Lemma E.1 to every old factor using (7), and then use (6). Since every new vertex has positive degree and every simple potential is at least one, we obtain

$$\mu(A)\ge|\eta(H)|\mu(B/\operatorname{rad}(B)).\tag{9}$$

For a nonempty transporter, its distinct fixed shifts form a coset of $\eta(H)$. Thus their number is exactly $|\eta(H)|$, and all their source kernel algebras are B. Equation (9) is therefore a bound on the sum of child potentials in a branching tree, not just on a product along one path. This proves the fixed-shift branch bound used in the semisimple affine problem and the augmentation algorithms.

F Affine transport for semisimple unitary groups

In this appendix, k is the stated base field, $q_k=|k|$, and N denotes the faithful degree of the represented algebra or physical rows over k. In the main algorithm $k=\mathbb{F}_p$, so $q_k=p$ and this degree is $O(D_0)$. Auxiliary module dimensions are not included in that faithful degree.

A layer of a radical filtration can carry an affine action even when the original tensor action is linear. Its translations arise from lifts in the current stabilizer and form a cocycle. We must solve this affine problem without assuming the stabilizer extension splits.

A group-invariant quotient of an algebra bimodule need not be an algebra bimodule. Lemma F.3 embeds it into a bimodule modulo a central subspace. The cohomology calculation in Appendix G then converts the affine equations into the central-shift problem.

For a left G-module Q, a cocycle $z:G\to Q$ satisfies $z(gh)=z(g)+gz(h)$ for $g,h\in G$. Its affine action is $g:x\mapsto gx+z(g)$ for $x\in Q$.

Theorem F.1. *Let F be a semisimple star-algebra faithfully represented in degree N, and put $G = \mathrm{U}(F)$. Let Q be an explicit G-invariant subquotient of an algebra F-bimodule. Given a cocycle $z : G \to Q$ and $c \in Q$, the affine transporter from 0 to c can be computed deterministically in $q_k^{O(N)}\,\mathrm{poly}(\mathsf{L}, \log q_k)$ time. It is returned as fixed-shift cosets $g_j\,\mathrm{U}(F')$, with a common source star-subalgebra $F' \le F$ and actual generators. For a potential ν on semisimple algebras satisfying $1 \le \nu(F) \le q_k^{O(N)}$, the number t of returned cosets satisfies $t\nu(F'/\operatorname{rad} F') \le \nu(F)$. If $F' = F$, there is at most one coset and the affine source is fixed by all of G.*

F.1 Embedding an invariant quotient into a central quotient

Proposition F.2 (The joint degree-two image algebra)**.** *Let F be a semisimple k-algebra with involution, faithfully represented on a space W of dimension N, and suppose every noncommutative simple factor has matrix size at least 36. Put $G = \mathrm{U}(F)$, $V_0 = \mathrm{End}_k(W)$, and*

$$A = \mathrm{span}_k\{\mathrm{Ad}(g)|_{V_0} : g \in G\}.$$

After isomorphic whole-module occurrences have been identified, the nonsemisimple part of the action consists of the chain types listed in Table 1. Each such type i is a G-module $M_i \le V_0$ with a defining field K_i/k and a flag

$$0 \subset K_i I \subset M_{i,0} \subset M_i, \qquad M_{i,0}/(K_i I) = H_i, \quad M_i/M_{i,0} \cong K_i,$$

where H_i is simple of K_i-dimension h_i and both adjacent extensions are nonsplit. On the sum of one copy of each type, the image is

$$A_0 = k1 + \bigoplus_i R_i, \qquad R_i = \left\{ \begin{pmatrix} 0 & b & \gamma \\ 0 & D & e \\ 0 & 0 & 0 \end{pmatrix} : \; b \in K_i^{1\times h_i},\; D \in \mathrm{M}(h_i, K_i),\; e \in K_i^{h_i \times 1},\; \gamma \in K_i \right\}. \tag{10}$$

In particular $R_i R_j = 0$ for $i \ne j$, and all scalar endpoints share one k-valued diagonal character. The remaining nontrivial simple module types give separate semisimple factors of A. Repeated copies of a type have the same action, not independent image factors. The fields, blocks, and identifications are constructible in $q_k^{O(N)}\,\mathrm{poly}(\mathsf{L}, \log q_k)$ time using polynomial-dimensional linear algebra.

Proof. The representation-theoretic input specifies the constituents; the argument below also proves the radical entries and their joint image. For the table, $a \ge 36$ is the natural matrix size, K is the defining matrix field, and in the second-kind unitary row K/K_0 is quadratic. A form adjoint means the adjoint involution for the defining bilinear or Hermitian form. Each listed component contains I, and $M_{i,0}$ is its trace-zero subspace. The heart means the quotient $M_{i,0}/(K_i I)$.

Constituents, twists, and repetitions. After decomposing the center fields, the terms in V_0 are products of two natural or dual modules with specified Frobenius twists. Terms supported on different classical factors are external simple tensors. On one exchanged general-linear factor the equal-twist terms are adjoint modules, symmetric squares, or exterior squares. Orthogonal and symplectic equal-twist terms split by the form adjoint; only the component in the table can contain scalar endpoints. For a unitary factor, the involution twist identifies a dual natural module with the appropriate natural twist. The equal adjoint twist gives the Hermitian component in the table; the natural-square twist gives symmetric and exterior squares. All other distinct-digit terms are the corresponding simple twisted tensor products. The constituent assertions use [8, Section 3.2 and

Classical factor	Component M_i when $p \mid a$	K_i	h_i
Exchanged general linear	$\mathrm{M}(a, K)$ under conjugation	K	$a^2 - 2$
Orthogonal	Form-self-adjoint matrices, identified with symmetric forms	K	$a(a+1)/2 - 2$
Symplectic	Form-self-adjoint matrices, identified with alternating forms	K	$a(a-1)/2 - 2$
Second-kind unitary	Hermitian form of $\mathrm{M}(a, K)$	K_0	$a^2 - 2$

Table 1: All nonsemisimple degree-two chain types in odd characteristic and matrix size $a \geq 36$. If $p \nmid a$, the scalar line splits off and the corresponding component is semisimple. These components are G-invariant subspaces of genuine matrix bimodules; they need not themselves be algebra bimodules.

the proofs of Theorems 2.9 and 2.11]; the relevant statements appear on pp. 15 and 65 of its cited preprint. Writing ω_j for the jth fundamental weight, same-digit squares have restricted weights $2\omega_1$ and ω_2, and distinct digits use the Steinberg tensor-product description. Since p is odd these weights are restricted even over $\mathbb{F}_3$; the cutoff excludes low-rank coincidences. This constituent input does not assert the joint algebra (10).

Here are the identifications used before forming that algebra. Repeated natural-module copies give identical whole matrix components. Transpose identifies the two adjoint occurrences of an exchange pair. Entrywise Frobenius identifies field-conjugate whole components after restriction to k. In the unitary case multiplication by a nonzero trace-zero element of K identifies the Hermitian and skew-Hermitian fixed-field forms. The remaining Galois orbits of simple twisted constituents are grouped by their actual module isomorphisms; they contribute semisimple blocks. Consequently there is at most one nonsemisimple chain type for each independent classical factor. The scalar constituents, including any from commutative factors, all carry the same trivial k-character. Commutative factors have prime-to-p unitary groups and create no nonsemisimple chains.

Nonsplit extensions and coefficient fields. The trace pairing on each component in the table is nondegenerate over K_i and is invariant under the finite group. Its invariant vectors form the single line $K_i I$. If the simple heart split off as a submodule, its dimension, greater than half the component's dimension, would force the restricted pairing to be nondegenerate. Its orthogonal complement would have dimension two and only trivial composition factors. Its image would lie in a two-dimensional orthogonal group of prime-to-p order; with those composition factors every image element is unipotent, so the image is trivial. This contradicts the single invariant line. Thus the bottom adjacent extension is nonsplit, and duality proves the same for the top one. If $p \nmid a$, the trace splits off the scalar line instead.

The coefficient field of each heart is the full defining field of its chain, not an unspecified subfield. Here is a direct verification. For a general linear factor use $\mathrm{diag}(s, r, 1, \ldots, 1)$, where $s \in K^\times$ and $r = \pm 1$. The difference between its two traces on the adjoint heart is

$$2(s + s^{-1}) + 4(a - 2).$$

The same formula holds on a unitary adjoint heart over the fixed field K_0, now taking s in the norm-one subgroup of $K^\times$. For orthogonal or symplectic factors use two hyperbolic pairs with natural eigenvalues s, s^{-1}, r, r^{-1} and all remaining eigenvalues one. The difference on either degree-two heart is

$$4(s + s^{-1} + a - 4).$$

Subtracting the trivial composition factors does not change these trace differences. The set of $s+s^{-1}$ has at least $(|K|-1)/2$ values in the first-kind cases, and at least $(|K_0|+1)/2$ in the unitary case. If the field is a proper extension of k, these cardinalities exceed the order of every proper subfield. The ordinary defining-field traces therefore generate the full chain field over k.

To conclude, the k-image algebra on the absolutely irreducible heart has full matrix algebra as its defining-field span. Its radical must vanish, and it has one simple factor: a radical or additional central idempotent would persist in that full matrix span. It is a matrix algebra over a subfield of the chain field. Its ordinary matrix traces belong to that subfield, so the preceding calculation makes the subfield equal to the whole chain field. This also explains why field traces down to k, which could erase information, are not used in this step.

From each chain to the joint image. For one chain the semisimple quotient of its k-image algebra is $k \times \mathrm{M}(h_i, K_i)$. Choose its semisimple complement and split the flag in those coordinates. The actual image is

$$\left\{ \begin{pmatrix} \alpha & b_i & \gamma_i \\ 0 & D_i & e_i \\ 0 & 0 & \alpha \end{pmatrix} : \ \alpha \in k, \quad b_i, D_i, e_i, \gamma_i \text{ over } K_i \right\}. \tag{11}$$

To see equality rather than just containment, the nonsplit bottom extension forces a nonzero radical row from the heart to the trivial vertex. It is a simple right $\mathrm{M}(h_i, K_i)$-module, so the row is full. The top extension similarly gives the full column. Their products give every K_i-valued top-to-bottom entry. The two trivial diagonal actions are both the same $\alpha \in k$, because every group element fixes both scalar factors pointwise. The invariant flag excludes every other entry. This proves (11), including its radical multiplication.

On the sum of the distinct chains, the independent classical factors act independently. The differences $g-1$ for elements in one factor generate its augmentation image R_i, with zero action on every other chain. Hence their products across different factors vanish, and their sum with the common identity is exactly (10). Repeated, dual, and field-conjugate chains were already identified as whole modules, so this does not introduce spurious independent radical rows or columns on their copies. Every other nontrivial simple type has an annihilator distinct from these heart and trivial types. The corresponding simple quotient therefore separates by the Chinese remainder theorem and gives a semisimple direct factor of the image algebra. This proves the claimed description of the entire joint image.

For construction, generate A by linear-span closure under products of the supplied matrices $\mathrm{Ad}(g)$. There are at most N^4 independent matrices. Recover the described idempotents and matrix units by linear equations in that algebra, using the original center coordinates and the whole-component identifications above. Finite-field factorization can be performed by deterministic enumeration over an original center field, of degree at most N, with polynomial extension degree. Thus the cost is $q_k^{O(N)}$ times a polynomial, without enumerating $q_k^{\dim A}$ or representing the group faithfully in degree N^2. □

Lemma F.3 (Constructive degree-two embedding)**.** *Let F be a semisimple k-algebra with involution, faithfully represented on a space W of dimension N. Assume every noncommutative simple factor has matrix size at least 36, and put $G = \mathrm{U}(F)$. Let Q be a supplied G-invariant subquotient of an F-bimodule. There are an explicit F-bimodule B, a subspace $P_0 \le B^F$, and an injective G-equivariant linear map $j : Q \to B/P_0$. Their dimensions are polynomial in the explicit input length, and they can be computed in $q_k^{O(N)} \operatorname{poly}(\mathsf{L}, \log q_k)$ time. Any supplied linear action commuting with G extends to this construction through the Q multiplicity coordinate.*

Proof. Use Proposition F.2 on $V_0 = \mathrm{End}_k(W)$. Every simple F-module occurs in the faithful module W, so every simple algebra F-bimodule occurs in V_0. The algebra $F \otimes_k F^{\mathrm{op}}$ is semisimple; hence every F-bimodule embeds into a direct sum of copies of V_0. In particular the supplied group-invariant subquotient Q is an A-module. We use the actual group-generated image algebra here; no F-bimodule action on Q is assumed.

An explicit dual regular embedding. For each chain type i, write $V_i = M_i$ for the whole chain from Proposition F.2, and let B_i be a genuine matrix-bimodule summand of V_0 containing it. In the orthogonal, symplectic, and unitary cases, V_i is only a G-invariant subspace of B_i; it is not asserted to be an algebra bimodule. Its scalar line $K_i I$ nevertheless lies in B_i^F. Let e_0 select the common trivial vertex in (10), and let e_{ij} select the jth heart coordinate. The right ideal $e_0 A_0$ has coordinates $\alpha \in k$, $b_i \in K_i^{h_i}$, and $\gamma_i \in K_i$. For $v_i = (u_i, y_i, z_i) \in V_i$, define

$$P_* = \{(u_i, 0, 0)_i : \sum_i \mathrm{Tr}_{K_i/k}(u_i) = 0\}.$$

Viewed inside $\bigoplus_i B_i$, it is a subspace of the algebra commutant $(\bigoplus_i B_i)^F$. The pairing

$$\langle a, v\rangle = \sum_i \mathrm{Tr}_{K_i/k}(\text{first scalar coordinate of } av_i)$$

identifies

$$(e_0 A_0)^* \cong (\bigoplus_i V_i)/P_*, \qquad (e_{ij} A_0)^* \cong V_i/(K_i I). \tag{12}$$

The free row and top-to-bottom entries detect the heart and top coordinates by the nondegenerate field trace pairing. The common scalar detects exactly the remaining scalar quotient. Associativity of the pairing proves equivariance. This argument does not use $\mathrm{Tr}(1) \neq 0$; trace remains nonzero and nondegenerate when the extension degree is divisible by p. The first right ideal has k-dimension $1 + \sum_i [K_i : k](h_i + 1)$. There are h_i heart-coordinate right ideals of dimension $[K_i : k](h_i + 1)$ for each i. Hence both sides of the resulting decomposition of A_0^* have dimension

$$1 + \sum_i [K_i : k](h_i + 1)^2.$$

The other nontrivial semisimple blocks of A^* embed in copies of their already present degree-two modules. All trivial occurrences share the common vertex already included in A_0; they add no independent trivial algebra factor. If there are no chains, that single trivial factor instead embeds in a central matrix line. We obtain an explicit polynomial-size embedding

$$A^* \hookrightarrow B_*/P_*, \qquad B_* \text{ a genuine } F\text{-bimodule}, \quad P_* \leq B_*^F. \tag{13}$$

For an arbitrary A-module Q, use coinduction:

$$Q \hookrightarrow \mathrm{Hom}_k(A, Q), \qquad v \longmapsto [x \mapsto xv], \qquad (af)(x) = f(xa).$$

Evaluation at 1 proves injectivity. Tensoring (13) with the underlying vector space of Q yields

$$j : Q \hookrightarrow B/P_0, \qquad P_0 \leq B^F, \tag{14}$$

with polynomial auxiliary dimensions. A supplied action commuting with G extends by postcomposition on $\mathrm{Hom}_k(A, Q)$, hence acts only on the original Q multiplicity coordinate in (14). It preserves P_0 and retains every actual group correlation.

All matrix and field coordinates here are computed by polynomial-dimensional linear algebra in the explicitly described components. Once the image description is proved, its specified idempotents and matrix units are recovered by linear equations in the generated image algebra. Finite-field factorization can use deterministic enumeration over an original center field of degree at most N, with polynomial extension degree, within $q_k^{O(N)}$ work. No enumeration of $q_k^{\dim A}$ or $q_k^{\dim Q}$ occurs. □

F.2 Bounded matrix sizes and the branching potential

We now prove Theorem F.1. Bounded matrix sizes are handled by an actual coset enumeration before the preceding embedding or the cohomology statement is used. A multiplicative potential pays for this enumeration and for all subsequent fixed-shift choices.

For a bounded natural matrix size $2 \le a < 36$, enumerate the unitary factor modulo its algebra-central unitary group and replace that algebra factor by its center. This is an actual coset decomposition. Its center has prime-to-p unit group. The construction above and the large-classical cohomology lemma apply to the reduced algebra, not to the unreduced small group. In particular, no affine coboundary assertion is made for $\mathrm{SL}_2(3)$.

Let $\delta(F)$ be the sum of the minimal faithful degrees of its simple algebra factors. Define

$$\nu(F) = \mu(F) q_k^{512\delta(F)},$$

where μ is the multiplicative potential in Appendix E. Replacing $\mathrm{M}(a, \mathbb{F}_{q_k^f})$ by its center decreases δ by $f(a-1)$ and has unitary index at most $q_k^{fa^2}$. The fixed constant 512 pays every bounded-rank index, including exchange pairs. The semisimple faithful degree cannot increase under restriction. The fixed-shift μ inequality therefore also contracts ν. Hence the sum of child potentials is at most the parent potential and $1 \le \nu(F) \le q_k^{O(N)}$.

For every coset of the bounded-rank center subgroup, normalize only the target by the full affine action. Keep source 0 and the same restricted cocycle. Thus the source embedding below and its exact source algebra are shared by all target-normalized cosets. If a noncommutative factor was replaced, that source algebra is proper in the old algebra; it is not a no-loss step.

F.3 Turning the affine action into a linear transporter

For a left G-module Q, a cocycle is a function $z : G \to Q$ satisfying $z(gh) = z(g) + gz(h)$ for $g, h \in G$. It defines the affine action $g : x \mapsto gx + z(g)$. A coboundary has the form $z(g) = gb - b$ for a fixed vector b. After the embedding of Lemma F.3, the cohomology statement guarantees such a vector in the larger central quotient, where it is found by linear equations.

By Lemma G.1, the whole bimodule ambient space B satisfies $H^1(G, B) = 0$, and the central space P_0 satisfies $H^2(G, P_0) = 0$, after the bounded-rank reduction. The exact sequence gives

$$H^1(G, B/P_0) = 0.$$

Given the original cocycle $z : G \to Q$, solve on generators the linear equations

$$j(z(g)) = gb - b \quad \text{in } B/P_0. \tag{15}$$

They are consistent by the vanishing just proved. The map $x \mapsto j(x) + b$ converts the original affine action into ordinary unitary conjugation on the ambient central quotient. Apply Proposition D.1. It supplies every nonempty fixed-shift coset, one common exact source star-algebra F', and the fixed-shift branch bound. Together with the bounded-rank charge, this proves Theorem F.1. The

equation is imposed in the original actual group through its generators, so the result includes witness words and full stabilizers.

For clarity, this applies to an arbitrary affine cocycle, not merely a translated fiber already embedded in the original module. If a commuting solvable group also acts affinely, with linear part c and shift $z_C(c)$, extend its linear action functorially as above and set

$$d_C(c) = j(z_C(c)) + b - cb.$$

Affine commutation implies that $d_C(c)$ is G-fixed on the whole ambient quotient, and its cocycle identity is inherited from that of z_C. Homogenize this external affine action by one fixed coordinate and apply the linear repair once more. Two local repairs still have polynomial size.

In every augmentation recursion the grade is represented as a subquotient of the *original* ambient bimodule. The local coinduction repair is discarded after recovering original group words. Repeatedly feeding an expanded repair into the next repair is not part of the algorithm.

G Solving the affine cocycle equations

The preceding appendix reduces affine transport to finding a vector b satisfying $z(g) = gb - b$ on generators. This appendix proves the existence required for that linear system. For a finite group G and a left G-module B, the first cohomology group $H^1(G, B)$ is the space of cocycles modulo coboundaries. Thus $H^1(G, B) = 0$ says exactly that every such system has a solution. The second cohomology group $H^2(G, P)$ records the obstruction to lifting these equations across a central quotient with trivial coefficient module P.

We prove the needed vanishings for whole Hom modules and central quotients. These are the modules supplied by the embedding lemma; no vanishing for an isolated simple constituent is needed. For $g, h \in G$, a cocycle $z : G \to B$, and a vector $b \in B$, we use the convention

$$z(gh) = z(g) + gz(h), \qquad z(g) = (g - 1)b$$

for cocycles and coboundaries. Scalar extension commutes with these finite linear equations. We may therefore work over an algebraic closure and subsequently descend every required solution by linear algebra.

Lemma G.1. *Let F be a semisimple algebra with involution over a finite field of odd characteristic. Suppose each noncommutative matrix factor has matrix size at least 36 over its defining field; commutative factors are allowed. Put $G = U(F)$. If B is an algebra F-bimodule with action $b \mapsto ubu^{-1}$, then*

$$H^1(G, B) = 0.$$

For every finite-dimensional trivial G-module P in characteristic p, $H^1(G, P) = H^2(G, P) = 0$. In particular, for a G-invariant central subspace $P \subseteq B$,

$$H^1(G, B/P) = 0.$$

All statements permit restriction of scalars to $\mathbb{F}_p$. A supplied cocycle in the last module is made a coboundary by solving linear equations on the supplied group generators.

G.1 Special linear groups including the field of order three

The special linear subgroup controls the natural and dual coordinates used later inside all classical groups. Averaging over its diagonal torus removes the torus part of a cocycle. Root conjugation then leaves very few possible matrix entries. The field of order three requires a separate order-three calculation, which is included explicitly.

Let $e \geq 1$ and $r \geq 5$ be integers, put $K = \mathbb{F}_{p^e}$ and $G = \mathrm{SL}(r, K)$, and let A_i denote the natural representation with entries twisted by p^i, $0 \leq i < e$. Write 1 for the one-dimensional trivial module. In matrix coordinates, E_{ab} is the matrix unit with its sole nonzero entry at position (a, b), and e_a is the ath standard column vector. On a diagonal torus, ε_a denotes its ath coordinate character. We first prove

$$H^1(G, \mathrm{Hom}(U, V)) = 0, \quad U, V \in \{1, A_i, A_i^* : 0 \leq i < e\}. \tag{16}$$

The diagonal determinant-one torus T has order prime to p. Subtract a coboundary so that $z|_T = 0$. Write $x_{ab}(s) = I + sE_{ab}$. Then

$$z(tx_{ab}(s)t^{-1}) = tz(x_{ab}(s)).$$

Consequently a root value lies in the fixed space of $\ker(t \mapsto t_a/t_b)$.

Characters of T are exponent vectors modulo $|K| - 1$ and the all-ones vector. A weight supported on at most two coordinates is trivial on that root kernel only if it is a multiple of $\varepsilon_a - \varepsilon_b$ modulo $|K| - 1$. Indeed the root character is onto, and $r \geq 5$ leaves a coordinate outside both supports, forcing the all-ones adjustment to vanish. The natural and dual weights admit no such value. The weights $p^i\varepsilon_u - p^j\varepsilon_v$ with $i \neq j$ admit none either. For $A_i \otimes A_j$ or its dual, a surviving value would require $p^i + p^j = 0$ modulo $|K| - 1$, which occurs in odd characteristic only when $K = \mathbb{F}_3$ and $i = j = 0$. Thus only the equal-twist endomorphism module and this tensor square require further work. The trivial module has zero first cohomology because the root commutator relation makes $\mathrm{SL}(r, K)$ perfect.

For the endomorphism module, a Frobenius automorphism reduces to untwisted conjugation. Torus normalization gives

$$z(x_{ab}(s)) = D(s) + B(s)E_{ab} + C(s)E_{ba},$$

with D diagonal. The root cocycle identity makes C additive, and comparison in the two orders of parameters gives $sC(t) = tC(s)$. Hence $C(s) = cs$. Torus conjugation gives $C(\lambda s) = \lambda^{-1}C(s)$. For $|K| > 3$ a value $\lambda^2 \neq 1$ forces $c = 0$. For $K = \mathbb{F}_3$, use the order-three relation instead:

$$(1 + \mathrm{Ad}\, x_{ab}(1) + \mathrm{Ad}\, x_{ab}(1)^2)E_{ba} = E_{ab},$$

whereas this norm operator kills the diagonal space and E_{ab}. It again forces $c = 0$.

Now D is additive. Conjugating by a torus element with root value -1 fixes D, whereas additivity negates it, so $D = 0$. The remaining identities are

$$B(s + t) = B(s) + B(t), \qquad B(\lambda s) = \lambda B(s),$$

and therefore $B(s) = b_{ab}s$. Equivalently the cocycle defines the dual-number representation

$$\rho_\epsilon(g) = (I + \epsilon z(g))g, \qquad \epsilon^2 = 0,$$

which replaces the root parameter s by $s(1 + \epsilon b_{ab})$. Applying the relation $[x_{ab}(s), x_{bc}(t)] = x_{ac}(st)$ gives $b_{ac} = b_{ab} + b_{bc}$ for distinct indices. Thus $b_{ab} = d_b - d_a$ for scalars d_a, and the cocycle is the coboundary of $\mathrm{diag}(d_1, \ldots, d_r)$. This includes $p \mid r$ and $K = \mathbb{F}_3$.

It remains to examine $A \otimes A$ over $\mathbb{F}_3$. Odd characteristic splits it into symmetric and exterior squares. In the symmetric-matrix model $X \mapsto gXg^T$, write

$$z(x_{ab}(1)) = D + bS_{ab}, \qquad S_{ab} = E_{ab} + E_{ba}.$$

These are all root-kernel fixed vectors; all diagonal squares have trivial torus character over $\mathbb{F}_3$. The order-three norm gives $D_{bb} = 0$. The inverse-root cocycle identity gives

$$z(x_{ab}(-1)) = 2D + 2bS_{ab} + 2bE_{aa}.$$

Torus conjugation fixes D and negates S_{ab}, whence $D = bE_{aa}$. For distinct $a, c \neq b$, the commuting-root identity for $x_{ab}(1)$ and $x_{cb}(1)$ gives $b_{ab} = b_{cb}$. Thus the coefficient depends only on b. Its values are exactly the coboundary of a diagonal symmetric matrix.

In the exterior-square model, write $z(x_{ab}(1)) = b_{ab}e_a \wedge e_b$. The same commuting-root identity gives $b_{ab} = -b_{cb}$. Three distinct rows outside column b force all coefficients to be zero. Dual tensor squares follow by inverse transpose. This proves (16) on all root generators and hence on G.

G.2 Additive maps between the relevant modules

The radical-layer shifts can be merely prime-field linear. To determine which block entries a cocycle may have, we need the types of equivariant additive maps, including their Frobenius twists. This step avoids an incorrect assumption that every additive map is linear over the larger defining field.

An additive map of K-vector spaces is only $\mathbb{F}_p$-linear. After scalar extension from $\mathbb{F}_p$, the underlying additive natural module splits into its Frobenius twists. The same is true for symmetric and exterior squares. In odd characteristic those two squares are irreducible for the special linear group of degree at least five. Distinct-twist products $A_i \otimes A_j$ and $A_i \otimes A_j^*$ are irreducible by Steinberg's tensor-product theorem. Equal-twist endomorphism modules have only scalar and adjoint-heart constituents. These degree-two types differ from the natural type and from the opposite symmetric or exterior-square type.

It follows that an equivariant additive map vanishes unless these types match. In a matching natural, symmetric-square, or exterior-square type it is a scalar multiple of the appropriate entrywise Frobenius map. This rule applies also over $\mathbb{F}_3$; it does not assume that an arbitrary additive map is K-linear. The finite-point degree-two constituent statements are the standard defining-characteristic statements used in the representation-repair appendix; see [8, Section 3.2 and the defining-characteristic proofs of Theorems 2.9 and 2.11].

G.3 Symplectic and split even orthogonal groups

A large special linear subgroup occurs in a block diagonal subgroup of these groups. The preceding calculation removes the cocycle on that subgroup. The upper and lower unipotent groups then leave at most one scalar each; one explicit commutator relation eliminates the remaining scalar after a coboundary correction.

Let $m \geq 5$, let $G = \mathrm{Sp}_{2m}(K)$, and let W be its natural $2m$-dimensional module, split as $W = A \oplus A^*$ with $A = K^m$. For a twist index i, write W_i for the p^i-Frobenius twist of W and ρ_i for its representation. Its Siegel Levi contains

$$L = \{\mathrm{diag}(g, g^{-T}) : g \in \mathrm{SL}(m, K)\}.$$

Let z have values in $\mathrm{Hom}(W_i, W_j)$, with action $X \mapsto \rho_j(g) X \rho_i(g)^{-1}$. By (16), subtract a coboundary so that $z|_L = 0$. Use

$$u_+(B) = \begin{pmatrix} I & B \\ 0 & I \end{pmatrix}, \qquad u_-(C) = \begin{pmatrix} I & 0 \\ C & I \end{pmatrix}, \qquad B^T = B,\ C^T = C.$$

Write $z(u_+(B))$ in blocks P, Q, R, S. The lower-left block R is additive because upper-unipotent conjugation fixes it. L-equivariance and the preceding type rule force $R = 0$: the symmetric natural-square parameter cannot map to the opposite dual-square target. Then P and S are additive and vanish by their types. Finally Q is additive. It is zero for distinct twists and is

$$Q(B) = \lambda B^{p^i}$$

when $i = j$. A block-scalar coboundary with diagonal entries $\alpha I, \beta I$, $\beta - \alpha = \lambda$, removes it without changing $z|_L$.

Apply the same argument to the lower group. Only the equal-twist value μC^{p^i} can remain in its lower-left block. Choose

$$B = E_{11} - E_{22}, \qquad C = (e_1 + e_2)e_3^T + e_3(e_1 + e_2)^T.$$

Then $BCB = CBC = 0$ and $BC \neq 0$, and direct multiplication gives

$$[u_+(B), u_-(C)] = \mathrm{diag}(I + BC, I - CB) \in L. \tag{17}$$

In the equal-twist dual-number representation, the cocycle fixes L and the upper group, but replaces C by $(1 + \epsilon\mu)C$ in the lower group. Relation (17) therefore forces $\mu BC = 0$, so $\mu = 0$. The Levi and the two full unipotent groups generate the symplectic group. This proves vanishing for every pair of twists, including when a simple exterior-square heart itself has nonzero first cohomology.

For the split even orthogonal group use the same matrices with B,C skew-symmetric. Replace the symmetric-square parameter by the exterior-square parameter in the type argument. For the final commutator take

$$B = E_{12} - E_{21}, \qquad C = E_{23} - E_{32}.$$

Here $BC = E_{13}$, $CB = E_{31}$, and $BCB = CBC = 0$. The elementary orthogonal subgroup so generated has index prime to p in the full orthogonal group. Restriction on first cohomology is injective across that index, by averaging or restriction and corestriction. Thus the same vanishing holds for the full split orthogonal group.

G.4 Even-dimensional unitary groups

The same block argument applies with transpose replaced by conjugate transpose. Its only additional issue is that the antihermitian parameter space has paired Frobenius types over the prime field. We identify those types before applying the commutator.

Let K/K_0 be quadratic, write $^-$ for its involution, and let $G = \mathrm{U}_{2m}(K/K_0)$ with $m \geq 5$. For a matrix X over K, put $X^\dagger = \bar{X}^t$, its conjugate transpose. Its Siegel Levi is $\mathrm{diag}(g, \bar{g}^{-T})$ and contains $\mathrm{SL}(m, K)$. The upper and lower parameters satisfy $B^\dagger = -B$ and $C^\dagger = -C$. If $|K_0| = p^f$, the additive antihermitian parameter module extends from $\mathbb{F}_p$ to the sum of the f types $A_s \otimes A_{s+f}$; transpose identifies the paired orientations. In the upper-right block of $\mathrm{Hom}(W_i, W_j)$, the matching type occurs only for $i = j$ modulo $2f$. The same block calculation therefore leaves only λB^{p^i} in the equal-twist case, removed by a block-scalar coboundary.

Choose a nonzero $\theta \in K$ with $\bar{\theta} = -\theta$ and use

$$B = \theta(E_{11} - E_{22}), \qquad C = \theta\big((e_1 + e_2)e_3^T + e_3(e_1 + e_2)^T\big).$$

They satisfy the hypotheses of (17). Its diagonal matrix lies in the special-unitary Siegel Levi and kills the remaining lower scalar. The generated elementary special unitary group has index prime to p in the full unitary group, completing the even-dimensional case.

G.5 The remaining form coordinates

For odd-dimensional and nonsplit forms, Appendix H proves the stronger statement directly on the natural derived group. It chooses an even-dimensional isotropic half and a nondegenerate tail of dimension at most four, so the central sign and half-swap used in the cocycle argument belong to that derived group. Restriction and corestriction across the prime-to-p full/derived quotient then give the full-group assertion here. This replacement avoids assuming that a tail isometry or a half-swap lies in a derived group without checking its determinant and spinor norm.

G.6 Products and central quotients

We have proved the needed first-cohomology vanishing on the natural matrix modules. To pass from a bimodule to a quotient by a fixed central space, we also need zero second cohomology with trivial coefficients. For the classical factors used here, the relevant part of the Schur multiplier has order prime to the characteristic. We state the precise exception list and explain how the matrix-size bound excludes it.

For a general linear factor, restrict to its special linear subgroup; the determinant quotient has order prime to p, so (16) gives the same whole-Hom vanishing. For a product of classical factors, an irreducible bimodule component is either one of the whole small Hom modules above or an external tensor of natural modules for two different factors. In the latter case the direct-product formula for first cohomology gives zero, because both nontrivial natural factors have zero invariant space. Commutative unitary factors have order prime to p. Semisimplicity of the associative enveloping algebra decomposes any algebra bimodule as a direct sum of these modules, with multiplicities and field twists. This proves the first assertion of Lemma G.1.

Here is the precise multiplier input for its second assertion. Kessar–Koshitani–Linckelmann [10, Proposition 7.1], on printed page 10 of the accepted version of *Conjectures of Alperin and Broué for 2-blocks with elementary abelian defect groups of order 8*, lists the exceptional Schur multipliers in odd defining characteristic as

$$A_1(9), \quad {}^2A_3(3), \quad B_3(3), \quad G_2(3).$$

The generic multiplier is the center of the simply connected cover and has order prime to p. All classical exceptions are excluded by the matrix-size bound above. An independent primary comparison is Babai–Goodman–Kantor–Luks–Pálfy, *Short Presentations for Finite Groups*, Remark 3.2, printed page 86, and its twisted-group discussion on printed page 95[11].

The natural derived classical group is perfect and differs from its simple quotient by a central group of order prime to p. The verified multiplier statement gives zero degree-one and degree-two cohomology with trivial characteristic-p coefficients for both groups. The quotient of the full classical group by the natural derived group has order prime to p: it is a determinant, norm-one determinant, or determinant/spinor component quotient. Averaging in the Lyndon–Hochschild–Serre sequence

preserves both vanishings. Direct products preserve them by the low-degree Künneth formula; additional central factors of order prime to p have no effect.

Therefore $H^1(G,P) = H^2(G,P) = 0$ for any trivial P. If P is central in an algebra bimodule B, it is trivial under conjugation. The long exact sequence of $0 \to P \to B \to B/P \to 0$ now gives $H^1(G, B/P) = 0$. All coboundary equations can be solved on the supplied generators. This proves Lemma G.1 constructively.

G.7 Quotients of whole matrix modules

For completeness, the radical-layer applications can also use the following more direct quotient observation. It applies to a quotient of a whole sum of matrix modules, with its full composition chains retained. The independent row and column maps in the image algebra explain why the residual quotient is again a central quotient of such whole modules.

For clarity, a stronger quotient observation used in radical-layer applications has a different hypothesis from an arbitrary subquotient. Consider a sum of whole small Hom modules. Separate its semisimple summands and group all isomorphic nonsemisimple chains first. The latter have composition series $1, H_i, 1$, with distinct hearts after this grouping. The actual degree-two image algebra has full independently supported top-to-heart and heart-to-bottom maps for each heart, their top-to-bottom products, and one common scalar character on the trivial coordinates. This is the joint image algebra described in the representation-repair appendix; the independence follows by multiplying its surjection onto each chain by the corresponding heart idempotent on the left or right.

For an invariant submodule K of such a whole sum, its projection to heart i is $H_i \otimes K_i$ for a multiplicity subspace K_i. Applying the independently supported row maps shows that K also contains the corresponding bottom coordinates. Quotient first by these entire radical submodules. The removed copies become trivial top coordinates; the remaining copies are still whole chains. The residual kernel has zero heart projection. The independently supported column maps then force its surviving top coordinates to vanish. It lies entirely in trivial coordinates.

Thus every such quotient is a sum of whole chains and semisimple summands, modulo a trivial subspace. The preceding first- and second-cohomology vanishings apply to it. This argument does not apply to an arbitrary submodule such as an isolated simple heart.

Over $\mathbb{F}_p$, extend scalars to the algebraic closure, introducing the listed Frobenius twists, and group isomorphic whole chains before using heart idempotents. The quotient argument and cohomology vanishing then descend by finite linear equations. In particular, an actual additive subgroup need only be $\mathbb{F}_p$-linear; no unsupported defining-field linearity is assumed.

H Derived-group cohomology and actual physical sections

The actual section construction requires cohomology for the derived group itself. Vanishing for a prime-to-p overgroup does not imply vanishing for its subgroup. We supply the derived argument explicitly.

Theorem H.1 (Whole-Hom quotients)**.** *Let D be a product of the natural derived groups $\mathrm{SL}_a(K)$, $\mathrm{SU}_a(K/K_0)$, $\Omega_a(K)$ and $\mathrm{Sp}_a(K)$ in odd characteristic and natural dimension at least 36. Let B be a whole sum of Hom spaces between the supplied natural, dual, Frobenius-twisted natural and trivial physical modules. For every invariant $\mathbb{F}_p$-subspace $P \le B$,*

$$H^1(D,B) = H^1(D, B/P) = 0.$$

For trivial characteristic-p coefficients T, also $H^1(D,T) = H^2(D,T) = 0$. Coboundaries are found by linear equations on the actual generators.

Proof. The special-linear computation in Appendix G takes place directly on root generators of SL, including the order-three norm calculation. The symmetric-parameter calculation generates Sp itself. The skew-parameter and antihermitian calculations generate respectively Ω^+_{2m} and SU_{2m} before passing to their full groups. Thus all these whole-Hom vanishings already hold on the derived groups.

For the remaining form dimensions choose an even integer m, maximal at most the Witt index, and write

$$W = A \oplus A^{\dagger,*} \oplus Z, \qquad \dim A = m, \quad \dim Z \le 4.$$

Here $m \ge 6$, and $\dagger$ is transpose or conjugate transpose as appropriate. The subgroup $D_0 = \Omega^+_{2m}$ or SU_{2m} acts on the first two summands and fixes Z. Both the central $-I$ on the large part and a half-swap belong to D_0: even m gives determinant one and, in the orthogonal case, trivial spinor norm. The whole-Hom calculation already handles its large blocks. Cross blocks have zero H^1 by the central order-two element, and the tail blocks by perfectness. Thus normalize a given cocycle to zero on D_0.

Let H_Z be the tail form. The upper elementary radical consists of

$$u(X,B) = \begin{pmatrix} I & B - \frac{1}{2} X H_Z^{-1} X^\dagger & X \\ 0 & I & 0 \\ 0 & -H_Z^{-1} X^\dagger & I \end{pmatrix}, \qquad B^\dagger = -B.$$

Its central subgroup $u(0,B)$ lies in D_0. For a cocycle in $\mathrm{Hom}(W_i, W_j)$, where subscripts denote twists, commuting with this center forces blocks $(2,1), (2,3), (3,1)$ to vanish. The remaining diagonal blocks are additive functions of X. Under the embedded SL_m, the parameter is a sum of natural types and the diagonal targets have even degree. Its central $-I$ therefore kills those additive maps.

The natural-type intertwiner equations leave

$$R(X) = X_j D_1, \qquad W(X) = E_1 X_i^\dagger$$

in positions $(1,3)$ and $(3,2)$. The commutator of two upper elements has zero cocycle, so comparing the two multiplication orders gives

$$X_j (E_1 - D_1 H_i^{-1}) Y_i^\dagger = Y_j (E_1 - D_1 H_i^{-1}) X_i^\dagger.$$

Taking X and Y on distinct large rows and varying their tail coordinates forces $E_1 = D_1 H_i^{-1}$. The remaining $(1,2)$ block is $X_j D_1 H_i^{-1} X_i^\dagger$: subtracting it leaves an additive map from an odd natural parameter to an even target, hence zero. Direct multiplication shows that these are exactly the blocks of the coboundary of the constant tail matrix D_1. It is D_0-fixed, so subtracting it preserves the normalization. The half-swap kills the opposite radical too. These radicals and the even derived group generate the full derived form group.

For a natural coefficient module, the same central commuting equations leave an additive large-half value $X_i v_0$ and no tail value. It is the coboundary of the tail vector v_0. The opposite radical follows from the same half-swap. Duals follow from the actual form/twist identifications. This proves all whole-Hom cases without assuming tail-group equivariance.

Products are handled by the low-degree direct-product formula; an external natural tensor on distinct factors has zero fixed space on each nontrivial factor. The stable derived groups are

perfect and their characteristic-p Schur multiplier part is trivial by the precise exception list cited in Appendix G. Cutoff 36 excludes those exceptions. This proves trivial-coefficient H^1 and H^2.

Finally use the whole-chain image algebra of Proposition F.2. Theorem C.1 gives the same image for D, jointly with the trivial coordinate. Group isomorphic whole chains and split the separate semisimple factors. In a chain type i, the actual heart idempotent puts $H_i \otimes K_i$ inside P, and the independently supported rows put its corresponding bottom coordinates inside P. Quotient these entire radical submodules first. Their copies become trivial tops; the remaining copies are whole chains. The residual denominator has zero heart part. Supported columns then force its whole-chain top coordinates to vanish, so it is entirely trivial. The new numerator is a sum of whole chains, semisimple direct summands of whole Hom modules, and trivial modules. Each has zero H^1. Quotienting its final trivial denominator preserves this by the just-proved H^2 vanishing. This proves the original $H^1(D, B/P) = 0$. Scalar extension and descent are finite linear equations, so merely additive P is permitted. □

Theorem H.2 (An actual standard section)**.** *Let an actual physical group H preserve a flag in W. Suppose its diagonal image contains a recognized normal standard derived subgroup $\overline{D}$, with explicit preimages, and its action on each flag grade is a semisimple sum of the permitted physical modules. There is a computable flag-preserving coordinate gauge after which the block-diagonal copy D_0 of $\overline{D}$ is an actual subgroup of H. The entire physical state and its old observations are transported by the same gauge. The cost is $p^{O(\dim W)}$ times polynomial auxiliary work.*

Proof. Let V be the full flag-unipotent group in any coordinate splitting, and let H_D be the full preimage of $\overline{D}$ in H. Then $H_D \le V \rtimes D_0$ surjects onto D_0. Filter V by flag distance: $V_i/V_{i+1} = B_i$ is the whole direct sum of Hom spaces between grades at distance i.

Inductively the current $H_i = H_D \cap (V_i \rtimes D_0)$ surjects onto D_0. Put $R_i = \ker(H_i \to D_0)$ and let P_i be its image in B_i. Fibers over d give an actual cocycle $z_i(d)$ in B_i/P_i. By Theorem H.1, solve $z_i(d) = (d-1)b_i$ modulo P_i. Lift b_i to V_i and conjugate the entire physical state by that lift with the appropriate sign. The gauge need not lie in H.

Every fiber now has distance-i coordinate in P_i and can be corrected by an actual R_i word to coordinate zero. Thus the next H_{i+1} still surjects. At the final distance, D_0 itself lies in the gauged group. Undoing the gauge gives the desired actual subgroup.

No generic intersection is computed. Initially generate H_D by the supplied lifts and the full diagonal kernel. At each step compute R_i as the kernel of its supplied projection; correct every full generator and adjoin the entire additive-coordinate kernel from Lemma A.3. Projection plus kernel proves that these generate the stated next prefix. There are at most $\dim W$ distances and every equation is in a whole physical Hom quotient. □

The semisimplicity premise follows once the normal diagonal core is recognized: the radical of its represented algebra is stable under the completely reducible parent, and annihilates each irreducible parent module by nilpotence. Recognition of the natural types is a separate part of the structural construction, not a consequence of an abstract simple-group name.

I Constructing the physical structural certificate

This appendix concerns a *supplied* matrix group. It does not search for an unknown tensor stabilizer. All its kernels, components, and quotient maps are computed from the supplied generators. In particular, none of the constructions below uses the group-intersection algorithm that is subsequently obtained from the tensor algorithm.

Write $q = p^e$, where p is odd, and restrict scalars to $\mathbb{F}_p$ when necessary. A faithful representation of degree N over $\mathbb{F}_q$ then has degree $D = eN$, and $p^{O(D)} = q^{O(N)}$. Original scalar operators, forms, and other equations are retained as equations on the original matrices. Restriction of scalars does not authorize their removal.

I.1 Statement and external inputs

A *simple section* means a simple quotient of a subgroup; it need not be a composition factor of the whole group. A bounded physical node below means an actual row together with the image of its actual stabilizer, its ancestor projection words, and its tags. Bounds on different such images are not multiplied unless this is explicitly justified.

Theorem I.1 (All-input physical preparation, P1)**.** *Fix the original prime-field physical budget D_0. There are absolute constants c and C_* with the following property. Given generators of an actual group $H \le \mathrm{GL}(V)$, where $\dim_{\mathbb{F}_p} V = D \le D_0$, one can construct, in $p^{cD_0}\mathsf{L}^{O(1)}$ deterministic time, the following certificate.*

1. *An H-invariant physical flag, the actual group $R = O_p(H)$, and a faithful completely reducible image $G = H/R$ on its grades, of total degree D. The original flag, matrices, kernels, and lift words remain available.*

2. *A G-permuted direct sum $V_{\mathrm{gr}} = \bigoplus_{i=1}^{k} W_i$, with $\sum_i d_i = D$, $d_i = \dim_{\mathbb{F}_p} W_i$, such that the image P_i of G_{W_i} on W_i is irreducible and linearly primitive. The block permutation homomorphism and its full kernel N are supplied.*

3. *Actual normal subgroups $\overline{D} \le C \le N \trianglelefteq G$. The group C is a central product of quasisimple components, all diagonal links between their local actions are supplied, and*

$$|N/C| \le p^{cD}.$$

   The group $\overline{D}$ is the product of precisely the classical components of C whose simple central quotients have order greater than

$$T = p^{C_* D_0}. \tag{18}$$

   Every nontrivial local occurrence of $\overline{D}$ has verified stable-rank natural classical coordinates, with its actual field, multiplicity, duality, and original projection maps.

4. *On a block where $\overline{D}$ is trivial, either its whole local image is retained as a bounded physical node of order at most p^{cD_0}, or there is a deleted-permutation cover with a genuine action of its full block stabilizer. These covers induce to a genuine G-representation. All numerators, denominators, evaluation kernels, and subquotient maps are supplied. Original rows, grades, and added covers have total degree $O(D)$.*

5. *The truly small quotient N/C, the block permutation action, the individual bounded-node images, and the actions on natural and multiplicity coordinates are retained separately, with their correlations. Each outside carrier has the hereditary property that every nonalternating simple section of each of its subgroups has order at most T.*

The construction respects already retained invariant parent flags and block tags, by refinement within their grades and induction over their actual orbits. It requires invariance under the current supplied group, not canonicity under an unrestricted ambient change of basis. The same cutoff (18) *is used after every later restriction. For any fixed number of copies of the original physical rows, the statement holds after enlarging the absolute constants.*

Here are the external inputs, in the exact forms needed.

(a) The constructive permutation and quotient-group library of Kantor–Luks [9, §§3–4, P1–P3, P6, P10–P16; §§6–9] computes orders, membership, specified homomorphism kernels, normal-subgroup centralizers, series, simple-factor names, prime cores, and socles, with straight-line words, in polynomial permutation degree. We use point stabilizers and pointwise kernels, never an arbitrary set stabilizer or an arbitrary intersection of two supplied permutation groups.

(b) There is an absolute c_0 such that a primitive irreducible $P \leq \mathrm{GL}_d(p)$ of order greater than $p^{c_0 d}$ has a normal quasisimple subgroup L with $|P/L| \leq p^{5d}$. On an appropriate extension-field tensor decomposition, L acts on one factor as a natural classical group over its defining subfield, or as an alternating group on its deleted natural module. This is [12, Proposition 5.7]; the prime-field tensor statement and its completely reducible formulation are [13, Theorems 3.1–3.3]. The tensor field is the field specified in those theorems, not an arbitrarily chosen older field. For an alternating factor the correct degree statement is $m \leq d + 2$, since deletion can remove two dimensions.

(c) For $A \leq \mathrm{Sym}(t)$, the product of the orders of its composition factors other than alternating groups of degree greater than 4 is at most $24^{(t-1)/3} < 3^t$ [15, Proposition 6.1, with parameter 4]. We apply this assertion to subgroups as well as to A itself.

The natural-module content in (b) is essential: abstract simple-factor names alone would not give the certificate. The representation gap may also be read through the base-size theorem [14, Theorem 1]; it is not used as a constructive recognition oracle here. All further constructive reductions are given below.

I.2 Physical flags and the exact prime kernel

Lemma I.2. *An invariant composition flag of a supplied matrix group on a d-dimensional space over $\mathbb{F}_p$ can be found in $p^{O(d)}$ time without enumerating subspaces. The kernel on its grades is exactly its normal p-core.*

Proof. Enumerate nonzero vectors v. Close $\langle v \rangle$ under the supplied matrices by linear-span operations; equivalently, enumerate its vector orbit and take its span. The result $\mathbb{F}_p H v$ is an invariant subspace. If the module has a proper nonzero submodule, a seed in that submodule produces a proper span. Thus exhaustive seeds either find a proper submodule or certify irreducibility. Recurse on the submodule and quotient, retaining bases and quotient maps. The sum of dimensions at each split is unchanged, and there are at most $2d - 1$ recursion nodes. At a node there are at most p^d seeds and each closure costs polynomial work in p^d and the supplied input. This proves the claimed bound.

Let $\gamma : H \longrightarrow \mathrm{GL}(V_{\mathrm{gr}})$ be the diagonal action on the resulting grades. Its kernel is block unitriangular and hence a normal p-group. Conversely a normal p-subgroup $A \trianglelefteq H$ fixes a nonzero vector on every nonzero characteristic-p module on which it acts: for example this follows by induction through a central subgroup of order p. On an irreducible H-module its fixed space is H-invariant, so it is the whole module. Hence every normal p-subgroup lies in $\ker \gamma$, proving $\ker \gamma = O_p(H)$.

For the computational assertion, represent H simultaneously on the vector sets of V and V_{gr}. Their disjoint union has at most $2p^d$ points. The pointwise kernel of the graded part has generators whose same words are evaluated on the original matrices. It is exactly $\ker \gamma$, not merely an abstract isomorphic group. The diagonal image is faithful and completely reducible by construction. Refining a supplied invariant flag uses the same procedure inside each of its grades. □

I.3 Primitive blocks from orbit matroids

Lemma I.3. *For an irreducible supplied group $A \le \mathrm{GL}(U)$, linear imprimitivity can be detected and a block system found in $p^{O(\dim U)}$ time. Refinement gives primitive block stabilizers with actual transversal words.*

Proof. For each nonzero u, form $\Omega = u^A$. It spans U. Regard its distinct vectors as the elements of a represented vector matroid. Choose a vector basis $B \subseteq \Omega$. For each $x \in \Omega \setminus B$, join all the elements in its fundamental circuit, as read from the expansion of x in B. The connected components $\Omega_1, \dots, \Omega_s$ of the resulting graph are the matroid components. To see this directly, expansions show that the spans belonging to distinct components use disjoint subsets of B, so their sum is direct; conversely each fundamental circuit must lie in one direct-sum component. Thus

$$U = \bigoplus_{j=1}^{s} \mathrm{span}(\Omega_j).$$

An automorphism permuting Ω preserves its circuits and hence this decomposition. The action on the summands is transitive because the action on Ω is transitive.

Suppose a genuine nontrivial block system $U = \bigoplus_{j=1}^{t} U_j$ exists. Take a seed $u \in U_1 \setminus 0$. Its orbit lies in $\bigcup_j U_j$. A minimal dependence between orbit vectors cannot meet two distinct U_j: projection onto one summand would give a strictly smaller nonempty dependence. Thus every matroid component lies in one true block, and this seed gives a nontrivial decomposition. Conversely any nontrivial matroid decomposition found above is valid. Exhaustive seeds therefore certify primitivity if none succeeds.

The stabilizer of a found block is a point stabilizer in the explicit action on at most $\dim U$ blocks. If its action on that block had a proper nonzero invariant subspace, inducing that subspace by actual block transversals would give a proper nonzero A-submodule of U. Its action is therefore irreducible. Apply the same test to it and induce every refinement to its block orbit. There are at most $\dim U$ final blocks and only linearly many nodes in the dimension-additive refinement tree. Orbits and circuit graphs have size at most $p^{\dim U}$; reduce generator lists by stabilizer chains at each step. The total cost is $p^{O(\dim U)}$. □

Apply the lemma to each graded irreducible summand. Compute the action on $\{W_1, \dots, W_k\}$ directly from the matrices, and let N be its pointwise kernel. The restriction maps embed N in $\prod_i P_i$. The chosen blocks are G-invariant as a family; a known incoming coset representative transports the entire chosen family to the target. No independently chosen target decomposition has to be guessed. Existing block tags are handled by starting the refinement inside their stabilizers and inducing it over their actual orbits; invariant flags are refined inside their grades. Such data must be actual current-group invariant data, as stipulated in the input, rather than incompatible decompositions from discarded branches.

I.4 Extracting the actual layer

For a finite group A, write $F(A)$ for its Fitting subgroup and $E(A)$ for the product of its subnormal quasisimple components.

Lemma I.4 (Layer extraction)**.** *The actual components of a supplied matrix group of degree at most D, with their centers, simple quotients, and original words, can be computed in $p^{O(D)}$ time using the quotient-group library above.*

Proof. Compute $F = F(A)$ as the product of the prime cores, then

$$B = C_A(F), \qquad Z = Z(F).$$

The group $F(B)$ is characteristic in $B \trianglelefteq A$, and is normal nilpotent in A. Consequently $F(B) \leq F \cap B = Z$; the reverse inclusion holds because Z is central in B. A nilpotent normal subgroup of B/Z would lift to a nilpotent normal subgroup of B, since the kernel Z is central. It follows that $F(B/Z) = 1$. Its solvable radical is also trivial: a nontrivial solvable normal subgroup would have a nontrivial characteristic Fitting subgroup.

Compute $\mathrm{Soc}(B/Z)$. It is a direct product of nonabelian simple groups. Its individual simple factors can be obtained by a minimal-normal-subgroup computation *inside this direct product*, followed by its centralizer there, and iteration. All these are normal-subgroup operations in specified quotients. For one factor S_j let A_j be its full preimage in B and put $Q_j = A_j'$. Since A_j/Z is perfect,

$$A_j = A_j' Z, \qquad A_j' = [A_j' Z, A_j' Z] = [A_j', A_j'].$$

Thus Q_j is perfect, with simple central quotient S_j, and is subnormal in A. Distinct such groups commute: their commutators are central and a central commutator map out of a perfect group is trivial. Conversely every component centralizes $F(A)$, and its image in the radical-free group B/Z is a simple factor of its socle. Hence these are exactly the components.

Use the faithful action on at most p^D vectors for the quotient library. There is no need to factor a generic integer of magnitude p^{D^2} by trial division. All prime divisors of $|A|$ divide $p \prod_{j=1}^{D} (p^j - 1)$; these D integers each have magnitude at most p^D. Factoring them by exhaustive division, or a sieve of this size, is already within $p^{O(D)}$. The number of group operations is polynomial in D and $\log p$, apart from the permitted polynomial dependence on permutation degree. Every subgroup and quotient retains compressed original words. □

I.5 The unique dominant local component and its coordinates

Lemma I.5 (Homogeneous physical coordinates)**.** *Let P act irreducibly and primitively on W, and let $L \trianglelefteq P$. The restriction $W|_L$ is semisimple and homogeneous. If L is quasisimple, its faithful irreducible constituent X, the actual field $K = \mathrm{End}_L(X)$, and the evaluation isomorphism*

$$X \otimes_K M \longrightarrow W, \qquad M = \mathrm{Hom}_L(X, W), \qquad x \otimes f \longmapsto f(x) \tag{19}$$

can be found in $p^{O(\dim W)}$ time.

Proof. Let $A = \mathbb{F}_p[L] \subseteq \mathrm{End}_{\mathbb{F}_p}(W)$. Its radical J is invariant under conjugation by P, so JW is a P-submodule. Irreducibility and nilpotence force $JW = 0$, hence $J = 0$ because A is represented faithfully. The isotypic summands for A are permuted by P; primitivity forces there to be just one. Find a simple submodule X by the vector-span procedure, descending until irreducible. Compute its commutant by linear equations. Schur's lemma and the finite-division-ring theorem identify this commutant as a field K. It is an actual algebra of operators, so arithmetic, a K-basis of X, and a K-basis of M can be computed and verified inside the representation. Homogeneity and semisimplicity prove that (19) is an isomorphism; its rank is checked as part of the certificate. The kernel on X equals the kernel on every isomorphic copy and hence the kernel on W. Thus X is faithful.

The commutant field has at most $p^{\dim X}$ elements. Even constructing its entire addition and multiplication tables or finding a field generator by exhaustive enumeration is permissible. All Hom

spaces and represented algebras are obtained by polynomial-size linear equations and matrix-span closure, not by enumeration of their elements. Finally the center of a quasisimple L acts faithfully by scalars on X, giving

$$|Z(L)| \le |K| - 1 < p^{\dim W}. \tag{20}$$

□

Enlarge c_0 so that $c_0 > 12$ and the structural input applies. A block with $|P_i| \le p^{c_0 d_i}$ is declared locally bounded. Otherwise extract all components of P_i by Lemma I.4. The structural component L satisfies

$$|L/Z(L)| > p^{(c_0-6)d_i} > p^{5d_i}.$$

Every other component Q commutes with L and intersects it centrally. Its simple central quotient is therefore a section of P_i/L, of order at most p^{5d_i}. There is exactly one component above this last bound; call it L_i. This criterion is intrinsic. In particular $L_i \trianglelefteq P_i$, and conjugation carrying W_i to W_j carries L_i to L_j. One may construct it on one block of an orbit and transport it. Its normality makes the resulting subgroup independent of the chosen block transversal.

Apply Lemma I.5 to L_i. The field K_i thereby recovered is the field of its irreducible physical representation. In the classical case its constituent is the natural module over the actual matrix-entry defining field; any extension field and tensor factor in the existence theorem have become multiplicity in (19). For the alternating case the constituent is the deleted module over $\mathbb{F}_p$, and the entire remaining scalar and tensor action belongs to M_i. This removes any need to search for a maximal semilinear field or for an unspecified tensor basis.

Verifying classical natural type. Work on X over its computed K, of degree a. In these coordinates test the candidate groups

$$\mathrm{SL}_a(K), \qquad \mathrm{Sp}_a(K,B), \qquad \Omega_a(K,Q), \qquad \mathrm{SU}_a(K/K_0,h).$$

Compute invariant bilinear forms by the equations $g^t B g = B$, and, when K has an involution σ, invariant Hermitian forms by $g^{\sigma t} B g = B$, $B^{\sigma t} = B$. These are linear equations over K or its fixed field. Irreducibility makes every nonzero invariant form nondegenerate; verify rank and the required symmetry explicitly. In odd characteristic a symmetric form determines $Q(x) = B(x,x)/2$. In the unitary case retain the actual involution and its fixed field.

The candidate standard groups can be generated deterministically within $p^{O(d_i)}$. For SL use elementary matrices. For the isometry groups, construct a standard orthogonal, symplectic, or Hermitian basis by elimination. Enumerate the possible image of its first nonisotropic vector, or of its first hyperbolic pair, and complete each image to an isometric basis; recurse on the orthogonal complement. There are at most $|K|^{2a}$ choices at any stage. Basis completion is ordinary form elimination, with field enumeration for norm equations if needed. The resulting transversals, together with the recursively generated pointwise frame stabilizer, generate the full isometry group: every isometry first has its initial frame matched by a transversal and then fixes that frame. The symplectic group is already the required group. In stable dimension the derived groups of the orthogonal and full unitary groups are respectively Ω and SU, so the quotient library computes the required derived generators. Equivalently, in the orthogonal case one can use all reflections in nonisotropic vectors and then derive their group. These constructions have degree at most $|K|^a \le p^{d_i}$ and never enumerate the whole classical group.

For each candidate, test its generators in the actual image of L_i by constructive membership and compare exact orders. Inclusion and equal order prove equality. The structural theorem guarantees

that one of the natural candidates succeeds. Failure of a representation check is not an isomorphism rejection: it would contradict the promised structural premise and must not be used to discard a tensor branch. Small exceptional ranks can be tested by these same group computations and are not retained as critical components.

Repeated constituents and outside maps. Compute the actual algebra $\mathbb{F}_p[\overline{D}]$ on the *whole* graded physical representation, after $\overline{D}$ has been selected below. It is semisimple: apply the radical argument separately to every irreducible G-summand, using normality of $\overline{D}$. Its simple modules, commutant fields, isotypic projectors, and evaluation maps give all repeated occurrences. Duality is tested by the actual Hom equations against the contragredient matrices. When comparing coordinates over isomorphic finite fields, enumerate their field isomorphisms and solve the corresponding intertwiner equations; this retains the Frobenius record rather than assuming that a merely additive map is linear over an older field. The actual maps from a global component to its local groups are retained even when two presentations use different natural coordinates.

Completeness of these linked-occurrence tests uses the classical automorphism theorem, in its explicit inner–diagonal–field–graph form [18, §3, statements 3.2–3.6]. In the selected stable ranks, its natural projective actions differ only by a field map, a change of basis, and the natural/dual choice; the exceptional orthogonal triality case has already been excluded by the permanent cutoff. There is no additional central ambiguity in the actual intertwiners. Indeed two representations of the same perfect component whose projective matrices have been made equal differ by a scalar homomorphism from that component to the multiplicative field group. Such a homomorphism is trivial. Thus the finite field/dual tests and linear intertwiner equations recover all its linked natural occurrences, with their actual central kernels. For a self-dual occurrence, the computed invariant form supplies the adjoint; for a non-self-dual pair, evaluation gives the exchange adjoint. The associated full natural cover consequently acts on every occurrence by the recovered natural, dual, and field-twisted matrices, with the same diagonal links. It does not act independently on two linked rows.

For clarity, these computations also give a concrete normalizer *factorization cover*, without requiring chosen factorizations to multiply. On one homogeneous row $X \otimes_K M$, a normalizer element induces a field automorphism σ of the center of $\mathrm{End}_K(X) \otimes 1$. After removing σ, its induced automorphism of $\mathrm{M}_a(K)$ is inner. To compute an implementing matrix u, solve

$$uA_j = \alpha(A_j)u$$

for a basis A_j of this matrix algebra. A nonzero solution is invertible by simplicity of its natural module. Removing $u\sigma$ from the physical map leaves $1 \otimes v$, which is read by linear equations. Thus the map has the form $u \otimes v$ with their common field automorphism. Its factorization ambiguity is exactly

$$(u, v) \sim (cu, c^{-1}v), \qquad c \in K^{\times}. \tag{21}$$

Retain the group of *all* such factorizations, with its scalar kernel, rather than choosing a spurious homomorphic section. On different isotypic rows the same construction is performed with the actual permutation of the simple algebra factors. Formal dual rows carry inverse-transpose maps. All determinant, form, field, scalar, and ancestor-group conditions remain conditions in this cover. For the paired star realization use a row and its formal dual, retain the induced adjoint, and fix the augmentation on an unrelated scalar coordinate to one. This is the normalizer data used by the subsequent transfer and physical-section arguments; it is not an enlargement of the exact returned group.

This cover itself has an original-degree implementation. For every supplied parent generator choose one computed factorization; add generators of all scalar kernels in (21). The resulting group is the full inverse image of the parent: any other factorization differs from a lift word by one of those kernels. Represent it on the original row and on the direct sums of its natural and multiplicity spaces. If their K-degrees are $a, b \geq 1$, then $a + b \leq 2ab$, so the added prime-field faithful degree is at most twice the row degree. The action on these factor spaces, together with the original ancestor matrix, is faithful; field automorphisms act on them as actual prime-field linear maps. Thus its original projection and every scalar kernel are specified homomorphisms on an $O(D)$-degree representation. A pointwise kernel or a full lift is again a permitted original-vector permutation computation. This is also a constructive implementation of the cover used to act on compatible natural–multiplicity frames.

I.6 A specified small-quotient kernel gives the diagonal links

For a locally bounded block set $L_i = 1$ and $B_i = P_i$; for every other block set $B_i = P_i/L_i$. Then

$$|B_i| \leq p^{c_0 d_i}, \qquad \prod_i |B_i| \leq p^{c_0 D}. \tag{22}$$

Enumerate each quotient by coset breadth-first search, using membership in the actual L_i for coset equality and retaining representative words. For a bounded block this is enumeration of its whole bounded image. The regular quotient actions have total degree at most $\sum_i p^{c_0 d_i} \leq D p^{c_0 D}$. Restricting an element of N to the blocks gives the specified homomorphism

$$\rho : N \longrightarrow \prod_i B_i, \qquad M = \ker \rho. \tag{23}$$

On the disjoint union of those quotient domains, compute its pointwise kernel with the original faithful action retained. Equivalently enumerate its at most $p^{c_0 D}$ image tuples and use Schreier relators. Thus M is constructed exactly. The descriptive equality $M = N \cap \prod_{i:L_i \neq 1} L_i$ does *not* invoke a generic intersection primitive. Covariance of L_i makes $M \trianglelefteq G$.

Lemma I.6 (Exact central product and links)**.** *The group $C = M'$ is a normal central product of quasisimple groups and*

$$[N : C] \leq p^{(c_0+1)D}. \tag{24}$$

Its components and their local diagonal links can be constructed using only the specified quotient kernels and derived-subgroup operations.

Proof. The projection M_i of M to P_i is normal in P_i and contained in L_i. A normal subgroup of a quasisimple group is either central or the whole group: if its image in the simple central quotient is full, then $L_i = M_i Z(L_i)$ and perfectness gives $L_i \leq M_i$. Hence M_i is full or central. Put

$$Z_0 = M \cap \prod_i Z(L_i).$$

This is central in M. After ignoring central projections, M/Z_0 is a subdirect product of nonabelian simple groups.

Such a subdirect product is a direct product of full diagonal factors. Here is an induction proof. Project onto all but the last simple factor; by induction that image is a product of simple diagonal factors. The kernel of this projection, viewed in the last factor, is normal there and is either the whole factor or trivial. In the first case the last factor splits off. In the second case the last coordinate

is a surjective homomorphism from the preceding product to a simple group. The images of its direct factors commute and are normal; precisely one is nontrivial. It is an isomorphism on that factor, giving a diagonal link. This proves the assertion, including links through automorphisms.

In particular M/Z_0 is perfect. Thus $M = M'Z_0$ and

$$M' = [M'Z_0, M'Z_0] = [M', M'].$$

So $C = M'$ is perfect with central quotient the same product of simple diagonal factors. It is normal in G because M is. Moreover

$$|M/C| \le |Z_0| \le \prod_i |Z(L_i)| \le p^D.$$

Together with (22), this proves (24).

To find links concretely, use the natural projective action of each classical L_i, whose kernel is its scalar center; its degree is at most $|K_i|^{a_i} \le p^{d_i}$. For an alternating L_i use its faithful letter action, constructed in the next subsection (the constructions are independent). For every nontrivial simple image of C, compute

$$J_i = \ker(C \longrightarrow L_i/Z(L_i))$$

as a pointwise kernel on this explicit domain. Two indices are in the same diagonal class exactly when $J_i = J_j$. Mutual containment is tested by generator membership in the original faithful action.

For a diagonal class I, let A_I be the kernel of all the actions belonging to one representative from each *other* class, and set $Q_I = A_I'$. Modulo $Z(C)$, this group has exactly the one simple factor indexed by I. The same central-extension calculation proves Q_I is quasisimple. Different Q_I commute, since their commutators are central and each is perfect. Their product surjects onto $C/Z(C)$; its quotient in C is abelian, and is trivial because C is perfect. Therefore

$$C = \prod_I Q_I.$$

All the kernels here are kernels of displayed actions on disjoint unions of domains of size $p^{O(D)}$. Their derived generators are original words. Conjugation by G permutes the actual kernels J_i and hence the groups Q_I. This supplies exact outside permutations and all diagonal correlations, without guessing isomorphisms of abstract simple groups. □

I.7 Deleted-permutation recognition and a genuine full cover

Lemma I.7 (Constructive alternating cover)**.** *Suppose the dominant local component is alternating. One can construct its natural letter action and a deleted-permutation cover of its entire isotypic row, extending to the full actual block stabilizer. The degree inflation is at most $m/(m-2)$, where m is its alternating degree.*

Proof. The structural input gives $L \cong A_m$ on its deleted natural constituent X. Increase c_0 if necessary so that every locally large alternating case has $m \ge 8$; the finitely many smaller alternating groups have bounded order and are locally bounded. Put

$$P = \mathbb{F}_p^m, \quad P_0 = \{x : \textstyle\sum_j x_j = 0\}, \quad U_m = \begin{cases} P_0, & p \nmid m, \\ P_0/\langle \mathbf{1} \rangle, & p \mid m. \end{cases}$$

Write $r = \dim U_m = m - \delta$, with $\delta \in \{1, 2\}$. The ordinary dot product induces a nondegenerate symmetric form on U_m: its radical on P_0 is exactly $\langle \mathbf{1} \rangle$ when $p \mid m$ and is zero otherwise. The

constituent is absolutely irreducible and defined over $\mathbb{F}_p$, so $W = X \otimes_{\mathbb{F}_p} M$ in the previously computed coordinates.

Here is an explicit recognition that checks this representation rather than relying on an abstract factor name. Try the candidates $m = \dim X + 1, \dim X + 2$ compatible with the characteristic and $|L| = m!/2$. Solve for a nonzero invariant symmetric form on X and check rank. Enumerate projective-line seeds and their L-orbits. Retain orbits of size $\binom{m}{2}$; join two distinct orbit lines when their invariant-form pairing is nonzero. A successful orbit exists: on U_m it consists of

$$\langle e_i - e_j \rangle \quad (i < j).$$

These lines remain nonzero and distinct after the possible quotient by $\mathbf{1}$. An alleged equality modulo that line would equate a vector supported on at most four coordinates to a constant vector, impossible for $m \geq 8$ unless that constant is zero. Their pairings are nonzero exactly when the two unordered pairs meet, including in characteristic three. The resulting graph is $L(K_m)$.

Recover its letters without a graph-isomorphism oracle. For adjacent vertices ij, ik, the common-neighbor graph consists of a clique of the other $m - 3$ edges through i and the isolated vertex jk. The large component, with ij, ik, gives the star at i. Collect these stars for all adjacent pairs, discard duplicates, and check that there are m stars, each of size $m - 1$, every vertex lies in two stars, and adjacency is exactly intersection of the corresponding two-element endpoint sets. Check also that the supplied generator permutations preserve the stars. They now give a homomorphism $\pi : L \to \mathrm{Sym}(m)$. Require even generators and image order $m!/2$. Together with the checked order of L, this certifies a faithful action onto A_m.

Build the standard matrices of U_m from these actual permutations and solve the intertwiner equations with X. Keep a candidate only if an invertible intertwiner is found. For the promised irreducible modules, a nonzero intertwiner is invertible, which is also rank-checked. This last test rules out any accidental orbit configuration. There are at most $p^{\dim X}$ seeds and at most that many orbit points per seed, so all the work costs $p^{O(\dim X)}$.

Let P_W be the full actual block stabilizer image, in which L is normal. For $h \in P_W$, conjugation gives an automorphism of A_m. For $m \geq 8$, $\mathrm{Aut}(A_m) = S_m$, and $C_{S_m}(A_m) = 1$. Thus there is a unique permutation π_h such that

$$\pi_h \pi(s) \pi_h^{-1} = \pi(hsh^{-1}) \qquad (s \in L). \tag{25}$$

It can be found by orbit calculations on $[m] \times [m]$: for each choice of the image of one letter, the orbit of the corresponding pair under the two generator actions is the graph of an intertwining bijection exactly when both projections are bijective and the orbit has m points. Test the resulting bijection in (25). Uniqueness implies $\pi_{hh'} = \pi_h \pi_{h'}$.

Write $d(\pi_h)$ for the actual deleted-module matrix on X. The map $(d(\pi_h)^{-1} \otimes 1)h$ centralizes L. Absolute irreducibility gives a unique $c_h \in \mathrm{GL}_{\mathbb{F}_p}(M)$ with

$$h = d(\pi_h) \otimes c_h. \tag{26}$$

Read c_h from the matrix entries. Since the two tensor actions commute, $c_{hh'} = c_h c_{h'}$. The full parent action therefore lifts genuinely to

$$\widetilde{W} = \mathbb{F}_p^m \otimes M, \qquad \widetilde{h} = \mathrm{perm}(\pi_h) \otimes c_h.$$

It preserves $P_0 \otimes M$ and, if $p \mid m$, its subspace $\langle \mathbf{1} \rangle \otimes M$. Their indicated subquotient is exactly W, with action (26). On an axis $e_j \otimes M$, the stabilizer acts by c_h. Since $c_s = 1$ for $s \in L$, this image

factors through P_W/L and has order at most $p^{5\dim W}$. Scalars and field actions on multiplicity are part of this actual $\mathbb{F}_p$-linear map, so none has been discarded.

Finally

$$\dim \widetilde{W} = m \dim M = \frac{m}{m-\delta} \dim W \le \frac{4}{3} \dim W \qquad (m \ge 8). \tag{27}$$

The cover is made only for this single dominant alternating factor. No product of degree inflation factors from smaller tensor factors is introduced. □

For completeness, induction over a block orbit is literal. Fix transversals t_i from its representative block to W_i. If $gW_i = W_j$, then $t_j^{-1} g t_i$ belongs to the representative stabilizer. Define the map from the ith copy of its cover to the jth by the just constructed representation of $t_j^{-1} g t_i$. In a product $g'g$, these stabilizer expressions multiply with the middle transversals cancelling. Thus the induced matrices satisfy the group law. This also proves compatibility of all subquotient maps. Across distinct original blocks the degree bounds (27) add, giving total covered degree at most $4D/3$ and at most $4D/3$ coordinate axes.

I.8 Permanent critical selection and hereditary retirement

Choose C_* once, larger than $c_0 + 1$, the fixed permutation-cover constants, and a fixed natural-degree cutoff $a_{\mathrm{st}} \ge 80$, so that all retained classical Lie ranks exceed 36. If a classical simple quotient has natural matrix degree a over $K = \mathbb{F}_{p^f}$, then

$$|S| < |K|^{a^2} = p^{fa^2}, \qquad fa \le d_i.$$

Thus bounded a implies $|S| \le p^{aD_0}$; increasing C_* ensures that every component selected by (18) lies in stable rank. Exceptional small classical–alternating coincidences and orthogonal triality do not occur in the selected set.

Let $\overline{D}$ be the product of the classical Q_I in Lemma I.6 with $|Q_I/Z(Q_I)| > T$. Its definition is conjugation-invariant, so $\overline{D} \trianglelefteq G$. The following point ensures that it captures every genuinely large local classical action.

Lemma I.8. *If L_i is classical and $|L_i/Z(L_i)| > T$, then some selected component Q_I projects onto L_i.*

Proof. Let N_i be the image of N in P_i; it is normal there. If it does not contain L_i, then $N_i \cap L_i \le Z(L_i)$, so the simple quotient of L_i is a section of P_i/N_i. But P_i/N_i is an image of G_{W_i}/N, a subgroup of the block permutation group $G/N \le S_k$, $k \le D$. A nonalternating simple section of any subgroup of S_k has order at most $24^{(k-1)/3} < p^D$: apply the GMP bound to the subgroup in which that section occurs. This contradicts $|L_i/Z(L_i)| > T$.

Hence $N_i \ge L_i$. If the projection of C to L_i were central, its simple quotient would be a section of N/C, whose order is at most $p^{(c_0+1)D} < T$. This is again impossible. The projection of C is therefore all of L_i. In the diagonal decomposition exactly its class component Q_I projects onto it. The simple central quotients of Q_I and L_i are isomorphic, so that Q_I is selected. □

A perfect component has either full or trivial projection to any local L_i: its only other possible image is central and therefore trivial. It has trivial projection to a locally bounded block because M does. A selected component cannot project to an alternating local component in stable rank, or to a classical local component with smaller simple quotient. Consequently all of its nontrivial local representations are exactly the already certified natural occurrences, with their actual multiplicities

and diagonal links. Distinct components have disjoint active local supports. In particular we have not taken an algebraic envelope of several independent tensor factors acting on the same row, which could create an unjustified full linear group.

A locally bounded block is retained with its entire image of order at most $p^{c_0 d_i}$. A classical local block not supporting $\overline{D}$ has $|L_i/Z(L_i)| \le T$ by Lemma I.8. Its full image satisfies

$$|P_i| \le p^{5d_i}|L_i| < p^{6d_i}T \le p^{(C_*+6)D_0}. \tag{28}$$

Retain this *one* bounded image as one physical node. Formula (28) is not a bound on the product of all retired nodes. Alternating local blocks are handled by Lemma I.7.

Lemma I.9 (Hereditary outside bound)**.** *With the same fixed threshold T, no nonalternating simple section of a subgroup of a retired-node image, an alternating-cover outside carrier, the soft quotient, or a retained tag permutation image has order greater than T.*

Proof. First observe an elementary extension fact. If $B \trianglelefteq A$ and $J \le A$, then $J \cap B \trianglelefteq J$ and $J/(J \cap B)$ embeds in A/B. Every composition factor of J occurs in one of these two groups. Applying this observation also to subgroups of J shows that a simple section of A comes from a simple section of B or of A/B. A central kernel contributes only abelian factors.

For a locally bounded node its whole order is at most $p^{c_0 D_0} < T$. For a retired classical node apply the extension fact to $Z(L_i) \trianglelefteq L_i \trianglelefteq P_i$. Sections coming through the central kernel are abelian, those through $L_i/Z(L_i)$ have order at most $|L_i/Z(L_i)| \le T$, and those through P_i/L_i have order at most $p^{5d_i} < T$. This uses the small simple quotient, rather than the slightly larger full-image bound (28).

For the sum of alternating covers, the permutation of axes has degree at most $4D/3$. Its every subgroup has nonalternating simple sections of order less than $3^{4D/3}$ by GMP. The kernel fixing these axes has local coordinate images contained in the commuting multiplicity images $c(P_i)$, each of order at most p^{5d_i}. The product of these particular bounds is at most p^{5D}, since their exponents are charged to disjoint original blocks. The induced axis kernel embeds in that product, with all original correlations preserved. The extension fact now proves the claim for the cover carrier and every subgroup of it. The same GMP argument applies to the original block permutation image and any retained tag permutation image of linear degree. Finally the whole order of N/C is at most $p^{(c_0+1)D}$. The permanent choice of C_* dominates all these constants. □

The assertion concerns the recorded outside carriers individually and extensions assembled from their actual correlated maps. It does not assert small order for their arbitrary product, nor does it retire the active natural components themselves. Because the bound applies to every subgroup and quotient encountered, a later restriction cannot revive a critical nonalternating factor in an already retired carrier. Resetting T to the new local dimension would lose precisely this property and is not permitted.

I.9 Normalizer reduction and the critical commuting image

The following refinement supplies the exact terminal and frame-normalization data. Here L denotes the full natural classical cover on the selected rows (with its paired dual realization where needed), whereas $\overline{D}$ denotes the derived core. Full cover and derived core are kept distinct.

Lemma I.10 (Constructive normalizer and commuting-image reduction)**.** *Let V_{crit} be the direct sum of the selected nontrivial isotypic physical rows, written in the computed coordinates as $\bigoplus_j X_j \otimes_{K_j} M_j$.*

Put $f_j = [K_j : \mathbb{F}_p]$, $a_j = \dim_{K_j} X_j$, and $b_j = \dim_{K_j} M_j$, counting each actual isotypic row once. Then

$$|C_{\mathrm{GL}(V_{\mathrm{crit}})}(\overline{D})| \le p^{\sum_j f_j b_j^2} = p^{O(D_0)}. \tag{29}$$

The genuine factorization cover constructed above supplies all natural, multiplicity, field, and exchange actions. After its actual finite field, type, exchange, and form-similitude records have been fixed, the normalizer of the full cover has the form L times its actual centralizer. More precisely, if A_0 is the resulting actual normalizer subgroup and $L \le A_0$, then

$$A_0 = LC_{A_0}(L). \tag{30}$$

The centralizer's complete joint image on V_{crit} can be enumerated within $p^{O(D_0)}$, with its full kernel and original lifts retained.

Proof. Schur's lemma and the evaluation coordinates identify the commuting algebra on V_{crit} with $\prod_j \mathrm{End}_{K_j}(M_j)$. Its unit group is $\prod_j \mathrm{GL}_{b_j}(K_j)$, proving the first inequality. Correlations, forms, or fixed scalar conditions only shrink the actual commuting image.

For every selected row the critical simple quotient satisfies $p^{C_* D_0} < |S_j| < p^{f_j a_j^2}$. Also the mass of these rows is bounded by the original physical mass, or by a fixed multiple κD_0 if formal duals have been included:

$$\sum_j f_j a_j b_j \le \kappa D_0, \qquad \frac{b_j}{a_j} = \frac{f_j a_j b_j}{f_j a_j^2} < \frac{\kappa}{C_*}.$$

Consequently

$$\sum_j f_j b_j^2 < \frac{\kappa}{C_*} \sum_j f_j a_j b_j \le \frac{\kappa^2}{C_*} D_0.$$

This proves the exponential bound with the original degree; an artificial multiplicity tensor factor has not been charged as fresh physical mass.

We give the normalizer calculation on each factor. Fixing the permutation of simple algebra factors and its induced field action reduces its algebra automorphism to an inner one. The implementing u is computed by the Skolem–Noether equations already displayed, with the scalar ambiguity retained in the factor cover. In a linear factor the full natural group is $\mathrm{GL}(X)$, so u is already in that group. In a symplectic, orthogonal, or unitary factor, normalizing the derived natural group preserves its unique invariant form up to a scalar: indeed the pullback form is another invariant form, and the appropriate intertwiner space has dimension one. The factor-cover matrix u therefore has its explicit multiplier λ. Once that multiplier has been fixed to one, u belongs to the full natural isometry group. The residual factor $u^{-1}h$ centralizes the natural group by the commutant equations. Paired linear factors carry (u, u^{-t}); fixed exchange records give the same assertion there. For linked occurrences choose u on one representative occurrence of the component and transport it using the verified field/dual intertwiners. These transported maps implement the same conjugation on the actual component as h, so their residual quotients centralize the component on every linked row. They therefore centralize its represented algebra and the full natural cover contained in that algebra. Thus one obtains one ℓ per diagonal component, not independent natural matrices on linked rows. These verifications prove that each element of A_0 is ℓc in the ambient representation. If $L \le A_0$, then $c = \ell^{-1}h \in A_0$, proving (30).

All records here are actual actions or scalar multipliers, not labels assigned from abstract group names. The cover is useful because a multiplier can change on changing a scalar factorization; the scalar kernel is retained when taking its multiplier kernel. If a derived SL, SU, or Ω is used in place of the full natural group, also fix respectively the determinant and the applicable determinant/spinor

quotient records before the same conclusion. No such extra condition is silently inferred from a similitude multiplier.

There are only $O(\sqrt{D_0})$ selected isotypic rows: each has $f_j a_j \geq \sqrt{f_j C_* D_0} \geq \sqrt{C_* D_0}$, whereas their total mass is $O(D_0)$. Hence their permutation-image order is at most $O(\sqrt{D_0})! = p^{O(D_0)}$. The products of field-automorphism orders, twofold exchange orders, and scalar-record orders are also $p^{O(D_0)}$, since $\sum_j f_j \leq O(D_0)$ and $\prod_j f_j \leq 3^{\sum_j f_j/3}$. Thus, when these records must be listed in a terminal call, their complete joint image is small enough to list. Their kernels are specified homomorphism kernels, computed successively without imposing unrelated conditions on their images.

The group L is normal in A_0. Compute $C = C_{A_0}(L)$ by the normal-subgroup centralizer algorithm, then its restriction homomorphism $C \to \mathrm{GL}(V_{\mathrm{crit}})$. Its image has the bound (29), so breadth-first image search and Schreier relations compute its complete image and $C_{\mathrm{fix}} = \ker(C \to \mathrm{GL}(V_{\mathrm{crit}}))$, with original words. Since L is faithful on these rows, $L \cap C_{\mathrm{fix}} = 1$, and $[L, C_{\mathrm{fix}}] = 1$. Therefore

$$LC_{\mathrm{fix}} = L \times C_{\mathrm{fix}} \longrightarrow L, \qquad \ell c \longmapsto \ell \tag{31}$$

is an exact homomorphism with full kernel C_{fix}. Coset representatives for C/C_{fix} cover A_0 by cosets of LC_{fix}; duplicates caused by $L \cap C$ may be retained or removed by ordinary coset membership. This is the complete joint commuting-image normalization, rather than independent choices on separate value rows. □

If the original prefix group does not contain the full cover L, one uses the displayed factorization in the supplied overgroup generated by that prefix and L, and keeps the original prefix as its exact pullback/filter. It is false in general that an arbitrary subgroup of LC equals L times its own centralizer. No use of (30) is made without its containment hypothesis. The lists in the lemma are terminal lists, to be reunited before a complete result is returned. Their mere small size does not authorize opening one at every recursive node. For compatible multiplicity-frame normalization one instead uses its actual frame orbit and the row-loss charge of Lemma 3.2.

The same bounds extend Lemma I.9 to active-row commuting images and their finite outer records: each has total order $p^{O(D_0)}$ with a fixed constant dominated by the permanent C_*.

I.10 Physical scope, lifts, and complexity

The covers above represent G on its *grades*. Pulled back along $H \to G$, their new coordinates have trivial $O_p(H)$-action. They are not asserted to give an H-invariant complement in the original V, or a whole-module cover splitting its possibly nonsplit flag extension. Keep the original V and its flag in parallel with all grades and covers. Their total degree is at most $D + D + 4D/3$, before any fixed number of formal dual or scalar coordinates, and hence remains $O(D)$. The exact numerator and denominator maps identify the covered grades, while the original matrices record all extensions between grades.

Likewise, $\overline{D}$ is initially an actual subgroup of the diagonal image G. Its generator words have preimages in H, but arbitrary preimages need not define a homomorphic section. When a later literal physical argument needs such a section, it uses the proved derived-core lifting result of Appendix H, on the retained physical flag and with the entire prime kernel still present. No splitting is inferred merely from an abstract simple-factor name or from the existence of lift words.

Every group operation used in this appendix takes place on the vector action of the original rows or of the displayed $O(D)$-degree factorization cover, on a specified quotient of that action, or on a disjoint union of explicit quotient/projective/letter domains. All these domains have total

size $p^{O(D)}$. Vector searches have at most p^D seeds. Represented algebra and Hom computations have polynomial dimension, and their field enumerations have at most p^D entries. The number of primitive blocks, local components, diagonal classes, and refinement stages is polynomial in D and $\log p$. Schreier generators are compressed and reduced between stages. These observations prove the stated deterministic $p^{O(D_0)}\mathsf{L}^{O(1)}$ bound, including construction of all kernels, quotient maps, and original words.

The proof of Theorem I.1 is now complete. Its permanent threshold and hereditary bound concern supplied groups and physical carriers. At a completed branch reunion one first constructs the actual returned stabilizer, with its original words, and then applies this theorem to that supplied group while preserving the incoming parent data. The accounting for births of new bounded nodes and for complete branch reunions belongs to the recursive controller; it is not a consequence of multiplying the individual local-order bounds.

J Full-cover graph extension without circular group operations

The auxiliary action of a physical group need not extend to a chosen physical parabolic. The following graph construction removes that unnecessary requirement while controlling invisible extra actions.

Theorem J.1 (Graph extension)**.** *Let $H \le M_+ \rtimes G$, where G is its actual physical image on a faithful original space W_0 of degree D_0, and the linear action of G on the auxiliary prime-field space M is supplied. Set $P = H \cap M_+$. Suppose $P \le R \lhd H$ is a p-group, an actual linear standard section $D \le H$ satisfies $D \cap R = 1$ and $RD \lhd H$, and its canonical full cover L has the following properties:*

1. *$D \lhd L$, $[L : D]$ is prime to p and at most $p^{O(D_0)}$, and L is faithful and completely reducible on W_0;*
2. *on M, the action is a permitted degree-at-most-two module or invariant subquotient, agreeing with the actual D-section and fixing the homogeneous coordinate;*
3. *each automorphism of D induced by H modulo R extends to its supplied natural, dual, field or factor-permutation automorphism of L;*
4. *the degree-eight transfer of Theorem C.1 applies jointly, including those twists and trivial coordinates.*

Then H normalizes RL, the product $\widehat{H} = HL$ is a group with $[\widehat{H} : H] \le [L : D]$, and $\widehat{H}/P$ has a faithful permutation representation of degree $p^{O(D_0)}$. All actual membership, specified kernels and bounded-index return filters are computable with polynomial auxiliary work.

Proof. Act faithfully on the graph space $X = W_0 \oplus M \oplus k$, using homogeneous coordinates for the affine action. Let J_R be the augmentation ideal of the actual R-action. Since R is a finite p-group, its descending image flag on X is nilpotent. Every term is D-invariant and hence L-invariant by transfer. Let V be the full group trivial on its grades. The flag grades have tensor degree at most two, their Hom grades degree at most four, and the differences used in Theorem C.2 degree at most eight. That theorem shows L normalizes the actual R. The group V is used only in this proof and is never enumerated.

For $h \in H$, the graded action of h intertwines D with its supplied twist, because $hdh^{-1} = r_d \alpha_h(d)$. Joint transfer extends this intertwining to L. Thus L and $L' = hLh^{-1}$ have the same graded image. The map $L \to L_{\mathrm{gr}}$ is injective: the induced flag on the completely reducible faithful W_0 splits, so an element trivial on its grades is trivial on W_0.

Put $K = \langle R, L, L' \rangle$ and $S = K \cap V$. Then $R \lhd K$, $K = SL$, S is a normal p-group and $L \cap S = 1$. Modulo R, the common image $\overline{D}$ of D and hDh^{-1} is normal in K/R. It has trivial intersection with S/R, by graded faithfulness, and therefore centralizes S/R. For $s \in S$ this says $[D, s] \leq R$. Lemma C.3 upgrades it to $[L, s] \leq R$. Hence

$$K/R = (S/R) \times \overline{L}.$$

The image of L' is the graph of a homomorphism $L \to S/R$ trivial on D. It factors through the prime-to-p group L/D into a p-group, so is trivial. Thus $hLh^{-1} \leq RL$, and applying the same argument to h^{-1} proves normalization. Consequently HL is a group, with index $[L : L \cap H] \leq [L : D]$.

The original translation kernel P is D-invariant, hence L-invariant by degree-two transfer, so is normal in $\widehat{H}$. Consider its action on the disjoint union of the original vector set of W_0 and $\widehat{H}/H$. The combined kernel contains P. Conversely an element in that kernel fixes the base coset, hence belongs to H, and trivial physical action then puts it in the exact original kernel P. The combined kernel is precisely P and the domain has size $p^{O(D_0)}$. □

The added coset action is necessary. If $G = \mathrm{GL}_a(k)$ acts on an auxiliary natural module by $g \mapsto \det(g)g$, while the canonical full cover acts naturally, their discrepancy is a row-invisible scalar of prime-to-p order. It is false that the enlarged group's entire physical kernel must be a p-group.

J.1 Constructing the permitted operations

To test $x \in H$, first test its physical image in G, obtain a word in the actual projected generators, and lift that word to $h_0 \in H$. Compare the auxiliary linear actions. If they agree, xh_0^{-1} is a translation, and membership is its inclusion in P. The latter is a linear-subspace test. Compute P, if necessary, from lifted physical-image relations and conjugation closure.

Using this membership test, breadth-first search enumerates the bounded set $\widehat{H}/H$, with original representatives and generator actions. Equality is tested by one membership test in H. Combined with the original vector action this gives the explicit model from Theorem J.1. The only operations below the degree-two theorem are those of Appendix A: specified normal kernels, additive corrections, generated subgroups, normal closure, and point operations. The proof-only groups V, K, S and their written intersections are not constructed.

Lemma J.2 (Exact original-group filter)**.** *For a complete transporter coset aS in $\widehat{H}$, its intersection with H is computable by one point transporter on $\widehat{H}/H$.*

Proof. Let $x = H$. Then $as \in H$ if and only if $sx = a^{-1}x$. If the target is outside Sx, the answer is empty. Otherwise let $s_0 x = a^{-1}x$ and compute the full point stabilizer S_x by Schreier generators. The exact answer is as_0S_x. Its orbit has at most $[\widehat{H} : H]$ points. Original words retain the whole action kernel, including invisible translations. □

Independent character or augmentation-one filters use their supplied bounded image in the same manner. This proof does not call generic coset intersection. That general intersection is established only after the dynamic module theorem.

K Complete degree-two transport with physical outside actions

This appendix establishes the module algorithm used by the outer recursion. It has two parts: an invariant-bimodule equation with a normal kernel of p-image, and a scoped recursion over supported retired physical coordinates. Neither calls the outer cubic algorithm.

K.1 The pure invariant-bimodule equation

Theorem K.1 (Normal-kernel transport)**.** *Let F_0 be a semisimple star-algebra of original faithful degree D_E. Supply an actual group $H_0 = R_0 \rtimes \mathrm{U}(F_0)$ of physical degree D_H, with explicit section, projection and original lifts. Let V be a genuine F_0-bimodule with a linear H_0-action whose section acts by $u \cdot v = uvu^{-1}$. Let $P \le U \le V$ be invariant prime-field subspaces. Assume the image of R_0 on the same whole quotient $M = U/P$ is a p-group; the invisible kernel may be arbitrary. Complete transport in M costs*

$$p^{O(D_E+D_H)}\,\mathrm{poly}(\mathsf{L}, \log p).$$

Proof. For each actual star-subalgebra $E \le F_0$, keep $H_E = R_0 \rtimes \mathrm{U}(E)$ on the original V and original quotient M. Let $J = \operatorname{rad} E$ and choose a star-Levi F. Then the kernel $K_E = R_0 \rtimes \mathrm{U}(1+J)$ has p-image on M, and the quotient is the actual $\mathrm{U}(F)$ section. Its augmentation filtration

$$M_0 = M, \qquad M_{i+1} = \sum_{r \in K_E} (r-1)M_i$$

terminates. Each layer is a group-invariant subquotient of the original genuine F-bimodule; it need not itself be a bimodule.

After target normalization, keep the full prefix

$$G_i = \{h \in H_E : hv - v \in M_i\}, \qquad K_i = \ker(G_i \to \mathrm{U}(F)),$$

with surjective projection and $w - v \in M_i$. For $r \in K_i$, its increment $\tau_i(r) = rv - v$ modulo M_{i+1} is additive. Compute the complete image B_i, full kernel, and actual image-basis lifts. Conjugation by G_i makes B_i $\mathrm{U}(F)$-invariant. The quotient

$$Q_i = (M_i/M_{i+1})/B_i$$

has an actual affine $\mathrm{U}(F)$-action: choose any prefix lift of f and take its increment on v. Different lifts differ by B_i. Thus the cocycle is well defined even when this prefix extension does not split.

Theorem F.1, proved by the central-shift and cohomology appendices, returns all target cosets $f_j\,\mathrm{U}(F')$ with one common source star-subalgebra $F' \le F$. Its potential satisfies

$$t\nu(F'/\operatorname{rad} F') \le \nu(F), \qquad 1 \le \nu(F) \le p^{O(D_E)}.$$

For each f_j, lift to the actual G_i and correct its layer error by an actual K_i word.

If $F' < F$, put $E' = F' + J$. The full preimage in H_E is exactly $R_0 \rtimes \mathrm{U}(E')$. Recurse in its corresponding coset on the original full equation in M, not merely the current layer. A complete solution of that equation automatically satisfies every earlier layer condition, so the intermediate prefix may safely be dropped for this full restart. All branches use the same proper source algebra E'; its radical may grow but its next layers are still sections of the same original bimodule. The original R_0, kernels, and observations are retained.

If $F' = F$, the affine source is fixed and a nonempty target condition is its zero class. Correct the target through B_i. For every full generator g of G_i, correct its increment by a K_i word, and adjoin the entire $\ker \tau_i$. These elements generate exactly the next prefix: their projections still generate $\mathrm{U}(F)$ and the difference of any two matching lifts lies in the retained full kernel. At the last zero layer this is the whole stabilizer and the accumulated corrections give a witness.

Every proper restart contracts the same source-common ν on a nested subalgebra of F_0. The sum of child potentials is at most the parent's, so the entire proper-branch tree has $p^{O(D_E)}$ nodes

up to polynomial depth. No-loss layers advance a polynomial-length augmentation flag. Proper steps reduce concrete algebra dimension. Each node costs $p^{O(D_E+D_H)}$ with a uniform polynomial auxiliary factor. Local representation repairs are discarded after original-word recovery; their enlarged dimensions are not new faithful degrees. Complete branch reunion yields one full original coset. □

The kernel R_0 in this theorem is permitted to be nonsolvable only when its action on this same complete quotient is a p-group. It may not be reused as a physical p-kernel on an unrelated observation. An affine input requires a valid whole-ambient linear homogenization and the actual section/full-cover construction, not the assertion that a trivial line is a bimodule of an arbitrary simple algebra.

K.2 Supported physical representations

A supported degree-two state has one current literal cover of total degree $O(D_0)$, at most r original physical factor positions for a fixed constant r, and exact invariant numerator and denominator. Every value has at most two current critical natural positions. Retired positions have actual covered physical actions; they are not arbitrary modules declared critical-trivial. Original matrices, projections, fields, characters, lift words and the permanent hereditary threshold are supplied.

The index domain consists only of vectors of retired physical components. Active isotypic rows remain whole value positions. Their natural factors, multiplicities and type permutations are not new index masters. A call exports only its incoming forest and a complete original coset. Refinements made locally may cease to be invariant under a reunited caller and are projected away at return.

Theorem K.2 (Dynamic degree-two transport)**.** *For every fixed arity r, complete transport in a supported degree-two state is computable in*

$$p^{O_r(D_0)}\,\mathrm{poly}(\mathsf{L}, \log p),$$

with full original stabilizer, kernels, lifts and the incoming physical forest.

Proof. We give the preparation, recursion, terminal and cost arguments separately.

Preparation and exact representation. Prepare the whole actual entry group using Appendix I, within each retained parent. Induce a chosen stabilizer-invariant refinement around its actual parent orbit. A future critical perfect core cannot act nontrivially on a saved retired image or its $O(D_0)$ tag action, by the permanent hereditary simple-order threshold. Joint degree-one-plus-trivial transfer makes the corresponding full cover trivial on the same grades. For a saved nonliteral cover use the exact fiber product over its physical action, retaining every covering kernel.

A genuinely new internal physical flag is processed once. Old cross-block shears stay in exact translation/lift kernels, not as newly unprocessed physical radicals. In an affine state use the actual normalized ambient p-group on the current literal cover and its tensor action. If F_i is its augmentation flag, set

$$\widehat{U} = \mathrm{span}((x,1),(U,0)), \quad \widehat{P} = P \times 0, \quad Q_i = (\widehat{U} \cap (\widehat{P} + F_i))/\widehat{P}.$$

These are the induced layers of the actual affine direction. Omit the fixed homogeneous anchor-only quotient. Their active direction dimensions sum to at most $\dim(U/P)$. Translation alone never starts another physical augmentation pass.

When a factor is retired, extend evaluation by its covered coordinates. For A/B inside a physical cover C, its dual is $B^{\perp}/A^{\perp} \leq C^{*}$ as a subquotient. Tensor the numerators and quotient by the sum of terms containing an $A^{\perp}$ factor. Pull back the old numerator and the entire old denominator once. This preserves all correlations and symmetry equations without division by 3 or 6. Rank-check the induced isomorphism of current directions. Rebuild from the original observation and one current cover; do not tensor earlier repairs repeatedly.

Balanced evaluation recursion. Evaluation on all retired variables is injective on the represented quotient when its full ambiguity is kept. Its explicit domain has size $p^{O_r(D_0)}$; critical value vector sets are never enumerated. Process domain orbits sequentially by complete cosets and Lemma 2.1. In a transitive focus let b be the sum of component incidences in its at most r projections. Split a largest projection into two balanced parts. Its actual subset stabilizer is computed by the subset orbit, of size at most 2^b, followed by Schreier generators. This is a proved bounded orbit, not a general set-stabilizer primitive.

For each orbit branch, complete the first focus, normalize by its actual witness, then complete the second in the exact remaining affine fiber. Only after the full focus has been matched are branches reunited. Away from a fixed small terminal cutoff, each child has incidence count at most $\gamma_r b$, for a constant $\gamma_r < 1$, for example $1 - 1/(3r)$. If new retired positions appear during preparation, resume this balanced recursion before opening any local-image list.

Terminal action. A terminal focus uses at most r retired components and has only whole degree-two critical packets left. Let a critical natural factor have field degree f_i, natural dimension a_i, and actual multiplicity d_i. The permanent threshold and original mass give

$$f_i a_i^2 > C_* D_0, \qquad \sum_i f_i a_i d_i = O(D_0), \qquad \sum_i f_i d_i^2 = O(D_0).$$

Thus the entire actual commuting multiplicity image has order $p^{O(D_0)}$. There are $O(\sqrt{D_0})$ critical factors, so their field/type/exchange records also have order $p^{O(D_0)}$. Enumerate these actual finite images only in the complete terminal. In the same bounded list include the at most r local retired-component images and every independent tensor-value character not already in the verified pure action.

Apply Theorem J.1 to the normalized terminal state and work in its bounded-index overgroup containing the full cover L, retaining the original group as the exact return filter. Lemma I.10 expresses the record-fixed diagonal action in that overgroup as L times its actual centralizer, with field, duality, similitude and determinant records retained. No such product equality is asserted for an original subgroup that does not contain L. After fixing the entire large-isotypic centralizer image, its remaining part fixes all large rows and the selected retired arguments. Consequently it fixes the whole focused linear value ambient. Pullback retains the physical and symbolic kernels; any further affine displacement has p-image. Invisible nonsolvable kernels are retained, not declared physical p-groups.

This condition is executable. For the planned normal kernel on the entire focused quotient, compute its represented algebra and the ideal generated by $\rho(r) - 1$. The image is a p-group exactly when that ideal is nilpotent: augmentation proves one direction, and containment in $1 + I$ proves the other. Also check the pure generator action against the genuine bimodule action and verify both numerator and denominator invariance. Under the construction these checks hold; a certificate failure is not evidence of tensor nonisomorphism.

Use the whole-Hom section and graph extension to obtain the actual full-unitary section, keeping independent characters separate. Theorem K.1 completes the entire terminal quotient. Filter the full result back to the original bounded-index group by Lemma J.2. Reunite all terminal records before returning. At a physical augmentation return, lift its diagonal witness, correct through the exact physical translation image and restore the zero-increment kernel.

Cost. Let v be unretired original physical mass and b the incidence count. Retirement of mass c creates at most $A_r c$ incidences and expands the explicit evaluation domain by at most $p^{E_r c}$, for fixed constants. Put $\Lambda = b + \lambda_r v$ with $\lambda_r > A_r$. A balanced child has

$$\Lambda - \Lambda_{\text{child}} \geq (1 - \gamma_r)b + (\lambda_r - \gamma_r A_r)c.$$

A fixed multiple pays both the $2^{O(b+A_r c)}$ branch count and the evaluation expansion. Initially $\Lambda = O_r(D_0)$. For polynomial bookkeeping use the focus volume times a fixed power of the current direction dimension: domain partitions add volume, and exact sequential/augmentation fibers add direction dimensions. Zero-direction foci are direct corrections. Equal-direction unary steps require a genuinely new physical flag refinement, so have polynomial depth.

All expensive local-image lists occur only in completed terminal bases. Source searches and preparation add work; they do not leave an open list multiplying later births. Complete siblings add their costs, with their local refinements discarded at return. The potential therefore bounds the whole recursion by $p^{O_r(D_0)}$ times a uniform polynomial. This also proves the exact return contract inductively. □

K.3 Whole subspace frames and exact intersections

Let L be a supplied supported outside module, trivial on the current critical cores, and let $X \leq V \otimes L$ be a subspace of dimension $d = O(D_0)$ represented by an injection $f : E \to V \otimes L$. On $H \times \mathrm{GL}(E)$ solve

$$(h_V \otimes h_L)f = f'a.$$

It is the whole evaluation tensor in $E^* \otimes V \otimes \mathrm{cover}(L)$, with at most two critical positions. The total physical degree is $O(D_0 + d)$ and its arity is bounded. Theorem K.2 completes the call. Projecting its whole coset and stabilizer onto H is exactly the subspace transporter, since a exists precisely when $hX = X'$ and is unique for such an h. The frame is discarded at complete return.

Finally, prepare $G_1 \times G_2$ on two actual degree-d rows and transport I to t in $\mathrm{Hom}(W_2, W_1)$ under $(g_1, g_2)X = g_1 X g_2^{-1}$. Projection gives $G_1 \cap tG_2$, and the source stabilizer projects to $G_1 \cap G_2$. This proves generic actual matrix-group coset intersection in $p^{O(d)}$ only now. Similarly, for a completed child coset tK and an actual parent factor map π, transport the identity tuple in $\bigoplus_i \mathrm{Hom}(V_i, V_i)$ under $H \times K$ and project to H. The result is the exact preimage $\{h : \pi(h) \in tK\}$, with every invisible parent kernel.

The dependency order is therefore finite algebra and transfer/cohomology, actual sections and graph closure, pure and dynamic degree-two transport, then actual group intersection, and finally the outer component dynamic program. Preparation itself uses explicit small-quotient kernels and the cited normal-subgroup algorithms, not this later intersection theorem.

L Dominant fields from raw triples

A short tuple need not generate a single large field. The following construction uses all raw triples, while keeping the total work singly exponential in physical degree.

Theorem L.1 (Direct field gluing)**.** *Let k be a finite field of odd characteristic, K/k an extension of degree f, $A = K^a$, and $M : C \to \mathrm{End}_k(A)^\ell$ a whole k-linear pencil. The constant field K is distinguished in every raw algebra*

$$B_S = \mathrm{alg}_k(K, I, \{M_j(x) : x \in S, 1 \le j \le \ell\}).$$

Assume for every $|S| \le 3$ that B_S has a field quotient $L_S = B_S/P_S$ with K embedded, and $A_S = \mathrm{ann}_A(P_S)$ has K-dimension at least $a - r > a/2$. Then there is a field F/K, with

$$[F : K] = E \le \mathrm{lcm}(1, \dots, a) = \exp(O(a)),$$

and a k-linear map $\lambda : C \to F^\ell$, unique up to one common K-Frobenius, satisfying

$$\lambda(x) = 0 \quad \Longleftrightarrow \quad \mathrm{rank}_k[v \mapsto (M_j(x)v)_j] \le fr.$$

The diagram, field, map and all embeddings are computable in $|k|^{O(\dim_k C + fa)}$ times polynomial label work. Conjugation and actual label changes transport the whole construction equivariantly.

Proof. For $S \subseteq T$, $|T| \le 3$, the image of B_S on A_T is a subfield of L_T. Thus A_T is semisimple of one B_S-type. It must be the L_S type, because otherwise A_T and A_S have zero intersection and dimensions summing to more than $\dim A$. Hence

$$P_T \cap B_S = P_S, \qquad i_{ST} : L_S \hookrightarrow L_T$$

are actual compatible K-field embeddings, and $A_T \subseteq A_S$. Write $e_x = [L_x : K] \le a$. The raw pair algebra is generated by its two singleton algebras, so its field quotient is their compositum and $e_{xy} = \mathrm{lcm}(e_x, e_y)$; the same holds for triples.

Let $F = \mathbb{F}_{|K|^E}$ with $E = \mathrm{lcm}_x\, e_x$. Choose arbitrary singleton and pair embeddings into F. Their pairwise discrepancies are phases

$$d_{xy} \pmod{\gcd(e_x, e_y)}.$$

An embedding of a triple field shows

$$d_{xy} + d_{yz} = d_{xz} \pmod{\gcd(e_x, e_y, e_z)}.$$

For each rational prime $\ell \mid E$, choose a reference x_ℓ with maximal ℓ-adic valuation of e_x and prescribe

$$s_x = d_{x,x_\ell} \pmod{\ell^{v_\ell(e_x)}}.$$

The triangle relation proves every pair congruence at this prime. Integer CRT gives s_x modulo e_x. Twisting singleton embeddings by these phases aligns every pair and triple diagram. Any two solutions differ by one common phase modulo E, since at each prime their differences are restrictions of one constant at maximal valuation. Composita give unique extensions to pairs and triples.

Define $\lambda_j(x)$ as the coherently embedded quotient value of $M_j(x)$. For $z = x + y$, literal pencil linearity gives $B_{x,y,z} = B_{x,y}$ and $M_j(z) = M_j(x) + M_j(y)$. Applying the common quotient proves additivity. The same argument proves k-homogeneity. These entries generate all singleton fields and hence F over K.

If $\lambda(x) = 0$, every $M_j(x)$ kills the same A_x, so the whole stack has k-rank at most fr. If some value is nonzero, that operator acts invertibly on A_x, and its rank is at least $f(a - r) > fr$. This proves the exact kernel assertion without a label factor. In the noncentral-old-field case, decompose

each original k-linear operator into its K/k Frobenius-semilinear components. On the K-stable field core it commutes with K; independence of field embeddings makes every nonidentity component zero there. Untwisting outputs gives one whole K-linear stack with the same core and rank bound, and with the actual twisted row actions retained.

A genuine tensor transporter conjugates all raw algebras, unique dominant cores and quotient embeddings. The uniqueness of the gluing gives one common field isomorphism, hence a common Frobenius after source and target are placed in one F. An invertible label change merely replaces a spanning set of the same complete operator image. Thus for moving supported labels the construction is the bilinear value in $\mathrm{Hom}_k(L, F)$, not an independent value for each chosen coordinate. □

L.1 Constructing the large field without enumerating its elements

For $n \geq 2$, put $L(n) = \mathrm{lcm}(1, \ldots, n)$. The quotient $L(n)/L(\lceil n/2 \rceil)$ divides $\binom{n}{\lfloor n/2 \rfloor}$: each new prime power lies above $n/2$ and its extra prime exponent appears in that binomial coefficient. Therefore $\log_2 L(n) \leq n + \log_2 L(\lceil n/2 \rceil) = O(n)$.

Factor the small degrees $e_x \leq a$. For each rational prime $\ell \mid E$, put $d_\ell = \ell^{v_\ell(E)} \leq a$. Construct a degree-$d_\ell$ field K_ℓ/K by enumerating at most $|K|^{d_\ell}$ monic polynomials and testing irreducibility. The pairwise coprime degrees make

$$F = \bigotimes_{\ell \mid E, K} K_\ell$$

a field. Its tensor basis gives arithmetic polynomial in E, hence exponential only in a. This avoids finding a degree-E irreducible by enumeration over $|K|^E$.

To embed a small field of degree $d \leq a$, compute $\ker(\mathrm{Frob}_K^d - I)$ in F. It is the unique d-dimensional subfield. Enumerating its $|K|^d$ elements finds a root of the small defining polynomial. Reuse a basis for each repeated degree. Phase tests involve at most a powers on a small generator. Source and target diagrams use the same constructed F after comparing E; no expensive isomorphism search between two exponential-degree presentations is hidden.

There are at most $|k|^{3 \dim C}$ raw triples. Algebra/module work is on the original row, and the field arrays have total size $|k|^{O(\dim C + fa)} \, \mathrm{poly}(\ell)$. This proves the algorithmic estimate.

L.2 The raw-triple source alternative

For a represented raw algebra B, perform Krull–Schmidt over the prime working field k, including when its distinguished old field K is noncentral. Write $A = \bigoplus_i Q_i^{b_i}$, let $L_i = \mathrm{End}_B(Q_i)/\operatorname{rad}$ and set $g_i = [L_i : k]$, $\dim_k Q_i = g_i j_i$. Its centralizer has semisimple factors $\mathrm{M}(b_i, L_i)$ and ordinary prime-field natural degrees $d_i = g_i b_i$. If some d_i exceeds $f(a - r) > fa/2$, then $j_i = 1$, because $j_i \geq 2$ would exceed total physical mass fa. The resulting one-dimensional residue-field module is a genuine field quotient of B. Only now does restriction of that quotient embed the distinguished K: its kernel on the field K is zero. The associated annihilator core is K-stable, and its k-dimension at least d_i is therefore K-dimension greater than $a - r$. Uniqueness follows from the same mass bound. If no such degree exists, every centralizer natural row has prime-field degree at most $f(a - r)$, and the raw tuple is a geometric fission certificate. These alternatives are exhaustive and constructible by represented-algebra operations. No residue field is asserted to contain the old K before the field quotient has been obtained.

For moving labels the code kernel is an actual prime-field physical subspace. Match a proper nonzero kernel by the whole supported frame call, retain its star-conductor, and make a strict unbranched restriction. If the code is injective, for each global phase the value equation kills the

critical coefficient action; complete the whole remaining at-most-two-critical observation before reuniting phases. This is the uniform rule of Section 4; no uncharged phase branch enters outer recursion.

M Natural insertions and their exact quotient forcing

Throughout this appendix k has odd characteristic, K/k is a finite extension of degree f, and dimensions used for physical costs are over k. When old defining fields do not contain the original input field, we take $k = \mathbb{F}_p$, as in Section 1. All original field-linearity equations remain imposed. A label family always means the whole supplied family with its actual common action. In particular, the parameters of an additive translation are not made independent by writing them in separate coordinates.

M.1 A whole-module calculation of the insertion types

Let L be a product of the full classical covers furnished by Appendix C. Natural dimensions are in the stable range; bounded natural dimensions are handled by the bounded-factor procedures. A permitted one-leg module is a natural module, its dual, or one of its specified Frobenius twists, with its actual defining-field realization. Fixed multiplicity spaces may accompany any leg.

Lemma M.1 (The four-leg calculation)**.** *For permitted one-leg modules S, T_1, T_2, T_3, every map in $\mathrm{Hom}_L(S, T_1 \otimes T_2 \otimes T_3)$ is a sum of positionwise pairing insertions. Such an insertion maps S to one output leg by an actual one-leg intertwiner and fills the other two output legs with an invariant pairing tensor. A position is allowed precisely when these intertwiners and pairings exist. Defining-field and multiplicity parameters, and all linear relations among the insertions, are retained.*

Proof. Currying gives the canonical isomorphism

$$\mathrm{Hom}_L(S, T_1 \otimes T_2 \otimes T_3) \cong \mathrm{Hom}_L(T_3^* \otimes S, T_1 \otimes T_2).$$

Both modules on the right are whole degree-two words. Their joint image algebra is the one of Proposition F.2; this uses the joint transfer theorem, not a replacement of a finite group by its algebraic closure.

We spell out the consequence for intertwiners. Between identified nontrivial simple degree-two summands, an intertwiner is scalar over the actual endomorphism field. Between different types it is zero. On the two summands of an equal-word square, the two projections are $(1 + \tau)/2$ and $(1 - \tau)/2$, where τ switches its legs. Distinct-digit products have only the genuine straight or switched identifications specified by their one-leg field and duality maps. A map between trivial summands contracts both source legs invariantly and inserts an invariant tensor in the two target legs. All linear maps between the trivial multiplicity spaces are included here.

It remains to justify this assertion on a nonsplit component. In bottom–heart–top coordinates the actual image algebra contains a heart idempotent, all heart matrices, all supported heart-to-bottom rows, and all supported top-to-heart columns. For different heart types, an intertwiner vanishes on the source heart. Commuting with the supported rows makes it vanish on the source bottom. Commuting with the target heart idempotent and columns forces its image into the target bottom. It therefore factors from the source trivial top to the target trivial bottom. For identified hearts, subtract the matching scalar whole-chain map first; the same argument applies to the remainder. The argument also treats a chain and an independent trivial summand. With repeated copies it

applies matrixwise in the multiplicity coordinates. In particular, neither an isolated heart projection nor an isolated heart inclusion has been asserted.

Thus the degree-two intertwiners are generated by the following three pair diagrams: contract the two source legs and insert in the two target legs; match the source legs to the target legs in order; or match them in the switched order. Uncurrying gives respectively the three permitted positions in the statement. Conversely each such diagram is equivariant. The argument may be checked after scalar extension and descended: both the Hom space and the span of these explicit maps are kernels or images of linear systems over the original field. It uses division by 2 only, and the degree-two catalog includes the field of order three. It makes no assertion about arbitrary cubic subquotients. □

The lemma is constructive. Solve the actual one-leg and pairing intertwiner equations, form their three insertion maps, and compute their linear span and kernel. Solving the full Hom equations gives the same space by the proof. Every matrix in these calculations has polynomial dimension in the physical degree and explicit label input.

Here is the resulting exhaustive list for a three-critical packet. For three independent masters there is no insertion. For a paired master and an independent free master there is one pairing times a free-slot functional. For a full general-linear master, only a natural twist and its matching dual twist can pair: the mixed word V, V, V^* has two positions, and the pure three-natural or three-dual word has none. For orthogonal and symplectic masters the form identifies a natural with its dual, giving three positions when all twists match and the eligible subset otherwise. For a unitary master its dual is the involution-Frobenius twist: on an individual defining-field signature there are at most two eligible positions. These are exactly the one-pair and general-linear two-position patterns. There is no additional determinant identification in stable natural dimension. Together with Proposition 2.4, this classifies the actual translation image, rather than just a possible list of its composition factors.

M.2 One original-field bundle for all twists

Individual Frobenius signatures need not become separate recursive observations. The following formulation keeps the original k-tensor on its original physical rows.

Let $A = K^a$. In the first-kind case let $Q : A \times A \to K$ be its nondegenerate symmetric or alternating form and put $q_0 = \mathrm{Tr}_{K/k} Q$. In the unitary case let K/K_0 be quadratic with involution σ, let h be Hermitian and linear in its first variable, and put $q_0 = \mathrm{Tr}_{K/k} h$. We assume $k \subseteq K_0$ in this latter notation; restriction to the prime field ensures this. The form q_0 is nondegenerate. Indeed, after choosing a nonzero value of Q or h, varying a K-scalar in one argument tests the nondegenerate field-trace pairing. In the Hermitian case it is symmetric, since trace is invariant under σ.

Lemma M.2 (The full functional insertion bundle)**.** *Write b for Q or h, as appropriate. All permitted first-kind or unitary insertions in the original k-trilinear ambient lie in the image of*

$$J : S \longrightarrow \mathrm{Hom}_k(A^{\otimes 3}, k), \qquad S = \mathrm{Hom}_k(A, K)^3, \quad (32)$$

$$J(\ell_1, \ell_2, \ell_3)(x, y, z) = \mathrm{Tr}_{K/k}\big(\ell_1(x)b(y, z) + \ell_2(y)b(x, z) + \ell_3(z)b(x, y)\big).$$

The statement holds jointly over all labels, with their actual actions. If N is the actual translation image, the allowed parameter space is $R = J^{-1}(N)$, including $\ker J$*, and remains exactly this space. The ambient S is a sum of actual natural-dual copies with passive field multiplicity; it adds no physical argument.*

Proof. The commuting algebra of the full natural group is K. Using the nondegenerate form q_0 to identify a bilinear form with an endomorphism, the invariant k-bilinear forms are precisely

$$(y, z) \longmapsto \mathrm{Tr}_{K/k}(c\, b(y, z)), \qquad c \in K.$$

Every product of such a pairing with a k-linear functional u on the free slot occurs in (32) by setting $\ell = cu$. These products span all insertions by Lemma M.1. This also includes all descended Frobenius occurrences. Each space $\mathrm{Hom}_k(A, K)$ is a direct sum of f copies of the underlying natural-dual k-module: choose a k-basis of the codomain K. Its dimension is polynomial in $\dim_k A$. The trace formula is equivariant under the actual form-preserving group by substitution. Semilinear maps, similitudes, value characters, and label actions give the corresponding actual action on S; they are retained, rather than replaced by independent scalar choices. The preimage R preserves exactly N, regardless of whether N is K-linear. □

For a naturally K-valued packet, use the ordered k-coordinate values of its output, or equivalently all trace-dual output functionals. The lemma is then applied to that one whole ordered family. A correction from S may leave a narrower balanced or single-signature subspace. Its result is stored in the full original k-tensor ambient and is not projected back to the narrower subspace. In particular there is no same-size cubic call for each scalar-extension signature.

For a mixed general-linear word the corresponding whole-bundle notation is an actual k-bilinear map $B : A \times A \to A$ with

$$J(\alpha, \beta)(x, y) = \alpha(x)y + \beta(y)x, \qquad \alpha, \beta \in \mathrm{Hom}_k(A, K). \tag{33}$$

This follows by using the trace evaluation pairing between A and its dual in the preceding argument. For a genuinely different free master C, the one-pair parameters are maps $u : K \to C$ over k, not hypothetical vectors in a shared defining field.

M.3 Projected contractions and noncentral old fields

We first treat (32). Set $a \geq 64$ and $r = \lfloor a/32 \rfloor$. For each nonzero $x \in A$ use all marked nondegenerate K-subspaces H of dimension at most two containing x. A nonisotropic vector spans such an H. An isotropic vector has a partner y with $b(x, y) \neq 0$, and their two-dimensional Gram determinant is nonzero. This includes the alternating case, in which every nonzero vector uses a two-plane. Retain x and its actual auxiliary mark, rather than choosing a preferred partner. Let P_v be the orthogonal projection onto $H^{\perp}$ for $v = (x, H)$. There are at most $|K|^{2a}$ choices of marks for one vertex. The zero coefficient contributes zero operators and needs no mark.

For each original k-trilinear label define $L_{j,x}$ by

$$q_0(L_{j,x}y, z) = T_j(x, y, z), \qquad D_{j,v} = P_v L_{j,x} P_v.$$

Here $L_{j,x}$ is only k-linear. It need not commute with K. Define the raw algebras on the original row by

$$B_v = \mathrm{alg}_k(K, I, P_v, \{D_{j,v}\}_j), \qquad B_Z = \mathrm{alg}_k\{B_v : v \in Z\}. \tag{34}$$

For moving labels use the image of the whole label map instead of individual j's. The distinguished copy of K is a constant subalgebra, not an assertion that it is central in B_Z.

Lemma M.3 (Exact compression and scalar quotients)**.** *Adding* (32) *changes the compressed operator by*

$$D_{j,v} \longmapsto D_{j,v} + \ell_{1,j}(x)P_v. \tag{35}$$

Consequently all raw algebras (34) *are exactly unchanged. Suppose that every raw tuple with at most three vertices has a dominant field quotient $B_Z/P_Z = L_Z$ whose annihilator core $A_Z = \operatorname{ann}_A(P_Z)$ has K-dimension at least $a - r$. The gluing of Appendix L then gives a common field F/K, unique up to one global K-Frobenius, and a k-linear map*

$$\lambda : A \longrightarrow F^{\{j\}},$$

whose entries, together with K, generate F. Its value at x is independent of the projection mark, and adding an insertion changes $\lambda_j(x)$ by $\ell_{1,j}(x)$. The degree $[F : K]$ is at most $\operatorname{lcm}(1, \dots, a)$.

Proof. Since P_v is self-adjoint for q_0,

$$q_0(P_v L_{j,x} P_v y, z) = T_j(x, P_v y, P_v z).$$

Both terms in the insertion involving $b(x, P_v y)$ or $b(x, P_v z)$ vanish. The remaining term is

$$\operatorname{Tr}_{K/k}(\ell_{1,j}(x) b(P_v y, P_v z)) = q_0(\ell_{1,j}(x) P_v y, z).$$

For a Hermitian form this uses its first-variable-linear convention. This proves (35). Its increment and negative already belong to the algebra containing K and P_v, so the raw algebra is unchanged, including every additive correlation among labels.

The restriction of a quotient homomorphism $B_Z \to L_Z$ to K is injective: it is a unital homomorphism from a field. Thus L_Z contains the distinguished K, even if K is noncentral in B_Z. Its core is K-stable. For nested tuples, the two cores have total dimension greater than a and hence meet. The restriction of the larger field type is therefore the unique smaller dominant type. This is exactly the annihilator-core argument of Appendix L. All quotient embeddings fix K, and the raw union algebras give its required pair and triple compositum identities.

The image of P_v in a dominant field is an idempotent, hence 0 or 1. If it were 0, P_v would annihilate a core of dimension at least $a - r$ and would have rank at most r. This contradicts $\operatorname{rank}_K P_v \geq a - 2 > r$. Hence its image is 1, so the singleton quotient fields are generated by the compressed values and K.

For any k-linear L and any K-linear projection of codimension at most two,

$$\operatorname{rank}_k(L - PLP) \leq \operatorname{rank}_k((I - P)L) + \operatorname{rank}_k(PL(I - P)) \leq 4f.$$

Two vertices for the same x consequently differ by rank at most $8f$. For vertices at $x, y, x + y$, the additive defect $D_{j,x+y} - D_{j,x} - D_{j,y}$ has rank at most $12f$. These differences belong to their raw pair or triple algebra. A nonzero image in its field quotient acts invertibly on the common core and has rank at least $f(a - r) > 12f$. Every displayed defect therefore has zero field value. The same argument with two vertices proves homogeneity under k-scalars. This establishes independence of the mark and genuine k-linearity; it does not assert an equality of compressed raw algebras at $x, y, x + y$. The coherent embeddings give λ, and the preceding generation observation proves that its entries generate F over K. Under an insertion the same quotient diagram is used, and (35) has quotient value $\ell_{1,j}(x)$. □

If the original pencil is genuinely K-linear in its coefficient argument, the same H can be used at x and αx; its quotients fix K, giving K-linearity of λ. No such conclusion is needed for the whole original-field bundle. In particular an extracted scalar functional always belongs to $\operatorname{Hom}_k(A, K)$, whether or not it has the narrower form $Q(u, -)$.

The dominant-field hypothesis in the lemma is an exhaustive outcome, not an extra tensor assumption. Apply the raw centralizer alternative of Appendix L in the original k-representation. Either some tuple has every ordinary natural degree at most $f(a-r)$, or its unique degree exceeding this bound is a multiplicity of a one-dimensional field module. The latter provides the stated quotient and K-stable core. Testing all tuples and constructing their fields costs $|k|^{O(fa)}$ times a polynomial in the labels. The former tuple is a paid fission: at most six marked physical vectors cost $|K|^{6a}$, while each succeeding active row loses a fixed fraction. For projected operators the marked observation consists of the original marks, the projectors P_v, and all ordered classes $D_{j,v} + KP_v$. In the mark stabilizer this is an explicit invariant quotient of an ordered degree-two Hom module. The complete procedure of Appendix K therefore matches it under the actual prefix. It is a necessary observation: its scalar denominator may contain more than the actual correlated image of N. The full original tensor and N remain to be solved under its complete returned coset. Every genuine original transporter occurs in one of the marked target branches. Lemma M.7 places every returned critical perfect core in the centralizer of B_Z, so the source centralizer bound applies to the actual successor. In a unitary realization this is the centralizer of B_Z intersected with that of B_Z^*; the added equations cannot create a larger ordinary natural constituent than those of the original centralizer. To see this last assertion, restrict each ordinary simple module of the old centralizer to the subalgebra imposed by the additional equations. The kernel of their direct sum is contained in the old nilpotent radical. Hence every new simple type occurs among these restrictions, and its k-dimension is at most the largest old simple module dimension. One must not claim that a class transporter fixes every $D_{j,v}$ literally.

M.4 The general-linear and balanced one-pair patterns

For (33), put $L_{j,x}y = B_j(x,y)$. Mark a nonzero x and a K-linear functional ϕ with $\phi(x) = 1$, and put

$$P = I - x \otimes \phi, \qquad D_{j,(x,\phi)} = PL_{j,x}P.$$

The simultaneous natural and contragredient action transports this projection by conjugation. Since $Px = 0$, an insertion changes D by $\alpha_j(x)P$. Its rank is $a-1$ and its image in a dominant field is 1. The compression error has k-rank at most $2f$, the same-point difference at most $4f$, and the triple additive defect at most $6f$. The proof of Lemma M.3 applies without alteration with these smaller constants. Perform it on both $B(x,y)$ and $B(y,x)$ to recover both scalar functional arrays. Its actual faithful star realization retains A and A^* with the exchange involution, so every projection has its star-dual partner. A projection on A need not itself be self-adjoint. For a unitary two-position signature the same computation is valid in the actual unitary subgroup. Alternatively its entire original-field bundle is covered by (32).

For the one-pair pattern let X, Y be paired K-rows with nondegenerate K-bilinear $Q : X \times Y \to K$, and let the free row be C over k. The packet and its insertions are

$$T_j : X \times Y \to C, \qquad T_j(x,y) \longmapsto T_j(x,y) + u_j(Q(x,y)), \quad u_j \in \mathrm{Hom}_k(K,C). \tag{36}$$

If $\dim_k C \le \dim_k X$, contract a coefficient $c^* \in C^*$ and identify X with Y^* by the trace pairing. The insertion becomes a scalar operator $s_j(c^*)I$, where trace duality uniquely gives

$$c^*(u_j(z)) = \mathrm{Tr}_{K/k}(s_j(c^*)z) \qquad (z \in K). \tag{37}$$

Thus the uncompressed raw algebra containing K is already insertion-invariant. The direct gluing proof has zero additive rank defect. Its tuple marks cost $|k|^{O(\dim C)}$, paid by a constant-fraction

fission of the paired row. If all field values are scalar, (37) recovers an actual map $u_j : K \to C$ with no assumed K-structure on C. All twists and parameter correlations are those of the original insertion map.

These constructions exhaust the fixed-signature cases listed after Lemma M.1. Equal or unequal twists on one original first-kind or unitary master can instead be processed at once by its trace-functional bundle. Pure general-linear words with no pairing are already literal. The only remaining size case of (36) is treated next.

M.5 The augmented pencil for a larger free row

Proposition M.4 (Unbalanced one-pair alternatives)**.** *In* (36)*, suppose* $\dim_K X = \dim_K Y = a \geq 64$ *and* $c = \dim_k C \geq fa$*. The exact insertion quotient has one of the following transitions: a high-rank mark with cost* $|k|^{O(fa)}$ *paid by ordinary natural-degree loss; an unbranched proper physical conductor on a paired row; an unbranched proper conductor on the free row with its whole label action; or an exact degree-one parameter quotient. No mark in the larger free row is required.*

Proof. For $x \neq 0$ mark $z \in Y$ with $Q(x, z) = 1$ and set

$$P_{x,z}(y) = y - zQ(x, y), \qquad M_x(y) = (T_j(x, y))_j.$$

Every insertion vanishes in $M_x P_{x,z}$. The marked compressed stack is therefore exactly quotient-invariant. Its marks cost at most $|K|^{2a}$, independent of c. If some stack has rank at least $fa/64$, Lemma 4.1 applies to these actual marked operators. It pays their $O(fa)$ marking entropy. The marks and actual returned prefix remain imposed throughout the retained batch.

Otherwise put $r_0 = \lfloor fa/64 \rfloor$ and define the whole augmented linear pencil

$$\widetilde{M}_x : Y \longrightarrow C^{\{j\}} \oplus K, \qquad \widetilde{M}_x(y) = (M_x(y), Q(x, y)).$$

The last coordinate has rank f for $x \neq 0$, with kernel $\operatorname{im} P_{x,z}$. Rank–nullity on this coordinate gives $\operatorname{rank}_k \widetilde{M}_x \leq r_0 + f$ for every x, also at $x = 0$. Equivalently, the discarded part factors through the single map

$$K \longrightarrow C^{\{j\}}, \qquad s \longmapsto (T_j(x, zs))_j.$$

This has rank at most f jointly over all labels. Pulling s outside T_j would be invalid for a merely k-bilinear tensor.

An insertion $u = (u_j)_j$ acts on the augmented output by the invertible shear

$$(d, s) \longmapsto (d + u(s), s).$$

The genuine row, field, and label transformations give their corresponding invertible block-diagonal maps, with the carried characters. Thus the augmented pencil is covariant under every actual quotient transporter. Proposition B.2 gives a canonical pair $W \subseteq Y$, $E \subseteq C^{\{j\}} \oplus K$ such that

$$\widetilde{M}_x(W) \subseteq E \quad \text{for all } x, \qquad \operatorname{codim}_Y W + \dim_k E \leq 4(r_0 + f) \leq fa/8 < fa.$$

In particular $W \neq 0$. If $W \neq Y$, match this actual physical k-subspace by the complete supported subspace procedure. It is a proper conductor of the full paired natural core, which acts irreducibly on Y. There is no claim that W is K-linear and no unrestricted K-subspace transporter is substituted.

Suppose $W = Y$. Nondegeneracy of Q implies that the projection $E \to K$ is onto. Set

$$F_0 = E \cap C^{\{j\}}.$$

Every insertion shear acts identically on this kernel, so F_0 is a genuine covariant subspace for the actual free-row and label actions. If $F_0 \neq 0$ and labels are fixed, any first nonzero ordered coordinate projection is a nonzero physical subspace of C of dimension at most $\dim E < fa \leq c$. Match it completely to obtain a proper conductor. For moving labels retain the whole F_0 in the supported module $C \otimes L$. Each nonzero module for the full free natural core in this isotypic sum has dimension at least c, since that core is trivial on L. Thus $0 < \dim F_0 < c$ ensures that its full-core stabilizer is proper. Match this whole subspace by the supported frame procedure under the actual prefix, including its label action, and prepare the complete returned physical group by Appendix I. Its stabilizer is a proper subgroup because some element of the old actual pure core moves F_0. This gives strict unbranched progress even if the label p-action prevents F_0 from descending to the old diagonal quotient. This operation does not ask for a free transporter of an arbitrary auxiliary matrix space.

Finally, if $F_0 = 0$, projection identifies E with K. It is the graph of a unique k-linear map $u : K \to C^{\{j\}}$. Since it contains every $\widetilde{M}_x(y)$, the complete packet is $T_j(x, y) = u_j(Q(x, y))$. Its information is exactly the parameter class of u modulo the actual correlated allowed translation parameters. The degree-one procedure matches this class and restores all actual physical translation lifts and their kernel. It need not have u itself in the allowed translation space. These three low-rank cases are exhaustive. □

The augmented cover is used only to extract W, the shear-fixed F_0, or the graph parameter. Matching the whole E by an ordinary unsheared frame procedure would lose the displayed insertion action. Nor does this augmented cover assert a literal zero block of the original tensor.

M.6 Nonzero field values and exact scalar extraction

Under Lemma M.3 the intrinsic value of the insertion quotient is

$$\overline{\lambda} : A \longrightarrow (F/K)^{\{j\}}. \tag{38}$$

For moving labels the codomain is $\mathrm{Hom}_k(L, F/K)$, with the supplied whole L-action. The map is one linear map on the entire original coefficient row; the label length is not an exponent.

Lemma M.5 (Fixed-label value bound)**.** *Suppose the old field and label actions are fixed. If $t = \mathrm{rank}_k\, \overline{\lambda} > 0$, at most $\mathrm{char}(k)|k|^t \leq |k|^{2t}$ field phases can carry its source code to its target code. For each phase, imposing the actual ordered value equations retires at least t original k-dimensions from the pure coefficient action.*

Proof. The phase transporter, when nonempty, is a coset of the cyclic subgroup of $\mathrm{Gal}(F/K)$ preserving the source code U. Its image in $\mathrm{GL}_k(U)$ has order at most $|k|^t$. One elementary bound is as follows: the order of an element is the product of its semisimple order and a p-power from its largest Jordan block. For distinct irreducible factors of degrees d_i, the semisimple order divides $\prod_i (|k|^{d_i} - 1)$. If the largest block exponent is e, its p-power is at most $p^{\lceil \log_p e \rceil}$, which is at most $|k|^{d(e-1)}$ for that block when $e > 1$. The sum of these block degrees is at most t, giving the bound.

An element of the pointwise kernel on U sends each chosen field generator v among the coordinates of λ to $v + c_v$ with $c_v \in K$. Its pth power fixes each such generator and fixes K. It is therefore the identity on F. The cyclic pointwise kernel has order at most p, proving the phase bound. The pointwise kernel need not be trivial.

After a phase and one transporter have been fixed, the residual pure coefficient group satisfies $\mathrm{im}(g - I) \subseteq \ker_k \overline{\lambda}$. If its defining field is K_C, its image is a K_C-subspace, hence lies in the largest

K_C-subspace W contained in this kernel. The group is the identity on the original physical quotient C/W, whose dimension is at least t. Its remaining natural action is supported on W, with its original shears and kernel retained. The number of phase choices is therefore paid by this actual physical loss. □

For moving labels the displayed phase bound is not used: a phase can be compensated by a label transformation. The actual-prefix procedure in the next subsection instead uses a proper kernel conductor or complete injective terminals.

If (38) is zero, every representative field value lies in K. Since these values generate the common field together with K, the field is K itself, with no additional gluing phase. Apply the construction to every eligible coefficient orientation. For (32) these are all three orientations, using the order of the paired variables appearing in that formula. For (33) there are two orientations. Each orientation is tested for a bad tuple or nonzero quotient value before claiming scalar extraction for the complete packet. If all these tests have scalar values, they give a functional tuple modulo the insertion kernel. Write $\overline{S} = S/\ker J$, let $\overline{J} : \overline{S} \to M$ be the induced injective map, and denote the extracted value by $\overline{s}(T) \in \overline{S}$. Its exact laws are

$$\overline{s}(gT) = g\overline{s}(T), \qquad \overline{s}(T + Jr) = \overline{s}(T) + (r + \ker J). \tag{39}$$

The shift identity is used for $r \in J^{-1}(N)$ with $T + Jr$ in the current scalar-extraction class; no equivariant parameter lift or stronger shift law is required. Indeed the compression identity extracts exactly the functional in its free position; its other insertion terms vanish. In the stable trace-functional and two-position cases J is injective: if $Jr = 0$, the same compression gives $\ell(x)P_v = 0$ for every free coefficient x, and $P_v \neq 0$ forces every extracted functional to vanish. The faithful quotient $\overline{S}$ also covers redundant parameter presentations without requiring an equivariant choice of a lift to S. In a one-pair packet use trace duality (37) instead. The first law in (39) follows from the covariance of the whole raw diagram and the uniqueness of its scalar quotient values. It includes the inherited semilinear and character action on S; no choice of a normalized similitude is necessary.

Proposition M.6 (Complete insertion-pair encoding)**.** *Let $J : S \to M$ be an equivariant insertion map on the original literal ambient, let $N \subseteq \operatorname{im} J$ be the actual translation image, and put $R = J^{-1}(N)$ and $\overline{R} = R/\ker J \subseteq \overline{S}$. On the scalar-extraction class define*

$$C(T) = T - \overline{J}\,\overline{s}(T), \qquad v(T) = \overline{s}(T) + \overline{R}.$$

The pair $(C(T), v(T))$ is well defined on $T + N$ and is equivariant under the full actual prefix. For every source and target in this class,

$$\{g : gT - T' \in N\} = \{g : gC(T) = C(T'),\ gv(T) = v(T')\}. \tag{40}$$

It preserves the entire transporter and stabilizer, not just the existence of an equivalence. The assertion does not require $\overline{s}(T) \in \overline{R}$.

Proof. For $r \in R$, the second law of (39) shows $C(T + Jr) = C(T)$ and $v(T + Jr) = v(T)$. Its first law proves equivariance. These facts give one inclusion in (40). Conversely, matching its two entries gives

$$gT - T' = \overline{J}\big(g\overline{s}(T) - \overline{s}(T')\big) \in \overline{J}(\overline{R}) = N.$$

This also proves the equality of stabilizers. A residual discrepancy in N is corrected through the actual physical translation map onto N; all of its kernel is restored as in Lemma 2.1. If the physical prefix acts affinely with increments in N, use its quotient action and these same actual correction words. No normalized complement to the physical normal p-subgroup is required. □

In particular $C(T)$ is a literal original-field tensor even when the chosen correction is not physically available by itself. Its missing affine information is exactly $v(T)$. First match this degree-one value completely, and then process the whole $C(T)$ under that exact returned prefix. The projected small-rank estimates do not imply small rank of the uncompressed labelled stack: the omitted output borders can stack across labels. Literal forcing is applied afresh to $C(T)$ inside the same node, rather than making that inference.

If a lower-critical part M_2 shares the translation image with the high part, use the single map and denominator

$$J_{\mathrm{joint}} : M_2 \oplus S \longrightarrow M_2 \oplus M, \quad (x, s) \longmapsto (x, Js), \qquad R_{\mathrm{joint}} = J_{\mathrm{joint}}^{-1}(N).$$

Take its faithful parameter quotient by $\ker J_{\mathrm{joint}}$. The extraction there is $(T_2, \overline{s}(T_3))$, its literal remainder is $(0, C(T_3))$, and its residual value is the one joint class modulo $R_{\mathrm{joint}}/\ker J_{\mathrm{joint}}$. The original complete preimage R_{joint} and every physical lift kernel remain recorded. Its possible affine cocycle stays in the degree-two procedure. This is an immediate application of the preceding proof and preserves correlations that separate low-part and parameter solves would destroy.

M.7 Moving outside records and perfect critical cores

We now verify that these constructions apply to the actual carried state of Definition 1.2. Write H for its group, R_p for its actual normal physical p-kernel, and $\overline{H} = H/R_p$. Its certified full natural carrier $\mathrm{U}(E)$ is normal in this diagonal quotient. It need not have a normal complement in H. The structural preparation of Appendix I retains the old outside actions and their permanent hereditary bound. In particular, a new critical perfect quasisimple core has trivial image in an old outside record. The degree-two procedure of Appendix K is used throughout with its actual-prefix return and supported whole label model.

Lemma M.7 (Raw centralizers under a complete prefix)**.** *Match a marked projected observation completely, retaining its field, form, character, exchange, and label actions. The returned group normalizes the corresponding raw algebra. Every newly prepared critical perfect core centralizes this raw algebra. Thus the natural-degree bounds of its raw centralizer remain valid without enumerating the outside records or scalar-shift images.*

Proof. An invertible change of the whole label basis leaves its generated operator algebra unchanged. A nonzero scalar multiple of a generator has the same generated algebra when that scalar lies in its distinguished field. Semilinear row maps transport this field as a set. Orthogonal projection is covariant under a similitude, since both H and $H^{\perp}$ are transported and the multiplier cancels. The general-linear projection is covariant under its natural–dual pair. Exchange records transport the retained adjoint partner. These observations prove normalization of the raw algebra by the complete returned group.

In the kernel of its old outside records, all labels, fields, and characters are fixed. For a marked projected observation its remaining action has the form

$$gD_{j,v}g^{-1} = D_{j,v} + c_{j,v}(g)P_v.$$

This scalar-shift record is an additive homomorphism: conjugation fixes K and P_v, so the increments add under multiplication. A new critical perfect core has trivial image in the old outside by the hereditary bound, and has trivial image in this abelian additive group because it is perfect. It therefore fixes every raw generator by conjugation. The same argument with zero scalar increments treats literal marked observations. All statements are made on the actual diagonal quotient; physical p-lifts and the current section gauge are retained on returning to H. □

For completeness, in a fixed-record branch the correlated scalar image can alternatively be enumerated at paid cost. On $U = \operatorname{im} P_v$ the equations are $gD_jg^{-1} = D_j + c_j(g)I$. A Sylow p-subgroup maps onto their additive image and has a nonzero fixed vector $u \in U$. Applying $g - I$ to every relation among $u, D_1u, \dots, D_\ell u$ shows that $(c_j(g))_j$ belongs to a K-linear space of dimension at most $\dim_K U - 1$. Thus its actual, possibly only prime-field-linear image has size at most $|K|^{\dim_K U-1}$. At most three vertices still give $|K|^{O(a)}$. This enumeration is used only in the paid bad-tuple branch. For the full moving-record group the record may be crossed, so treating it directly as this additive homomorphism would be incorrect.

Lemma M.8 (Exact proper-subspace conductor)**.** *Let W be an intrinsic nonzero proper physical k-subspace of a certified full natural graded row, on which R_p acts trivially. More generally the same assertion applies to a covariant F_0 in a supplied one-sided module whose action descends to $\overline{H}$, with its normal pure action $\rho(E)$. Its pure stabilizer has the explicit star-algebra*

$$E_{F_0} = \{e \in E : \rho(e)F_0 \subseteq F_0,\ \rho(e^*)F_0 \subseteq F_0\}.$$

For a physical row use $\rho(e) = e$ and $F_0 = W$. The complete subspace transporter gives an unbranched proper conductor whenever the full pure core does not preserve the subspace. Its new outside quotient embeds in the old diagonal outside quotient, and the entire actual R_p-extension is retained.

Proof. The displayed set is closed under sums, products, identity, and star, so it is a represented star-subalgebra computable by linear equations. For a unitary element g, $g^* = g^{-1}$. Consequently $g \in \mathrm{U}(E_{F_0})$ if and only if $gF_0 = F_0$, proving equality with the pure stabilizer. The complete diagonal subspace stabilizer $\overline{H}_{F_0}$ normalizes this algebra and has

$$\overline{H}_{F_0} \cap \mathrm{U}(E) = \mathrm{U}(E_{F_0}).$$

Its quotient by this intersection embeds in $\overline{H}/\mathrm{U}(E)$. Pulling back to H therefore retains the whole normal p-extension, not merely a selected section. The unitary radical of E_{F_0} is absorbed into this kernel, with its actual lifts, and its semisimple natural factors are obtained by represented algebra operations. A nonzero proper physical subspace cannot be invariant under a full irreducible natural core. For a one-sided module the assertion of properness has the explicitly stated noninvariance hypothesis. Hence the conductor is proper in the asserted cases. No numerical natural-degree loss is needed for this unbranched strict restriction. □

A whole label subspace need not descend to H/R_p: R_p can act nontrivially on its labels. In that case the expression H_{F_0}/R_p is not used. The complete supported frame call returns the actual H_{F_0}, its intersection with R_p, and all lifts; structural preparation is performed on this actual group. The small nonzero F_0 of Proposition M.4 still gives strict subgroup progress by its dimension argument. No outside-quotient embedding is required for this alternative.

Apply these facts to a nonzero common-field value $\lambda : C \to F \otimes L^*$, or $\overline{\lambda} : C \to (F/K) \otimes L^*$. Every genuine transporter has a common field isomorphism on the first factor and the actual invertible label action on the second. Its physical kernel is therefore intrinsic. If this kernel is nonzero and proper, match exactly this k-subspace and use Lemma M.8; no field phase is selected and no claim of K_C-linearity is made.

If the value is injective, use the common field constructed for both source and target by Appendix L. For each fixed field phase θ, solve the complete value equations by the supported whole-Hom procedure, with F as passive explicit coefficient multiplicity. If the old field action can move, use $k = \mathbb{F}_p$, enumerate all prime-field Frobenius powers of F, and impose the actual restriction of θ to

K in this same call. There are at most $f[F : K] = \exp(O(fa))$ such phases. All characters remain in the old correlated label model.

In the returned stabilizer for a fixed phase, a pure coefficient element fixing the labels satisfies $\lambda g_C = \lambda$, so injectivity gives $g_C = I$. A possible new critical perfect factor also has trivial old outside image by the hereditary bound; it cannot survive as a diagonal tie with that outside. Its coefficient projection is therefore trivial by the same equation. Thus the entire current three-slot observation has at most two critical slots under this returned prefix. Complete that whole observation, including its affine ambiguity and every inherited matched-prefix condition, by Appendix K before proceeding to the next phase. Reunite only these completed terminal cosets in the original group. The phase costs add; they are not exported as recursive choices. This argument uses the supplied support realization of L. Pure-core triviality of an arbitrary unrepresented label action would not suffice.

M.8 Physical progress and the complete interface

Every operation above keeps the original physical filtration, current numerator and denominator, completed observation family, and actual translation lift map. The alternatives are now exhaustive. A packet with no insertion is literal. Each permitted insertion pattern has a paid high-rank or bad-tuple alternative, a strict unbranched physical conductor, a nonzero field-value transition, or scalar extraction. The unbalanced one-pair case additionally has its exact pure-insertion parameter terminal. Scalar extraction gives the complete pair of Proposition M.6, including the joint lower-critical residual when necessary.

This last transition consumes the current insertion denominator once. It is a change of representation inside the node: the literal first entry is subsequently processed under the complete prefix obtained from the residual value. A new affine denominator can arise only after strict advancement of the retained physical layer or a proper physical conductor or paid fission. It cannot be recreated by dropping the pair and restarting an abstract cubic quotient problem. This is the consumed phase in Definition 1.2, whose finite progress is counted in Section 6.

All coordinate spaces used here have polynomial size in the original physical degree and explicit labels, except the glued field coordinates. Those occur only inside complete local operations of cost $|k|^{O(D_0)}\,\mathrm{poly}(\mathsf{L})$. Marks in the projected same-row cases have entropy $O(fa)$ and their bad tuples have a constant-fraction row loss. Balanced one-pair marks are bounded by the free-row degree, which is no greater than the paired-row degree. Unbalanced marks use only the paired rows and their high-rank loss. Proper conductors and canonical pair encoding export no target lists. Together with the actual-prefix lemmas, these are precisely the insertion alternatives and return properties used by Sections 4 and 6.

N Scope and mathematical dependencies

The algorithm uses established external results only at the following points: finite represented-algebra structure and module isomorphism; star-norm correction and unitary generators; deterministic permutation and specified quotient-group operations; the large primitive linear-group structure theorem and its standard natural representation alternatives; the stated stable degree-two constituent catalog and characteristic-prime-to-p multiplier facts; and constructive noncommutative rank. Their exact applications and bibliographic references are given where they enter the proofs.

The remaining reductions are proved here. The finite-torus interpolation is an equality of actual joint image algebras and treats small odd fields. Its actual p-subgroup and fixed-coset consequences do not assume an algebra group. The whole-Hom quotient and derived-tail arguments

give the physical section. The graph construction keeps its new invisible prime-to-p actions and uses only bounded point filters below the dynamic transporter. The semisimple shift potential and pure normal-kernel recursion close the invariant-subquotient equation. The dynamic algorithm preserves exact fibers, physical ancestry and fixed arity. Only afterward is general matrix-group coset intersection derived.

At the outer level, the common-group component theorem preserves one correlated translation action. Compatible-frame normalization charges only newly created multiplicity. The raw-triple field construction uses no short primitive tuple and enumerates no elements of the exponentially large auxiliary field. All insertion variants stay on the original prime-field tensor rows; their canonical extraction is taken in the faithful parameter quotient when the insertion map has a kernel. The low-rank exit needs a boundary small relative to the two operator rows, not the coefficient row. The final numerical argument counts the whole weighted tree and explicitly handles its unique heavy continuations.

These restrictions are part of the theorem's proof. In particular, the argument does not authorize arbitrary cubic quotient inputs, arbitrary unsupported outside modules, independent local shift spaces in place of a correlated denominator, enumeration of the whole retired outside group, or a fresh unrestricted same-size recursive call after only an algebraic core drop. It gives no assertion in characteristic two or for arity growing with the input. The manuscript is a proposed research proof and should be read with the usual independent scrutiny for an unpublished result.

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